

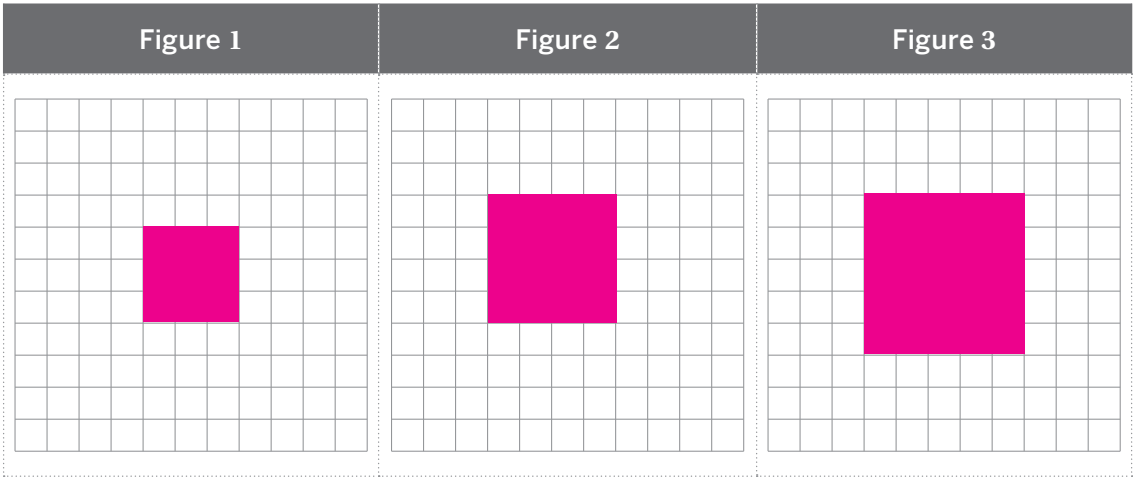
Additional Practice

A1.1.01

1. Figures 1–3 represent a square growing in size.

Figure	Number of Tiles
1	9
2	16
3	25

a Draw three figures to match the patterns in the table.



b How many tiles will there be in figure 4?

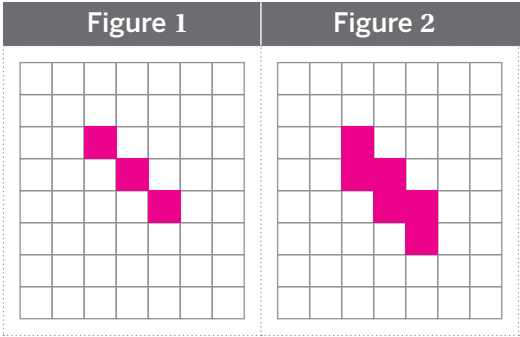
36 tiles

2. Sketch figure 1 and figure 2 of a visual pattern that doubles every stage.

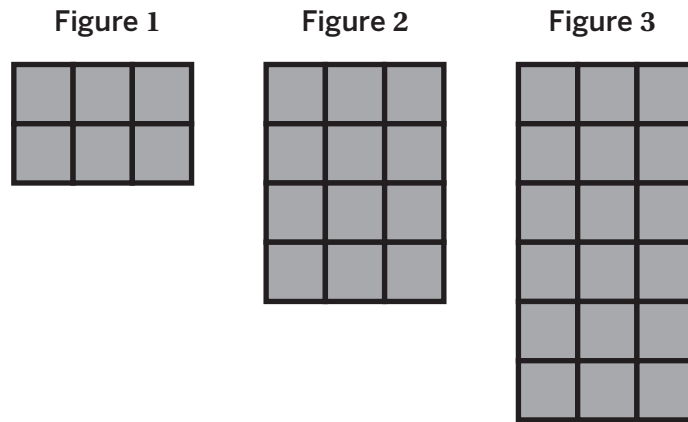
Responses vary.

3. Create an expression that could represent the number of tiles in figure 3.

Responses vary. $2 \cdot 2(3)$



4. Here is a visual pattern.



- a** Describe what figure 4 will look like.

Responses vary. I think figure 4 will be a rectangle. The rectangle will have 3 columns of tiles and 8 rows of tiles.

- b** How many tiles will there be in figure 4?

24 tiles

Problems 5–6: This table shows the number of tiles in figures 1–3.

Figure	Number of Tiles
1	6
2	12
3	18

5. Will figure 5 have $24 + 6$ tiles? Circle one. Explain your thinking.

☒ Yes

☐ No

☐ I'm not sure

Explanations vary. The difference between figure 1 and figure 2 is 6. The difference between figure 2 and figure 3 is 6. It makes sense that the next figure would grow by 6 tiles, so figure 4 will have 24 tiles and figure 5 will have $24 + 6$ tiles.

6. How many tiles will there be in figure 6?

36 tiles

Name: _____ Date: _____ Period: _____

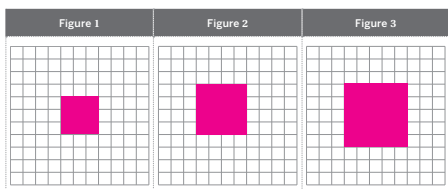
Additional Practice

A1.1.01

1. Figures 1–3 represent a square growing in size.

Figure	Number of Tiles
1	9
2	16
3	25

- a. Draw three figures to match the patterns in the table.



- b. How many tiles will there be in figure 4?

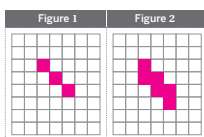
36 tiles

2. Sketch figure 1 and figure 2 of a visual pattern that doubles every stage.

Responses vary.

3. Create an expression that could represent the number of tiles in figure 3.

Responses vary. $2 \cdot 2(3)$



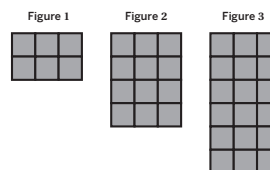
Unit 1 Lesson 1

1

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Name: _____ Date: _____ Period: _____

4. Here is a visual pattern.



- a. Describe what figure 4 will look like.

Responses vary. I think figure 4 will be a rectangle. The rectangle will have 3 columns of tiles and 8 rows of tiles.

- b. How many tiles will there be in figure 4?

24 tiles

Problems 5–6: This table shows the number of tiles in figures 1–3.

Figure	Number of Tiles
1	6
2	12
3	18

5. Will figure 5 have $24 + 6$ tiles? Circle one. Explain your thinking.

☒ Yes ☐ No ☐ I'm not sure

Explanations vary. The difference between figure 1 and figure 2 is 6. The difference between figure 2 and figure 3 is 6. It makes sense that the next figure would grow by 6 tiles, so figure 4 will have 24 tiles and figure 5 will have $24 + 6$ tiles.

6. How many tiles will there be in figure 6?

36 tiles

Unit 1 Lesson 1

2

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.BF.A.1.A
2	2	HSF.BF.A.1.A
3	2	HSF.BF.A.1.A
4	2	HSF.BF.A.1.A
5	2	HSF.BF.A.1.A
6	2	HSF.BF.A.1.A

Notes:

Additional Practice

A1.1.02

1. Here is a sequence that has a constant ratio.

200, 100, 50, ...

What is the next term? Explain your thinking.

25. Explanations vary. I divided each term by 2 because each term is half as big as the one before it.

Problems 2–4: Fill in the blanks to complete each sequence. Each sequence has a constant difference.

2. 10, 13, 16, 19, 22, 25

3. 5, 11, 17, 23, 29, 35

4. 1.5, 1.0, 0.5, 0.0, -0.5, -1.0

5. Which sequence has a constant ratio of 4?

- ☒ A. 16, 64, 256, ...
- ☐ B. 24, 20, 16, ...
- ☐ C. 80, 20, 5, ...
- ☐ D. 4, 8, 12, ...

Problems 6–8: Here is the start of a sequence: 5, 9, 13, . . .

6. Determine the next two terms for the sequence.

17, 21

7. What will be the 6th term in the sequence? Show or explain your thinking.

25. Explanations vary. I noticed that each term adds 4 to the previous term. Since the 5th term is 21, I added 4 to that and got 25.

8. What rule does the sequence follow?

- A. Constant ratio of 2
- B. Constant ratio of 4
- C. Constant difference of 2
- D. Constant difference of 4**

9. A sequence has a first term of 12 and a constant ratio of $\frac{1}{2}$. What are the first four terms of the sequence?

- A. 12, $12\frac{1}{2}$, 13, $13\frac{1}{2}$
- B. $\frac{1}{2}$, $12\frac{1}{2}$, $24\frac{1}{2}$, $36\frac{1}{2}$
- C. 12, 6, 3, $\frac{3}{2}$**
- D. 12, 24, 36, 48

Name: _____ Date: _____ Period: _____

Additional Practice

A1.1.02

1. Here is a sequence that has a constant ratio.

200, 100, 50, ...

What is the next term? Explain your thinking.

25. *Explanations vary. I divided each term by 2 because each term is half as big as the one before it.*

Problems 2–4: Fill in the blanks to complete each sequence. Each sequence has a constant difference.

2. 10, 13, 16, 19, 22, 25

3. 5, 11, 17, 23, 29, 35

4. 1.5, 1.0, 0.5, 0.0, -0.5, -1.0

5. Which sequence has a constant ratio of 4?

- ☒ A. 16, 64, 256, ...
☐ B. 24, 20, 16, ...
☐ C. 80, 20, 5, ...
☐ D. 4, 8, 12, ...

Unit 1 Lesson 2

3

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Name: _____ Date: _____ Period: _____

Problems 6–8: Here is the start of a sequence: 5, 9, 13, ...

6. Determine the next two terms for the sequence.

17, 21

7. What will be the 6th term in the sequence? Show or explain your thinking.

25. *Explanations vary. I noticed that each term adds 4 to the previous term. Since the 5th term is 21, I added 4 to that and got 25.*

8. What rule does the sequence follow?

- A. Constant ratio of 2
☒ B. Constant ratio of 4
☐ C. Constant difference of 2
☐ D. Constant difference of 4

9. A sequence has a first term of 12 and a constant ratio of $\frac{1}{2}$. What are the first four terms of the sequence?

- A. 12, $12\frac{1}{2}$, 13, $13\frac{1}{2}$
☒ B. $\frac{1}{2}$, $12\frac{1}{2}$, $24\frac{1}{2}$, $36\frac{1}{2}$
☐ C. 12, 6, 3, $\frac{3}{2}$
☐ D. 12, 24, 36, 48

Unit 1 Lesson 2

4

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSF.BF.A.1.A
2	2	HSF.BF.A.1.A
3	2	HSF.BF.A.1.A
4	2	HSF.BF.A.1.A
5	2	HSF.BF.A.1.A
6	2	HSF.BF.A.1.A
7	2	HSF.BF.A.1.A
8	2	HSF.BF.A.1.A
9	2	HSF.BF.A.1.A

Notes:

Additional Practice

A1.1.03

1. This sequence has a constant ratio of 5. Fill in the missing terms.

$\frac{1}{5}$, 1, 5, 25, 125, 625

2. This sequence has a constant difference of 7. Fill in the missing terms.

-6, 1, 8, 15, 22

3. Here is the start of a sequence: -2, 1, 4, ...

- a Write a rule for this sequence.

Constant difference of 3

- b Write the next three terms of this sequence.

7, 10, 13

Problems 4–5: Here is the start of a sequence: 1, 6, ...

4. Write a rule the sequence could follow and the next three terms.

Rule: Constant difference of 5 Terms: 11, 16, 21

5. Write a *different* rule the sequence could follow and the next three terms.

Rule: Constant ratio of 6 Terms: 36, 216, 1296

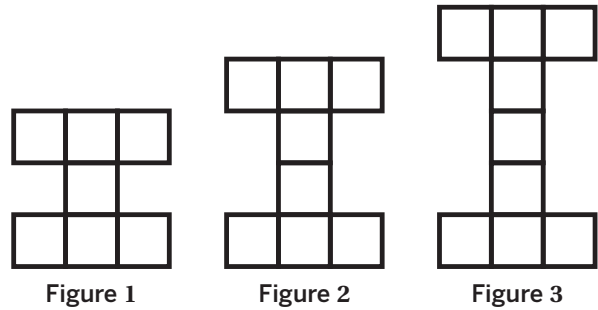
6. Brenden is saving money to buy a new phone. He started with \$200 and is saving an additional \$50 each month. Write a recursive definition to model the situation.

First term: 200 Rule: Constant difference of 50

Problems 7–9: Here is a visual pattern.

7. Complete the table with the number of tiles in each figure.

Figure	Tiles
1	7
2	8
3	9



8. Write a recursive definition to model the situation.

First term: 7

Rule: Constant difference of 1

9. How many tiles would be in figure 10? Show or explain your thinking.

16 tiles. *Explanations vary.* There will be $7 + 10 - 1$ or 16 tiles. The pattern started with 7 tiles in the first figure and adds 1 tile each time. So, figure 1 was $7 + (1 - 1)$, figure 2 was $7 + (2 - 1)$, figure 3 was $7 + (3 - 1)$, and so on. Therefore, figure 10 will be $7 + (10 - 1)$ or 16 tiles.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.1.03

1. This sequence has a constant ratio of 5. Fill in the missing terms.

$\frac{1}{5}$, 1, 5, 25, 125, 625

2. This sequence has a constant difference of 7. Fill in the missing terms.

-6, 1, 8, 15, 22

3. Here is the start of a sequence: -2, 1, 4, ...

a Write a rule for this sequence.

Constant difference of 3

b Write the next three terms of this sequence.

7, 10, 13

Problems 4–5: Here is the start of a sequence: 1, 6, ...

4. Write a rule the sequence could follow and the next three terms.

Rule: Constant difference of 5

Terms: 11, 16, 21

5. Write a different rule the sequence could follow and the next three terms.

Rule: Constant ratio of 6

Terms: 36, 216, 1296

6. Brenden is saving money to buy a new phone. He started with \$200 and is saving an additional \$50 each month. Write a recursive definition to model the situation.

First term: 200

Rule: Constant difference of 50

Unit 1 Lesson 3

5

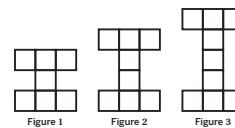
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Name: _____ Date: _____ Period: _____

Problems 7–9: Here is a visual pattern.

7. Complete the table with the number of tiles in each figure.

Figure	Tiles
1	7
2	8
3	9



8. Write a recursive definition to model the situation.

First term: 7

Rule: Constant difference of 1

9. How many tiles would be in figure 10? Show or explain your thinking.

16 tiles. Explanations vary. There will be 7 + 10 - 1 or 16 tiles. The pattern started with 7 tiles in the first figure and adds 1 tile each time. So, figure 1 was 7 + (1 - 1), figure 2 was 7 + (2 - 1), figure 3 was 7 + (3 - 1), and so on. Therefore, figure 10 will be 7 + (10 - 1) or 16 tiles.

Unit 1 Lesson 3

6

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.BF.A.2, HSF.LE.A.1.B
2	1	HSF.BF.A.2, HSF.LE.A.1.B
3	1	HSF.BF.A.2
4	2	HSF.BF.A.2
5	2	HSF.BF.A.2
6	2	HSF.BF.A.2
7	1	HSF.LE.A.1.B
8	2	HSF.BF.A.2
9	2	HSF.BF.A.2

Notes:

Additional Practice

A1.1.04

1. Determine whether each sequence is arithmetic, geometric, or neither.

Sequence	3, 10, 17, 24	1, 10, 100, 1000	12, 6, 3, $\frac{3}{2}$	1, 4, 9, 25
Arithmetic, Geometric, or Neither	Arithmetic	Geometric	Geometric	Neither

2. Complete each arithmetic sequence with its missing terms.

- a -10, -4, **2**, 8, **14**, 20
- b 14, **114**, 214, **314**, 414
- c **3**, 4.25, 5.50, **6.75**, **8**
- d 12, **5**, -2, -9, **-16**

3. Complete each geometric sequence with its missing terms.

- a 10, 20, **40**, 80, **160**, 320
- b 12, **6**, 3, **$\frac{3}{2}$** , $\frac{3}{4}$
- c **$\frac{4}{3}$** , 4, 12, **36**, **108**
- d 1, **5**, 25, 125, **625**

4. For each sequence, identify the type of sequence, the first term, and a rule to produce a new term from the previous term.

Sequence 1	Sequence 2	Sequence 3												
<table><tr><th>Term</th><th>Value</th></tr><tr><td>1</td><td>-2</td></tr><tr><td>2</td><td>5</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>19</td></tr><tr><td>5</td><td>26</td></tr></table>	Term	Value	1	-2	2	5	3	12	4	19	5	26		
Term	Value													
1	-2													
2	5													
3	12													
4	19													
5	26													
Type: Arithmetic First term: -2 Rule: Add 7 to the previous term	Type: Geometric First term: 1 Rule: Multiply the previous term by 5	Type: Arithmetic First term: 4 Rule: Subtract 2 from the previous term												

5. Complete the table.
- a Circle if the sequence is arithmetic, geometric, or neither.
- b Write the first 4 terms of the sequence.

	First term: 2 Rule: Multiply the previous term by 4	First term: 1 Rule: Add 1 to the previous number and square it	First term: -12 Rule: Add 4 to the previous number
a	Arithmetic Sequence <u>Geometric Sequence</u> Neither	Arithmetic Sequence Geometric Sequence <u>Neither</u>	<u>Arithmetic Sequence</u> Geometric Sequence Neither
b	2, 8, 32, 128	1, 4, 25, 676	-12, -8, -4, 0

Name: _____ Date: _____ Period: _____

Additional Practice

A1.1.04

1. Determine whether each sequence is arithmetic, geometric, or neither.

Sequence	3, 10, 17, 24	1, 10, 100, 1000	12, 6, 3, $\frac{3}{2}$	1, 4, 9, 25
Arithmetic, Geometric, or Neither	Arithmetic	Geometric	Geometric	Neither

2. Complete each arithmetic sequence with its missing terms.

- a. $-10, -4, \underline{2}, 8, \underline{14}, 20$
 b. $14, \underline{114}, 214, \underline{314}, 414$
 c. $\underline{3}, 4.25, 5.50, \underline{6.75}, \underline{8}$
 d. $12, \underline{5}, -2, -9, \underline{-16}$

3. Complete each geometric sequence with its missing terms.

- a. $10, 20, \underline{40}, 80, \underline{160}, 320$
 b. $12, \underline{6}, 3, \underline{\frac{3}{2}}, \underline{\frac{3}{4}}$
 c. $\underline{\frac{4}{3}}, 4, 12, \underline{36}, \underline{108}$
 d. $1, \underline{5}, 25, 125, \underline{625}$

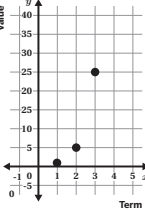
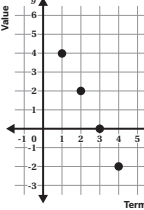
Unit 1 Lesson 4

7

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Name: _____ Date: _____ Period: _____

4. For each sequence, identify the type of sequence, the first term, and a rule to produce a new term from the previous term.

Sequence 1		Sequence 2	Sequence 3											
<table><thead><tr><th>Term</th><th>Value</th></tr></thead><tbody><tr><td>1</td><td>-2</td></tr><tr><td>2</td><td>5</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>19</td></tr><tr><td>5</td><td>26</td></tr></tbody></table>	Term	Value	1	-2	2	5	3	12	4	19	5	26		
Term	Value													
1	-2													
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4	19													
5	26													
Type: Arithmetic First term: -2 Rule: Add 7 to the previous term	Type: Geometric First term: 1 Rule: Multiply the previous term by 5	Type: Arithmetic First term: 4 Rule: Subtract 2 from the previous term												

5. Complete the table.

- a. Circle if the sequence is arithmetic, geometric, or neither.
 b. Write the first 4 terms of the sequence.

	First term: 2 Rule: Multiply the previous term by 4	First term: 1 Rule: Add 1 to the previous number and square it	First term: -12 Rule: Add 4 to the previous number
a.	Arithmetic Sequence <u>Geometric Sequence</u> Neither	Arithmetic Sequence Geometric Sequence <u>Neither</u>	<u>Arithmetic Sequence</u> Geometric Sequence Neither
b.	2, 8, 32, 128	1, 4, 25, 676	-12, -8, -4, 0

Unit 1 Lesson 4

8

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSF.LE.A.1.B
2	2	HSF.LE.A.2
3	2	HSF.LE.A.2
4	2	HSF.LE.A.1.B, HSF.LE.A.2
5	2	HSF.LE.A.1.B

Notes:

Additional Practice

A1.1.05

1. Hannah's family decides to save money for a vacation. They start by saving \$100 in the first month, and each subsequent month they save 1.5 times the amount they saved the previous month. Select *all* the expressions that represent how much money they will save after 5 months.

☐ A. $100 + 1.5^5$

☐ B. $100 + 1.5 + 1.5 + 1.5 + 1.5 + 1.5$

☐ C. $100 \cdot 5^{1.5}$

☒ D. $100 \cdot 1.5 \cdot 1.5 \cdot 1.5 \cdot 1.5 \cdot 1.5$

☒ E. $100 \cdot 1.5^5$

2. The tables show the number of red and yellow globs each day.

Day	0	1	2	3	4
Red Globs	50	70	90	110	130

Day	0	1	2	3	4
Yellow Globs	5	10	20	40	80

- a Complete each table with the number of globs there will be on Day 4. Show or explain your thinking.

Explanations vary. In the first table, the number of red globs is increasing by adding 20 to the previous day. In the second table, the number of yellow globs is increasing by multiplying the number of globs from the previous day by 2, or doubling it.

- b Will there be more red or yellow globs on Day 10? Show or explain your thinking.

Yellow globs. Explanations vary. There will be more yellow globs on Day 10 because these numbers are growing at a faster rate.

- c Which group of globs grows by a constant difference? Show or explain your thinking.

Red globs. Explanations vary. The red globs are growing by a constant difference because they are increasing by the same number of globs every day.

3. Scientists tracked a population of birds in a park. They represented this number using the equation $n = 24 \cdot 1.25^t$, where n represents the number of birds and t represents the number of years since the scientists started counting.

- a Explain what the 24 and the 1.25 represent in this situation.

The 24 represents the starting population of birds. The 1.25 represents the growth factor of the bird population each year.

- b What was the bird population after 5 years. Show or explain your thinking.

Around 74 birds. Explanations vary. $n = 24 \cdot 1.25^5 = 73.2421875$ or approximately 74

Problems 4–5: Here is a table representing a pattern.

4. Which equation represents the table?

- A. $y = 50 + 10x$
- B. $y = 50 \cdot 10x$
- C. $y = 50 - 10x$**
- D. $y = 50 \cdot \left(\frac{1}{10}\right)^x$

x	y
0	50
1	40
2	30
3	20
4	10

5. Explain your thinking.

Explanations vary. The starting value is 50 and the constant difference is -10 .

6. Kai is saving pennies, and starts with 20 pennies. Each week, the number of pennies Kai has triples.

- a** Complete the table to show the amount of pennies Kai saved after n weeks.

Week, n	0	1	2	3	...	n
Number of Pennies	20	60	180	540	...	$20 \cdot 3^n$

- b** Write an explicit expression for this situation. **$20 \cdot 3^n$**
- c** Use your expression to determine how many pennies Kai saved in the 8th week. Show or explain your thinking.

131,220 pennies. Explanations vary. $20 \cdot 3^8 = 131220$ pennies

Name: _____ Date: _____ Period: _____

Additional Practice

A1.1.05

1. Hannah's family decides to save money for a vacation. They start by saving \$100 in the first month, and each subsequent month they save 1.5 times the amount they saved the previous month. Select *all* the expressions that represent how much money they will save after 5 months.

- ☐ A. $100 + 1.5^5$ ☐ B. $100 + 1.5 + 1.5 + 1.5 + 1.5 + 1.5$
☐ C. $100 \cdot 5^{1.5}$ ☒ D. $100 \cdot 1.5 \cdot 1.5 \cdot 1.5 \cdot 1.5 \cdot 1.5$
☒ E. $100 \cdot 1.5^5$

2. The tables show the number of red and yellow globs each day.

Day	0	1	2	3	4
Red Globs	50	70	90	110	130

Day	0	1	2	3	4
Yellow Globs	5	10	20	40	80

- a. Complete each table with the number of globs there will be on Day 4. Show or explain your thinking.
Explanations vary. In the first table, the number of red globs is increasing by adding 20 to the previous day. In the second table, the number of yellow globs is increasing by multiplying the number of globs from the previous day by 2, or doubling it.
- b. Will there be more red or yellow globs on Day 10? Show or explain your thinking.
Yellow globs. Explanations vary. There will be more yellow globs on Day 10 because these numbers are growing at a faster rate.
- c. Which group of globs grows by a constant difference? Show or explain your thinking.
Red globs. Explanations vary. The red globs are growing by a constant difference because they are increasing by the same number of globs every day.
3. Scientists tracked a population of birds in a park. They represented this number using the equation $n = 24 \cdot 1.25^t$, where n represents the number of birds and t represents the number of years since the scientists started counting.
- a. Explain what the 24 and the 1.25 represent in this situation.
The 24 represents the starting population of birds. The 1.25 represents the growth factor of the bird population each year.
- b. What was the bird population after 5 years. Show or explain your thinking.
Around 74 birds. Explanations vary. $n = 24 \cdot 1.25^5 \approx 73.2421875$ or approximately 74

Unit 1 Lesson 5

9

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Name: _____ Date: _____ Period: _____

Problems 4–5: Here is a table representing a pattern.

x	y
0	50
1	40
2	30
3	20
4	10

4. Which equation represents the table?

- A. $y = 50 + 10x$
 B. $y = 50 \cdot 10x$
☒ C. $y = 50 - 10x$
 D. $y = 50 \cdot \left(\frac{1}{10}\right)^x$

5. Explain your thinking.

Explanations vary. The starting value is 50 and the constant difference is -10 .

6. Kai is saving pennies, and starts with 20 pennies. Each week, the number of pennies Kai has triples.

- a. Complete the table to show the amount of pennies Kai saved after n weeks.

Week, n	0	1	2	3	...	n
Number of Pennies	20	60	180	540	...	$20 \cdot 3^n$

- b. Write an explicit expression for this situation. $20 \cdot 3^n$
 c. Use your expression to determine how many pennies Kai saved in the 8th week. Show or explain your thinking.
131,220 pennies. Explanations vary. $20 \cdot 3^8 = 131220$ pennies

Unit 1 Lesson 5

10

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.SSE.A.1
2	2	HSF.BF.A.1.A
3	2	HSA.SSE.A.1
4	1	HSA.SSE.A.1
5	1	HSA.SSE.A.1
6	2	HSF.BF.A.1.A, HSF.LE.A.2

Notes:

Additional Practice

A1.1.06

1. Match each sequence to its explicit expression.

Sequence

Explicit Expression

a. 3, 12, 48, 192

c $3 \cdot 2^{(n-1)}$

b. 3, 5, 7, 9

a $3 \cdot 4^{(n-1)}$

c. 3, 6, 12, 24

d $3 + 4(n-1)$

d. 3, 7, 11, 15

b $3 + 2(n-1)$

2. Write an explicit expression for each pattern.

Figure	# of Tiles
1	11
2	14
3	17
4	20

$11 + 3(n-1)$ (or equivalent)

Figure	# of Tiles
1	5
2	20
3	80
4	320

$5 \cdot 4(n-1)$ (or equivalent)

3. Select *all* the expressions that could represent the number of tiles in figure n of this pattern.

☒ A. $3 + 2n$

☐ B. $4 + 1(n-1)$

☐ C. $6n - 1$

☒ D. $5 + 2(n-1)$

☐ E. $4n + 1$



Figure 1

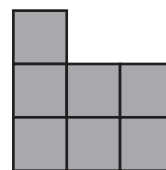


Figure 2

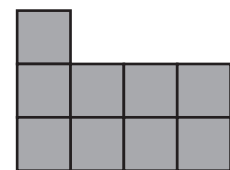
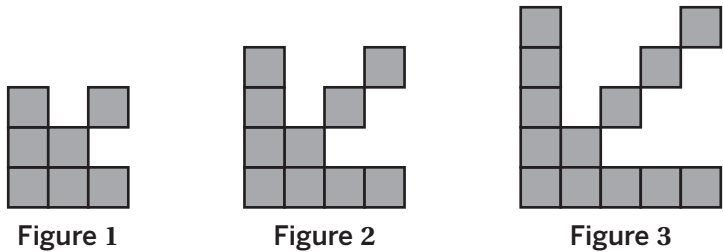


Figure 3

Problems 4–6: Here is a visual pattern.

4. Complete the table with the number of tiles in each figure.

Figure	Number of Tiles
1	7
2	10
3	13
4	16
...	...
12	40



5. Write an explicit expression for the number of tiles in figure n .

- $4 + 3n$ (or equivalent)
- $7 + 3(n - 1)$ (or equivalent)

6. Use one of your expressions to determine the number of tiles in figure 12.
Show or explain your thinking.

40 tiles. Work varies.

- $4 + 3(12) = 4 + 36 = 40$
- $7 + 3(12 - 1) = 7 + 3(11) = 7 + 33 = 40$

7. The first four terms in a sequence are 2, 6, 18, 54. Write two explicit expressions for term n of this sequence.

Expression 1: $2 \cdot 3^{(n-1)}$ (or equivalent)

Expression 2: $\frac{2}{3} \cdot 3^n$ (or equivalent)

Name: _____ Date: _____ Period: _____

Additional Practice

A1.1.06

1. Match each sequence to its explicit expression.

Sequence

a. 3, 12, 48, 192

Explicit Expression

c. $3 \cdot 2^{(n-1)}$

b. 3, 5, 7, 9

a. $3 + 4^{(n-1)}$

c. 3, 6, 12, 24

d. $3 + 4(n-1)$

d. 3, 7, 11, 15

b. $3 + 2(n-1)$

2. Write an explicit expression for each pattern.

Figure	# of Tiles
1	11
2	14
3	17
4	20

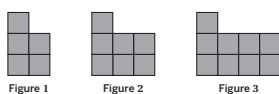
$11 + 3(n-1)$ (or equivalent)

Figure	# of Tiles
1	5
2	20
3	80
4	320

$5 \cdot 4^{(n-1)}$ (or equivalent)

3. Select all the expressions that could represent the number of tiles in figure n of this pattern.

- ☒ A. $3 + 2n$
☐ B. $4 + 1(n-1)$
☐ C. $6n - 1$
☒ D. $5 + 2(n-1)$
☐ E. $4n + 1$



Unit 1 Lesson 6

11

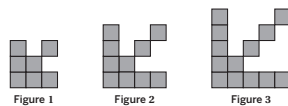
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Name: _____ Date: _____ Period: _____

Problems 4–6: Here is a visual pattern.

4. Complete the table with the number of tiles in each figure.

Figure	Number of Tiles
1	7
2	10
3	13
4	16
...	...
12	40



5. Write an explicit expression for the number of tiles in figure n .

- $4 + 3n$ (or equivalent)
- $7 + 3(n-1)$ (or equivalent)

6. Use one of your expressions to determine the number of tiles in figure 12. Show or explain your thinking.

- 40 tiles. Work varies.
 $4 + 3(12) = 4 + 36 = 40$
 $7 + 3(12-1) = 7 + 3(11) = 7 + 33 = 40$

7. The first four terms in a sequence are 2, 6, 18, 54. Write two explicit expressions for term n of this sequence.

Expression 1: $2 \cdot 3^{(n-1)}$ (or equivalent)

Expression 2: $\frac{2}{3} \cdot 3^n$ (or equivalent)

Unit 1 Lesson 6

12

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSF.BF.A.1.A
2	2	HSF.BF.A.2
3	2	HSF.BF.A.2
4	2	HSF.BF.A.2
5	2	HSF.BF.A.2
6	1	HSF.BF.A.1.A
7	2	HSF.BF.A.2

Notes:

Additional Practice

A1.2.10

1. Which value of x is a solution to the inequality $-x - 4 > 0$?

☒ A. $x = -5$ B. $x = -1$
 C. $x = 0$ D. $x = 2$

2. What is the solution to the inequality $2x - 3 \leq x$?

A. $x \geq 3$ ☒ B. $x \leq 3$
 C. $x \geq -3$ D. $x \leq -3$

3. Here is the inequality $\frac{8x-5}{3} \geq 2x + 1$. Select *all* of the values that are a solution to the inequality.

☐ A. $x = -1$ ☐ B. $x = 0$
☐ C. $x = 2$ ☒ D. $x = 4$
☒ E. $x = 5$ ☒ F. $x = 6$

4. Lin is solving the inequality $8x - 6 < 3x + 19$. She solves the equation $8x - 6 = 3x + 19$ and gets $x = 5$. Select *all* the statements that show how the solution to the equation $8x - 6 = 3x + 19$ helps Lin solve the inequality $8x - 6 < 3x + 19$.

☐ A. She can substitute 5 into the inequality to determine the correct solution.
☒ B. She can substitute 4 or 6 into the inequality to determine the correct solution.
☐ C. She can substitute -1 or 1 into the inequality to determine the correct solution.
☒ D. She knows that the solution must be either all the values less than 5 or all the values greater than 5.
☐ E. She knows that the solution must be either all the values less than or equal to 5 or all the values greater than or equal to 5.

5. Write three values of c that make this inequality true: $-4(c + 2) < 8$.

Responses greater than -4 are considered correct.

6. Solve each inequality.

a $2x - 3 \leq 21$

$x \leq 12$

b $-4x - 1 \leq 7$

$x \geq -2$

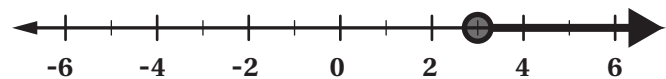
c $-5 + \frac{x}{3} > 2$

$x > 21$

d $-6x + 5 < 3x - 31$

$x > 4$

7. Which inequality is represented by the graph?



- A. $5x - 2 > 13$
 B. $5x - 2 \leq 13$
 C. $-5x + 2 \leq 13$
D. $-5x + 2 \leq -13$

8. Tyler argues that $x = 0$ is a solution to the inequality $-4x < -24$ because it is possible to divide both sides by -4 and get $x < 6$. Diego argues that $x = 0$ is not a solution to the inequality. Who is correct? Explain your thinking.

Diego. Explanations vary. Substituting 0 into the original inequality results in $0 < -24$, which is not a true statement.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.2.10

1. Which value of x is a solution to the inequality $-x - 4 > 0$?

☒ A. $x = -5$ B. $x = -1$
C. $x = 0$ D. $x = 2$

2. What is the solution to the inequality $2x - 3 \leq x$?

A. $x \geq 3$ ☒ B. $x \leq 3$
C. $x \geq -3$ D. $x \leq -3$

3. Here is the inequality $\frac{8x-5}{3} \geq 2x+1$. Select *all* of the values that are a solution to the inequality.

☐ A. $x = -1$ ☐ B. $x = 0$
☐ C. $x = 2$ ☒ D. $x = 4$
☒ E. $x = 5$ ☒ F. $x = 6$

4. Lin is solving the inequality $8x - 6 < 3x + 19$. She solves the equation $8x - 6 = 3x + 19$ and gets $x = 5$. Select *all* the statements that show how the solution to the equation $8x - 6 = 3x + 19$ helps Lin solve the inequality $8x - 6 < 3x + 19$.

☐ A. She can substitute 5 into the inequality to determine the correct solution.
☒ B. She can substitute 4 or 6 into the inequality to determine the correct solution.
☐ C. She can substitute -1 or 1 into the inequality to determine the correct solution.
☒ D. She knows that the solution must be either all the values less than 5 or all the values greater than 5.
☐ E. She knows that the solution must be either all the values less than or equal to 5 or all the values greater than or equal to 5.

Unit 2 Lesson 10

33

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Name: _____ Date: _____ Period: _____

5. Write three values of c that make this inequality true: $-4(c+2) < 8$.

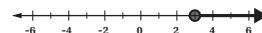
Responses greater than -4 are considered correct.

6. Solve each inequality.

a. $2x - 3 \leq 21$ b. $-4x - 1 \leq 7$
 $x \leq 12$ $x \geq -2$
c. $-5 + \frac{x}{3} > 2$ d. $-6x + 5 < 3x - 31$
 $x > 21$ $x > 4$

7. Which inequality is represented by the graph?

A. $5x - 2 > 13$
B. $5x - 2 \leq 13$
C. $-5x + 2 \leq 13$
☒ D. $-5x + 2 \leq -13$



8. Tyler argues that $x = 0$ is a solution to the inequality $-4x < -24$ because it is possible to divide both sides by -4 and get $x < 6$. Diego argues that $x = 0$ is not a solution to the inequality. Who is correct? Explain your thinking.

Diego. Explanations vary. Substituting 0 into the original inequality results in $0 < -24$, which is not a true statement.

Unit 2 Lesson 10

34

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSA.REI.B.3
2	2	HSA.REI.B.3
3	2	HSA.REI.B.3
4	2	HSA.REI.B.3
5	2	HSA.REI.B.3
6	2	HSA.REI.B.3
7	2	HSA.REI.B.3
8	2	HSA.REI.B.3

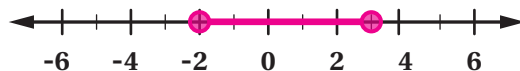
Notes:

Additional Practice

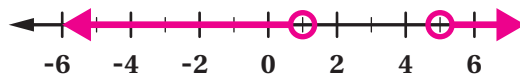
A1.2.11

Problems 1–3. Graph all the solutions to each inequality or equation.

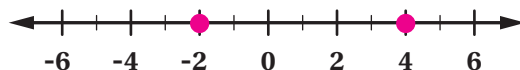
1. $|x| \leq 2$



2. $|x - 3| > 2$

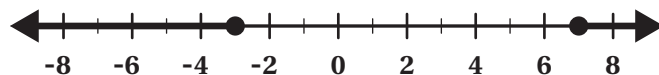


3. $|x - 1| = 3$

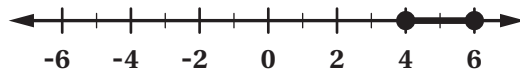


4. What value of t would make $|x - t| \geq 5$ match the graph?

$t = 2$



5. Which inequality's solutions are shown on the graph?



A. $|x - 1| \leq 5$

B. $|x - 5| \leq 1$

C. $|x + 5| \geq 1$

D. $|x + 1| \geq 5$

6. Piper is thinking of a number somewhere between -6 and 6 . Here are some guesses from Piper's friends:



No guess was more than 3 away from Piper's number. Only one guess was correct. What is Piper's number? Explain your thinking.

0. Explanations vary. Piper's friends guessed -3 , -2 , 0 , and 3 . The only number that is no more than 3 away from the other numbers is 0 .

Name: _____ Date: _____ Period: _____

Additional Practice

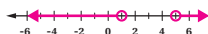
A1.2.11

Problems 1–3. Graph all the solutions to each inequality or equation.

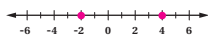
1. $|x| \leq 2$



2. $|x - 3| > 2$

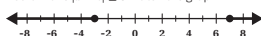


3. $|x - 1| = 3$

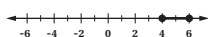


4. What value of t would make $|x - t| \geq 5$ match the graph?

$t = 2$



5. Which inequality's solutions are shown on the graph?



A. $|x - 1| \leq 5$

B. $|x - 5| \leq 1$

C. $|x + 5| \geq 1$

D. $|x + 1| \geq 5$

Unit 2 Lesson 11

35

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Name: _____ Date: _____ Period: _____

6. Piper is thinking of a number somewhere between -6 and 6 . Here are some guesses from Piper's friends:



No guess was more than 3 away from Piper's number. Only one guess was correct. What is Piper's number? Explain your thinking.

0. *Explanations vary. Piper's friends guessed -3 , -2 , 0 , and 3 . The only number that is no more than 3 away from the other numbers is 0.*

Unit 2 Lesson 11

36

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSA.REI.B.3
2	2	HSA.REI.B.3
3	2	HSA.REI.B.3
4	2	HSA.CED.A.1
5	2	HSA.CED.A.1
6	3	HSA.CED.A.1

Notes:

Additional Practice

A1.2.13

Problems 1–3: Selena can spend up to \$25 on bagels and croissants at the bakery. A bagel costs \$2 and a croissant costs \$3.

- b represents the number of bagels.
- c represents the number of croissants.

1. Which inequality represents the situation?

- ☒ A. $2b + 3c \leq 25$
- ☐ B. $2b + 3c \geq 25$
- ☐ C. $3b + 2c \leq 25$
- ☐ D. $3b + 2c \geq 25$

2. Explain how you know that $b = 2$ and $c = 3$ are solutions to this situation.

Explanations vary. When I substitute $b = 2$ and $c = 3$ into $2b + 3c \leq 25$, I get $2(2) + 3(3)$ is less than or equal to 25. Since 13 is less than 25, $b = 2$ and $c = 3$ are solutions.

3. Determine a different option for the number of bagels and the number of croissants that Selena can buy at the bakery.

Responses vary. 3 bagels and 2 croissants.

4. Write an ordered pair for a point that is *not* a solution to $x + 5y \geq 42$. Explain your thinking.

Responses and explanations vary. (1, 8). When I substitute $x = 1$ and $y = 8$ into the inequality $x + 5y \geq 42$, I get $1 + 40 \geq 42$, which is not true. Since 41 is not greater than or equal to 42, the ordered pair (1, 8) is not a solution to the inequality.

5. Anthony is at an art store and needs to buy green and red paint tubes for a project. A green paint tube costs \$6 and a red paint tube costs \$7. Anthony cannot spend more than \$49 on green and red paint at the art store.

- g represents the number of green paint tubes.
- r represents the number of red paint tubes.

Anthony says that the inequality $6g + 7r \geq 49$ represents all the green and red paint tubes that he can buy at the art store. Is Anthony correct? Explain your thinking.

No. Explanations vary. The inequality should be $6g + 7r \leq 49$. Anthony cannot spend at least \$49. Rather, Anthony can spend up to and including \$49 for the paints at the art store.

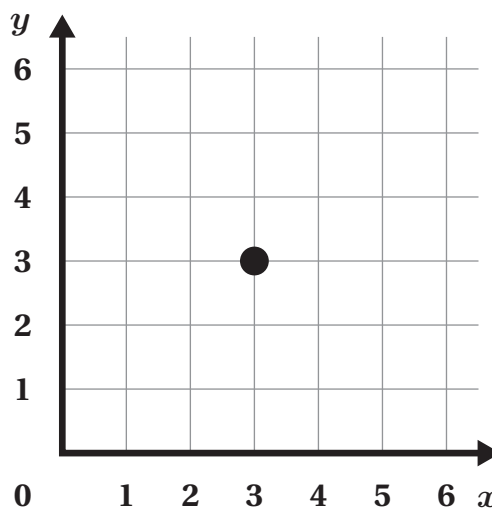
6. Determine if each ordered pair in the table is a solution to the inequality. Circle Yes or No in the table.

$$2x + 7y \leq 67$$

Ordered Pair	Is the ordered pair a solution?
(2, 4)	<u>Yes</u> or No
(3, 7)	<u>Yes</u> or No
(4, 9)	Yes or <u>No</u>

7. Is the point on the graph a solution to the inequality $x + 0.5y \geq 4$? Explain your thinking.

Yes. Explanations vary. When I substitute (3, 3) into the inequality, I get $3 + 0.5(3) \geq 4$, which is a true statement.



Name: _____ Date: _____ Period: _____

Additional Practice

A1.2.13

Problems 1–3: Selena can spend up to \$25 on bagels and croissants at the bakery. A bagel costs \$2 and a croissant costs \$3.

- b represents the number of bagels.
- c represents the number of croissants.

1. Which inequality represents the situation?

- ☒ A. $2b + 3c \leq 25$
☐ B. $2b + 3c \geq 25$
☐ C. $3b + 2c \leq 25$
☐ D. $3b + 2c \geq 25$

2. Explain how you know that $b = 2$ and $c = 3$ are solutions to this situation.

Explanations vary. When I substitute $b = 2$ and $c = 3$ into $2b + 3c \leq 25$, I get $2(2) + 3(3)$ is less than or equal to 25. Since 13 is less than 25, $b = 2$ and $c = 3$ are solutions.

3. Determine a different option for the number of bagels and the number of croissants that Selena can buy at the bakery.

Responses vary. 3 bagels and 2 croissants.

4. Write an ordered pair for a point that is not a solution to $x + 5y \geq 42$. Explain your thinking.

Responses and explanations vary. (1, 8). When I substitute $x = 1$ and $y = 8$ into the inequality $x + 5y \geq 42$, I get $1 + 40 \geq 42$, which is not true. Since 41 is not greater than or equal to 42, the ordered pair (1, 8) is not a solution to the inequality.

Unit 2 Lesson 13

39

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Name: _____ Date: _____ Period: _____

5. Anthony is at an art store and needs to buy green and red paint tubes for a project. A green paint tube costs \$6 and a red paint tube costs \$7. Anthony cannot spend more than \$49 on green and red paint at the art store.

- g represents the number of green paint tubes.
- r represents the number of red paint tubes.

Anthony says that the inequality $6g + 7r \geq 49$ represents all the green and red paint tubes that he can buy at the art store. Is Anthony correct? Explain your thinking.

No. Explanations vary. The inequality should be $6g + 7r \leq 49$. Anthony cannot spend at least \$49. Rather, Anthony can spend up to and including \$49 for the paints at the art store.

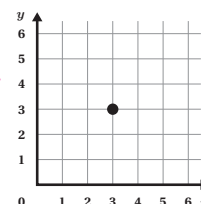
6. Determine if each ordered pair in the table is a solution to the inequality. Circle Yes or No in the table.

$$2x + 7y \leq 67$$

Ordered Pair	Is the ordered pair a solution?
(2, 4)	☒ Yes or ☐ No
(3, 7)	☒ Yes or ☐ No
(4, 9)	Yes or ☒ No

7. Is the point on the graph a solution to the inequality $x + 0.5y \geq 4$? Explain your thinking.

Yes. Explanations vary. When I substitute (3, 3) into the inequality, I get $3 + 0.5(3) \geq 4$, which is a true statement.



Unit 2 Lesson 13

40

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.CED.A.3
2	2	HSA.CED.A.3
3	2	HSA.CED.A.3
4	2	HSA.CED.A.3
5	2	HSA.CED.A.3
6	1	HSA.CED.A.3
7	1	HSA.CED.A.3

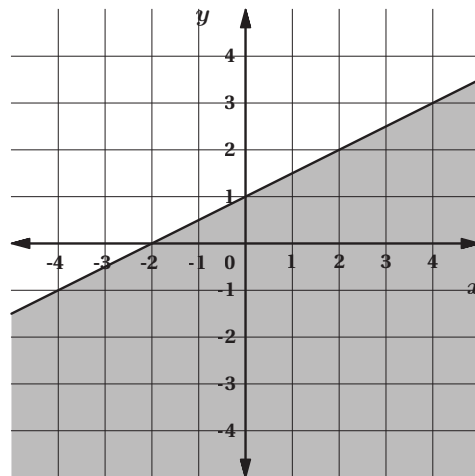
Notes:

Additional Practice

A1.2.14

1. Here is the graph of the inequality $x - 2y \geq -2$. Select *all* the points that are solutions.

- ☐ A. $(-3, 4)$
☒ B. $(-1, -2)$
☒ C. $(0, 0)$
☐ D. $(1, 3)$
☒ E. $(2, 2)$
☒ F. $(4, -1)$

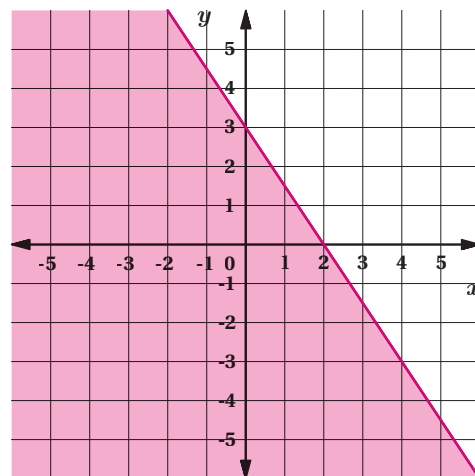


2. Here is the graph of the equation $3x + 2y = 6$.

- a Select *all* the points that are solutions to the inequality $3x + 2y \leq 6$.
- ☒ A. $(-1, 1)$
☒ B. $(2, 0)$
☒ C. $(0, 2)$
☐ D. $(4, 3)$
☐ E. $(5, -4)$

- b Graph all the solutions to $3x + 2y \leq 6$. Explain your thinking.

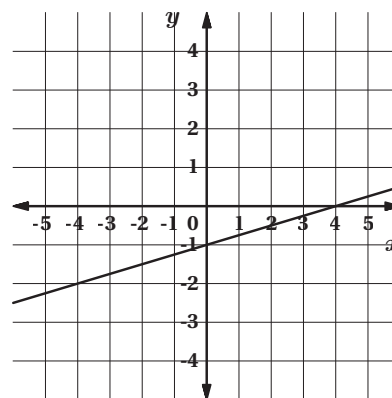
Explanations vary. The line is solid because the inequality sign is $\leq$. When I plugged $(0, 0)$ into the inequality, I got a true statement, so I shaded that side of the line.



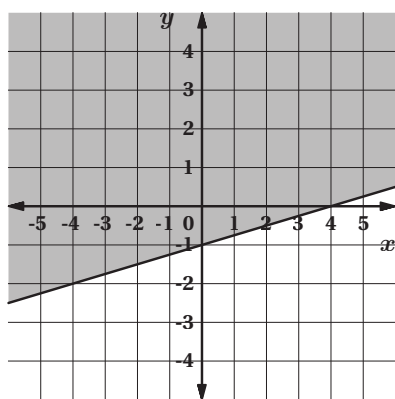
3. Select *all* the ordered pairs that are solutions to the inequality $8x - 3y > 24$.

- ☐ A. $(0, 0)$
☐ B. $(0, 3)$
☐ C. $(1, -1)$
☒ D. $(3, -8)$
☐ E. $(3, 0)$
☒ F. $(4, 1)$

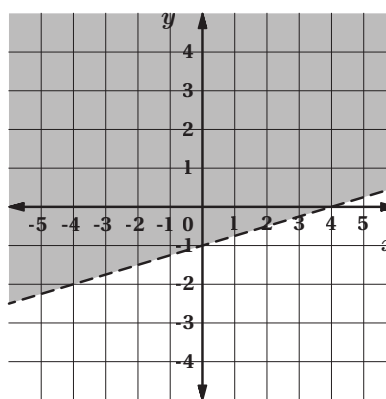
4. Here is the graph of the equation $x - 4y = 4$. Which of the following represents the solution set to the inequality $x - 4y \leq 4$?



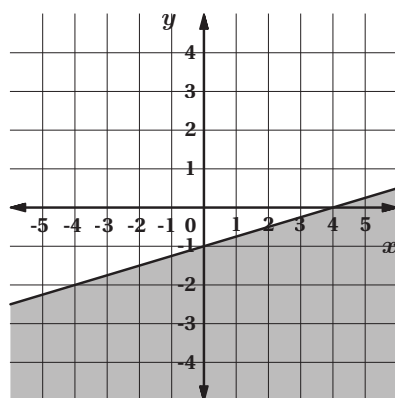
A.



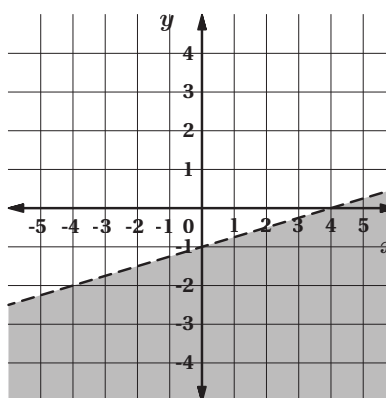
B.



C.



D.



5. Here is the inequality $3x + 7y > -6$. Priya argues that $(-4, 2)$ is a solution to the inequality, and Shawn argues that $(3, -2)$ is a solution. Who is correct? Circle one.

Priya

Shawn

Both are correct.

Explain your thinking.

Explanations vary. I substituted the values of both points into the inequality:
 $3(-4) + 7(2) = 2$ is greater than -6 and $3(3) + 7(-2) = -5$ is greater than -6 .

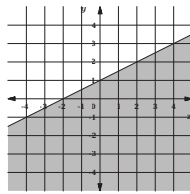
Name: _____ Date: _____ Period: _____

Additional Practice

A1.2.14

1. Here is the graph of the inequality $x - 2y \geq -2$. Select all the points that are solutions.

- ☐ A. $(-3, 4)$ ☒ B. $(-1, -2)$
☒ C. $(0, 0)$ ☐ D. $(1, 3)$
☒ E. $(2, 2)$ ☒ F. $(4, -1)$



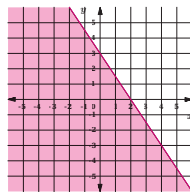
2. Here is the graph of the equation $3x + 2y = 6$.

- a. Select all the points that are solutions to the inequality $3x + 2y \leq 6$.

- ☒ A. $(-1, 1)$ ☒ B. $(2, 0)$
☒ C. $(0, 2)$ ☐ D. $(4, 3)$
☐ E. $(5, -4)$

- b. Graph all the solutions to $3x + 2y \leq 6$. Explain your thinking.

Explanations vary. The line is solid because the inequality sign is $\leq$. When I plugged $(0, 0)$ into the inequality, I got a true statement, so I shaded that side of the line.



3. Select all the ordered pairs that are solutions to the inequality $8x - 3y > 24$.

- ☐ A. $(0, 0)$ ☐ B. $(0, 3)$
☐ C. $(1, -1)$ ☒ D. $(3, -8)$
☐ E. $(3, 0)$ ☒ F. $(4, 1)$

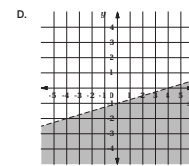
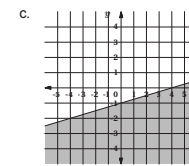
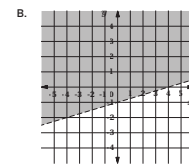
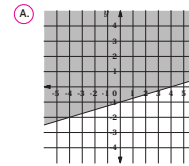
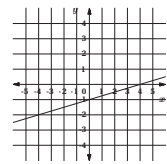
Unit 2 Lesson 14

41

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Name: _____ Date: _____ Period: _____

4. Here is the graph of the equation $x - 4y = 4$. Which of the following represents the solution set to the inequality $x - 4y \leq 4$?



5. Here is the inequality $3x + 7y > -6$. Priya argues that $(-4, 2)$ is a solution to the inequality, and Shawn argues that $(3, -2)$ is a solution. Who is correct? Circle one.

Priya Shawn

Both are correct

Explain your thinking.

Explanations vary. I substituted the values of both points into the inequality: $3(-4) + 7(2) = 2$ is greater than -6 and $3(3) + 7(-2) = -5$ is greater than -6 .

Unit 2 Lesson 14

42

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.D.12
2	1	HSA.REI.D.12
3	2	HSA.REI.D.12
4	2	HSA.REI.D.12
5	2	HSA.REI.D.12

Notes:

Additional Practice

A1.3.01

1. Which would *not* be a good survey question?
 - A. How many servings of fruits and vegetables do you eat daily?
 - B. What are your favorite healthy snacks?
 - C. Why do people dislike eating healthy foods?**
 - D. How much water do you drink daily?

2. Determine which type of data these questions produce.

Question	Categorical or Quantitative?
What is your favorite type of music?	Categorical
How many cups of coffee or tea do you drink daily?	Quantitative
What is your average screen time in hours per day?	Quantitative

3. Select *all* the questions that would produce categorical data.
 - ☒ A. What mode of transportation do you use most often?
 - ☐ B. How many cups of coffee or tea do you drink daily?
 - ☒ C. Which season do you prefer the most?
 - ☒ D. Where do you spend your time after school?
 - ☐ E. How many pets do you have?
4. Maya claims that students who play musical instruments have better grades than those who do not. Write two survey questions that Maya could ask to investigate the claim.

Question 1: **Responses vary. Do you play a musical instrument?**

Question 2: **Responses vary. What was your GPA last semester?**

5. Liam wants to know about the types of movies his classmates prefer. Write a survey question that would give him *categorical* data about his classmates' movie preferences.

Responses vary. Which is your favorite type of movie? (Comedy, Action, Romance, Science Fiction, Horror, Documentary, or Other)

6. A survey asks students how many hours they spend on homework each week and whether they prefer group work or individual work. Identify what data is collected *and* whether it is categorical or quantitative.

Hours spent on homework — Quantitative

Prefer working in Groups or as Individuals — Categorical

Problems 7–8: Here are some responses to the question: *What is your grade level in school?*

8th Grade

9th grade

10th grade

7. Brett is not sure whether the data is categorical or quantitative. Explain why this type of data is unclear.

Explanations vary. Although grade level in school is a numerical value, it represents a category in this case. Therefore, it is categorical data.

8. What is another question that might generate data that is unclear?

Responses vary. What place did you get in the track meet on Saturday?

Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.01

- Which would not be a good survey question?
 - How many servings of fruits and vegetables do you eat daily?
 - What are your favorite healthy snacks?
 - ☒ Why do people dislike eating healthy foods?
 - How much water do you drink daily?

- Determine which type of data these questions produce.

Question	Categorical or Quantitative?
What is your favorite type of music?	Categorical
How many cups of coffee or tea do you drink daily?	Quantitative
What is your average screen time in hours per day?	Quantitative

- Select *all* the questions that would produce categorical data.

- ☒ A. What mode of transportation do you use most often?
- ☐ B. How many cups of coffee or tea do you drink daily?
- ☒ C. Which season do you prefer the most?
- ☒ D. Where do you spend your time after school?
- ☐ E. How many pets do you have?

- Maya claims that students who play musical instruments have better grades than those who do not. Write two survey questions that Maya could ask to investigate the claim.

Question 1: *Responses vary. Do you play a musical instrument?*

Question 2: *Responses vary. What was your GPA last semester?*

Unit 3 Lesson 1

47

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Name: _____ Date: _____ Period: _____

- Liam wants to know about the types of movies his classmates prefer. Write a survey question that would give him *categorical* data about his classmates' movie preferences.
Responses vary. Which is your favorite type of movie? (Comedy, Action, Romance, Science Fiction, Horror, Documentary, or Other)

- A survey asks students how many hours they spend on homework each week and whether they prefer group work or individual work. Identify what data is collected and whether it is categorical or quantitative.
Hours spent on homework — Quantitative
Prefer working in Groups or as Individuals — Categorical

Problems 7–8: Here are some responses to the question: *What is your grade level in school?*

8th Grade 9th grade 10th grade

- Brett is not sure whether the data is categorical or quantitative. Explain why this type of data is unclear.

Explanations vary. Although grade level in school is a numerical value, it represents a category in this case. Therefore, it is categorical data.

- What is another question that might generate data that is unclear?

Responses vary. What place did you get in the track meet on Saturday?

Unit 3 Lesson 1

48

Additional Practice

Practice Problem Analysis

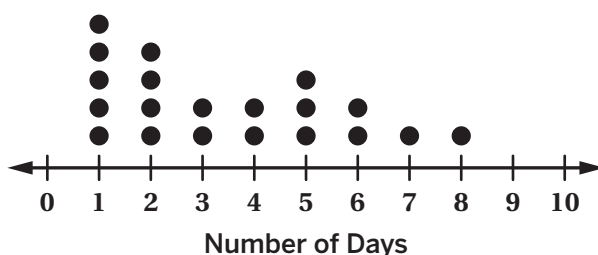
Problem	DOK	Standard(s)
1	1	6.SP.A.1
2	1	HSS.ID.A, HSS.ID.B
3	1	HSS.ID.B
4	2	6.SP.A.1
5	2	6.SP.A.1
6	1	HSS.ID.A, HSS.ID.B
7	2	HSS.ID.A, HSS.ID.B
8	2	HSS.ID.A, HSS.ID.B

Notes:

Additional Practice

A1.3.04

1. The dot plot shows the number of days in May that a city had a low temperature less than 40°F , for each year from 2010 to 2024. In how many years was the low temperature in May less than 40°F for 5 days or more?

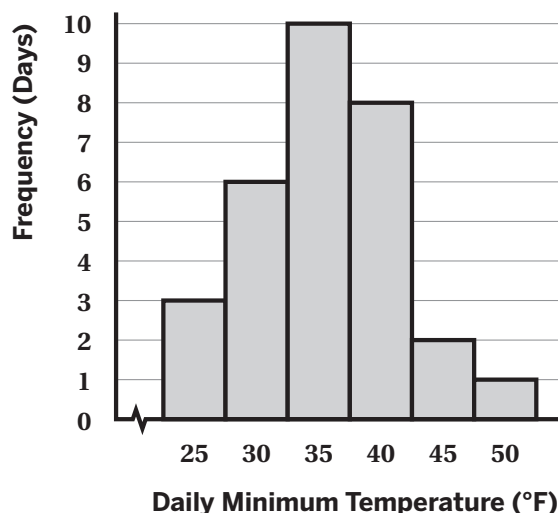


- A. 3
- B. 5
- C. 7**
- D. 8

Problems 2–3: Elena created a histogram about the different minimum temperatures in her hometown, in degrees Fahrenheit, for each day in March. The histogram below shows how many days the temperatures were in each temperature range.

2. Select *all* of the statements that must be true.

- ☐ A. There were 5 days when the minimum temperature was in the range of $30\text{--}34^{\circ}\text{F}$.
- ☒ B. There were 11 days when the minimum temperature was 40°F or higher.
- ☒ C. The most common minimum temperature range was $35\text{--}39^{\circ}\text{F}$.
- ☐ D. There are 31 days of minimum temperature data collected.

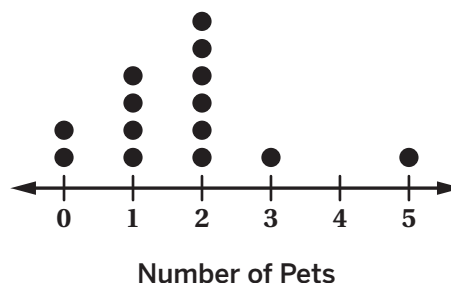


3. Elena claims that half the minimum temperatures are 37°F or less and half are 37°F or greater. Tyler claims that half of the minimum temperatures are 39°F or less and half are 40°F or greater.

Who do you agree with, if either? Explain your thinking.

Explanations vary. I don't agree with either. I can't tell how many of each specific temperature are within each bin of the histogram, so I can't be sure if Elena is right. Tyler is incorrect because there are 19 temperatures that are 39°F or less and only 11 that are 40°F or greater.

Problems 4–6: Sharla asked her scout troop about the number of family pets they each had. She collected the data on the dot plot shown.



4. How many members of Sharla's scout troop were surveyed?

14

5. How many of the surveyed members have two or more pets?

8

6. Write one true statement that can be made using the dot plot above that is different from Problems 4 and 5.

Responses vary. The number of troop members who have 2 pets is 2 less than the number of other troop members combined.

7. Jerome collected the data below about the length of the stop light cycle, in seconds, at three different intersections in his town over the course of an hour.

60	62	85	120	75	106
90	110	90	90	78	120
80	82	115	95	92	95

Would this data be best displayed using a dot plot or histogram?
Explain your thinking.

Histogram. Explanations vary. I would use a histogram due to the large data set and the range of times from 60 to 120 seconds.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.04

1. The dot plot shows the number of days in May that a city had a low temperature less than 40°F, for each year from 2010 to 2024. In how many years was the low temperature in May less than 40°F for 5 days or more?

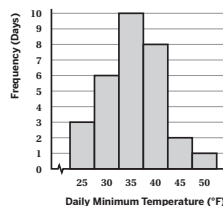


- A. 3
B. 5
C. 7
D. 8

Problems 2–3: Elena created a histogram about the different minimum temperatures in her hometown, in degrees Fahrenheit, for each day in March. The histogram below shows how many days the temperatures were in each temperature range.

2. Select *all* of the statements that must be true.

- ☐ A. There were 5 days when the minimum temperature was in the range of 30–34°F.
☒ B. There were 11 days when the minimum temperature was 40°F or higher.
☒ C. The most common minimum temperature range was 35–39°F.
☐ D. There are 31 days of minimum temperature data collected.



3. Elena claims that half the minimum temperatures are 37°F or less and half are 37°F or greater. Tyler claims that half of the minimum temperatures are 39°F or less and half are 40°F or greater.

Who do you agree with, if either? Explain your thinking.

Explanations vary. I don't agree with either. I can't tell how many of each specific temperature are within each bin of the histogram, so I can't be sure if Elena is right. Tyler is incorrect because there are 19 temperatures that are 39°F or less and only 11 that are 40°F or greater.

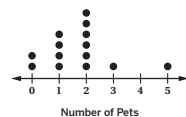
Unit 3 Lesson 4

53

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Name: _____ Date: _____ Period: _____

Problems 4–6: Sharla asked her scout troop about the number of family pets they each had. She collected the data on the dot plot shown.



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14

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8

6. Write one true statement that can be made using the dot plot above that is different from Problems 4 and 5.

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60	62	85	120	75	106
90	110	90	90	78	120
80	82	115	95	92	95

Would this data be best displayed using a dot plot or histogram?

Explain your thinking.

Histogram. Explanations vary. I would use a histogram due to the large data set and the range of times from 60 to 120 seconds.

Unit 3 Lesson 4

54

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.A.1
2	1	HSS.ID.A.1
3	2	HSS.ID.A.1
4	1	HSS.ID.A.1
5	1	HSS.ID.A.1
6	2	HSS.ID.A.1
7	2	HSS.ID.A.1

Notes:

Additional Practice

A1.3.05

Problems 1–2: The following sample from a data set represents the number of minutes students spend walking to school each morning.

7, 9, 9, 10, 15, 15, 16, 18, 18, 20

1. What is the median time spent walking to school?

15 minutes

2. Approximately what percentage of students spend more than 18 minutes walking to school?

25% of students

3. A food truck owner tracks the number of customers they have each hour they are open for a month. They determine the median number of customers per hour is 62. What information does this tell the food truck owner about the number of customers per hour?

A. The average number of customers per hour for the food truck is 62.

B. Most of the time, the food truck has 62 customers per hour.

C. Twenty-five percent of the time, the food truck has 62 customers per hour.

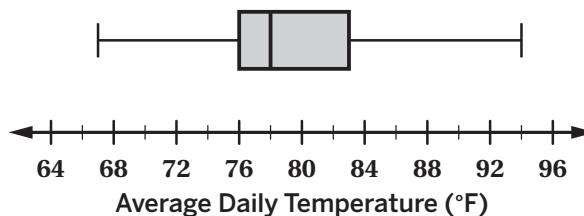
D. Half of the time, the food truck has 62 or fewer customers per hour, and half the time, the food truck has 62 or more customers per hour.

4. Calculate the median of the following data set.

32, 35, 35, 41, 42, 44, 49, 51, 56, 60

43

5. The box plot represents the distribution of average daily temperatures of a town during 20 days of summer. Determine each of these statistics:



Minimum	Q1	Median	Q3	Maximum
67	76	78	83	94

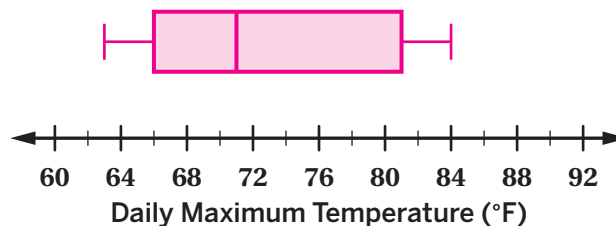
Problems 6–7: The table summarizes the daily maximum temperatures of a major city, in degrees Fahrenheit, for the first 18 days of May.

70	75	84	82	70	72	66	63	64
70	66	64	70	81	81	84	82	79

6. Determine each of these statistics:

Minimum	Q1	Median	Q3	Maximum
63	66	71	81	84

7. Create a box plot that represents the data.



Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.05

Problems 1–2: The following sample from a data set represents the number of minutes students spend walking to school each morning.

7, 9, 9, 10, 15, 15, 16, 18, 18, 20

1. What is the median time spent walking to school?

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4. Calculate the median of the following data set.

32, 35, 35, 41, 42, 44, 49, 51, 56, 60

43

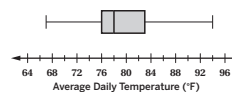
Unit 3 Lesson 5

55

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Name: _____ Date: _____ Period: _____

5. The box plot represents the distribution of average daily temperatures of a town during 20 days of summer. Determine each of these statistics:



Minimum	Q1	Median	Q3	Maximum
67	76	78	83	94

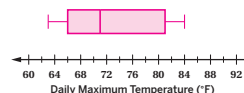
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70	75	84	82	70	72	66	63	64
70	66	64	70	81	81	84	82	79

6. Determine each of these statistics:

Minimum	Q1	Median	Q3	Maximum
63	66	71	81	84

7. Create a box plot that represents the data.



Unit 3 Lesson 5

56

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.A.2
2	1	HSS.ID.A.1
3	2	HSS.ID.A.2
4	1	HSS.ID.A.1
5	2	HSS.ID.A.1
6	2	HSS.ID.A.1
7	2	HSS.ID.A.1

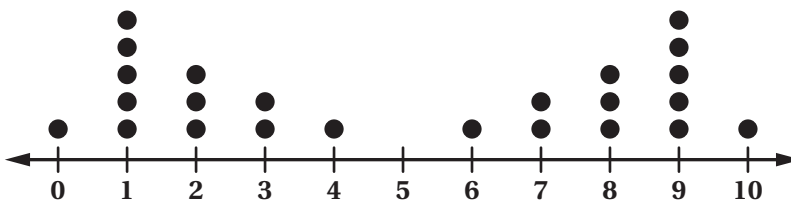
Notes:

Additional Practice

A1.3.06

1. Which term best describes this dot plot?

A. Bell-shaped
☒ B. Bimodal
 C. Skewed
 D. Uniform

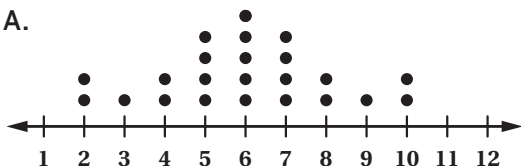


2. A dot plot has a uniform distribution. Which of the following is always *true*?

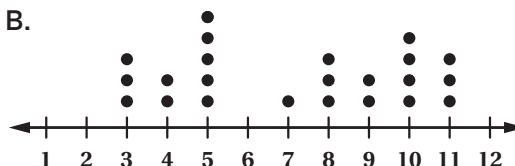
A. There are more data values near the center.
☒ B. The data values are evenly distributed.
 C. There are more data values on the left or on the right side of the center.
 D. There are few data values near the center of the data, with two peaks on the left and right of the center.

3. Select *all* dot plots that have a symmetric, or approximately symmetric, distribution.

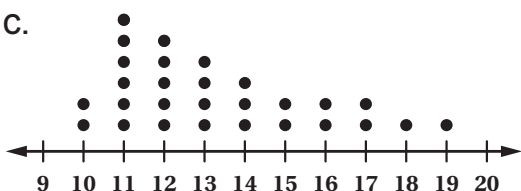
☒ A.



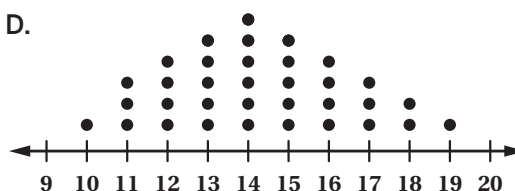
☐ B.



☐ C.

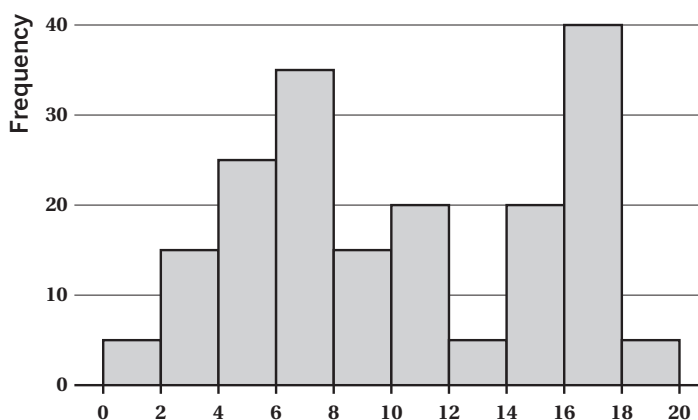


☒ D.

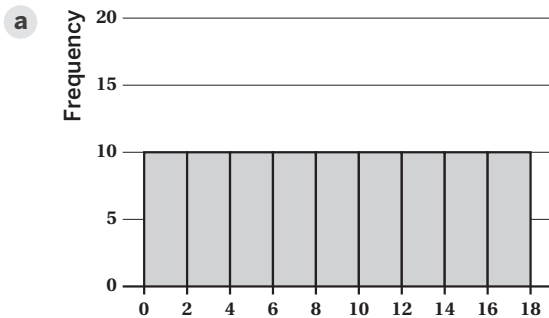


4. Which describes the shape of the distribution of data shown in the histogram?

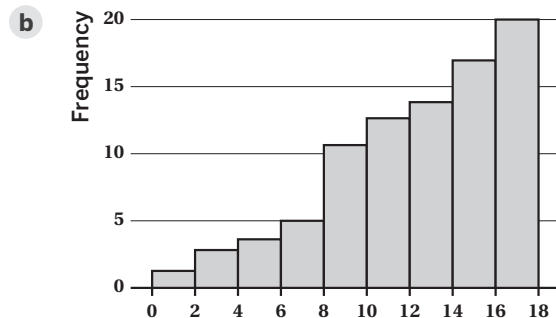
☒ A. Bimodal
 B. Bell-shaped
 C. Skewed
 D. Symmetric



5. Describe the shape of each distribution shown.



This data set is uniformly distributed and symmetric.



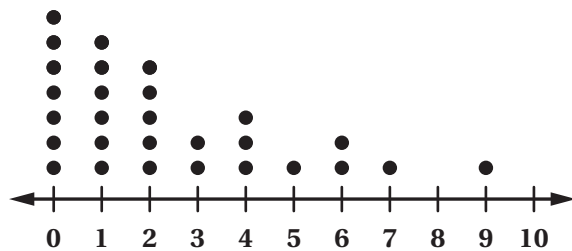
This data set is skewed.

6. A histogram has a symmetric distribution. Which of the following could be possible shapes of the distribution? Select *all* that apply.

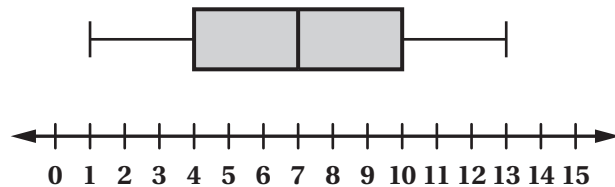
- ☒ A. Bell-shaped
- ☒ B. Bimodal
- ☐ C. Skewed
- ☒ D. Uniform

7. Which statements correctly describe how the data is distributed in the given dot plot? Select *all* that apply.

- ☐ A. The data is bimodal
- ☒ B. The data is skewed.
- ☐ C. The data is distributed symmetrically.
- ☒ D. The data is clustered mostly in the lower values of 0–2.
- ☐ E. The data is uniform.



8. Consider the box plot shown. Lin claims that the distribution of the data is both symmetric and uniform. Andre argues that the distribution is symmetric, but not uniform. Who is correct? Explain your thinking.



Andre. *Explanations vary.* The data is not uniform because there are values from 1–13 with 50% of them within the range of the values of 4–10.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.06

1. Which term best describes this dot plot?

A. Bell-shaped
☒ B. Bimodal
 C. Skewed
 D. Uniform

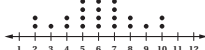


2. A dot plot has a uniform distribution. Which of the following is always true?

A. There are more data values near the center.
☒ B. The data values are evenly distributed.
 C. There are more data values on the left or on the right side of the center.
 D. There are few data values near the center of the data, with two peaks on the left and right of the center.

3. Select *all* dot plots that have a symmetric, or approximately symmetric, distribution.

☒ A.



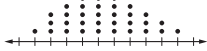
☐ B.



☐ C.

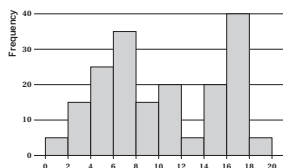


☒ D.



4. Which describes the shape of the distribution of data shown in the histogram?

☒ A. Bimodal
 B. Bell-shaped
 C. Skewed
 D. Symmetric



Unit 3 Lesson 6

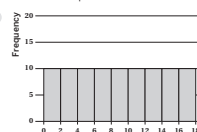
57

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Name: _____ Date: _____ Period: _____

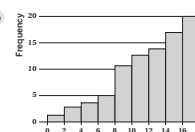
5. Describe the shape of each distribution shown.

a.



This data set is uniformly distributed and symmetric.

b.



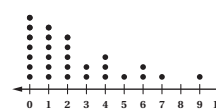
This data set is skewed.

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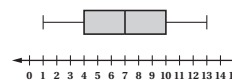
☒ A. Bell-shaped
☒ B. Bimodal
☐ C. Skewed
☒ D. Uniform

7. Which statements correctly describe how the data is distributed in the given dot plot? Select *all* that apply.

☐ A. The data is bimodal.
☒ B. The data is skewed.
☐ C. The data is distributed symmetrically.
☒ D. The data is clustered mostly in the lower values of 0–2.
☐ E. The data is uniform.



8. Consider the box plot shown. Lin claims that the distribution of the data is both symmetric and uniform. Andre argues that the distribution is symmetric, but not uniform. Who is correct? Explain your thinking.



Andre. *Explanations vary.* The data is not uniform because there are values from 1–13 with 50% of them within the range of the values of 4–10.

Unit 3 Lesson 6

58

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.A.1
2	1	HSS.ID.A.1
3	1	HSS.ID.A.1
4	1	HSS.ID.A.1
5	1	HSS.ID.A.1
6	1	HSS.ID.A.1
7	1	HSS.ID.A.1
8	2	HSS.ID.A.1

Notes:

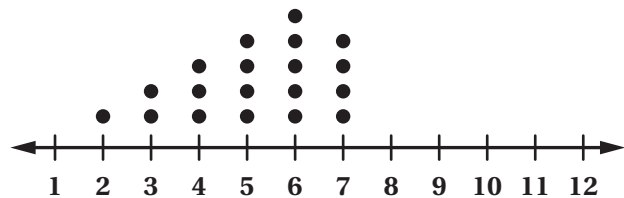
Additional Practice

A1.3.07

1. Consider the following data set: 13, 22, 5, 4, 7, 13, 8, 11, 10, 41. What is the mean?

A. 10
B. 10.5
C. 13
D. 13.4

2. Consider the data shown in the dot plot. What is the median?

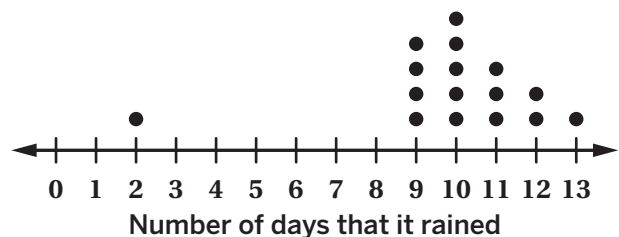


A. 4
B. 4.5
C. 5
D. 6

3. For each distribution shape, determine whether it is more appropriate to use the mean or median as a measure of center.

a Uniform Mean	b Skewed Median
c Symmetric Mean	d Bell-shaped Mean

4. The number of days that it rained in one month for several cities is displayed in the dot plot. Which is greater, the mean or the median? Explain your thinking using the shape of the distribution.



Median. Explanations vary. Because the distribution is skewed right, the mean will be less than the median.

5. The data set represents the scores of Han's assignments. Which measure of center would you use to represent Han's scores?

56, 78, 78, 80, 83, 85, 88, 94, 97, 100

Mean

Median

Explain your thinking.

Explanations vary. Han's score of 56 is an extreme value compared to the rest of his scores. The median would be less affected by the extreme value than the mean.

6. The following data set represents the number of pounds of paper recycled by 12 different classrooms in a school year. What effect does eliminating the least value, 8, from the data set have on the mean and the median?

8, 45, 48, 51, 52, 58, 60, 64, 65, 68, 69, 70

A. Only the mean will increase.

B. Only the median will increase.

C. Both the mean and median will increase, and the mean will increase more.

D. Both the mean and median will increase, and the median will increase more.

7. Here is a data set: 2, 13, 4, 9, 1, 0, 11, 5, 5, 24, 2, 8.

Mai claims that the median would be a more representative measure of center. Han thinks that the mean would be the best measure of center. Who is correct? Explain your thinking.

Mai. Explanations vary. This data set contains an extreme value of 24. This value has a greater impact on the mean than it does the median.

8. Here is a data set: 45, 45, 46, 46, 50, 50, 50.

Shawn claims that the values 50, 55, 55, 60, and 10 are added to the data set, the median will be greater than the mean. Do you agree? Explain your thinking.

Yes. Explanations vary. The new data set is skewed left with an extreme value of 10, which has a greater effect on the median than the mean. The median of the new data set is 50 and the mean is about 46.8.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.07

1. Consider the following data set: 13, 22, 5, 4, 7, 13, 8, 11, 10, 41. What is the mean?

A. 10
B. 10.5
C. 13
D. 13.4

2. Consider the data shown in the dot plot. What is the median?

A. 4
B. 4.5
C. 5
D. 6

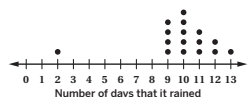


3. For each distribution shape, determine whether it is more appropriate to use the mean or median as a measure of center.

a. Uniform **Mean**
b. Skewed **Median**
c. Symmetric **Mean**
d. Bell-shaped **Mean**

4. The number of days that it rained in one month for several cities is displayed in the dot plot. Which is greater, the mean or the median? Explain your thinking using the shape of the distribution.

Median. Explanations vary. Because the distribution is skewed left, the mean will be less than the median.



Unit 3 Lesson 7

59

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Name: _____ Date: _____ Period: _____

5. The data set represents the scores of Han's assignments. Which measure of center would you use to represent Han's scores?

56, 78, 78, 80, 83, 85, 88, 94, 97, 100

Mean **Median**

Explain your thinking.

Explanations vary. Han's score of 56 is an extreme value compared to the rest of his scores. The median would be less affected by the extreme value than the mean.

6. The following data set represents the number of pounds of paper recycled by 12 different classrooms in a school year. What effect does eliminating the least value, 8, from the data set have on the mean and the median?

8, 45, 48, 51, 52, 58, 60, 64, 65, 68, 69, 70

A. Only the mean will increase.
B. Only the median will increase.
C. Both the mean and median will increase, and the mean will increase more.
D. Both the mean and median will increase, and the median will increase more.

7. Here is a data set: 2, 13, 4, 9, 1, 0, 11, 5, 5, 24, 2, 8.

Mai claims that the median would be a more representative measure of center. Han thinks that the mean would be the best measure of center. Who is correct? Explain your thinking.

Mai. Explanations vary. This data set contains an extreme value of 24. This value has a greater impact on the mean than it does the median.

8. Here is a data set: 45, 45, 46, 46, 50, 50, 50.

Shawn claims that the values 50, 55, 55, 60, and 10 are added to the data set, the median will be greater than the mean. Do you agree? Explain your thinking.

Yes. Explanations vary. The new data set is skewed left with an extreme value of 10, which has a greater effect on the median than the mean. The median of the new data set is 50 and the mean is about 46.8.

Unit 3 Lesson 7

60

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.A.2
2	1	HSS.ID.A.2
3	2	HSS.ID.A.2
4	2	HSS.ID.A.2
5	2	HSS.ID.A.2
6	2	HSS.ID.A.2
7	3	HSS.ID.A.2
8	3	HSS.ID.A.2

Notes:

Additional Practice

A1.3.08

1. The data set represents the grams of protein in all the breakfast bars in a box.

8, 8, 8, 8, 8, 8, 8, 8

- a What is the mean?

8

- b What is the standard deviation?

0

2. Which of the following data sets has the same mean as the data set 3, 3, 3, 3, 3, 3?

A. -3, -3, -3, -3, -3, -3

B. -2, -2, -2, 4, 4, 4

C. 1, 1, 1, 5, 5, 5

D. 1, 4, 1, 4, 1, 4

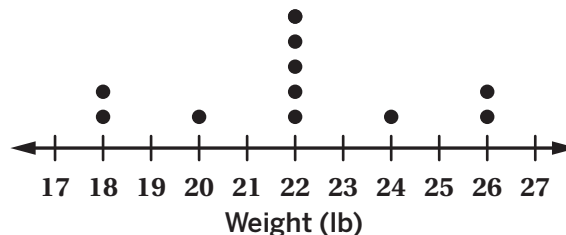
3. The dot plot represents the weights of 11 watermelons. Determine which of the following *best* estimates the standard deviation of the weights of the watermelons.

A. 2 pounds

C. 22 pounds

B. 10 pounds

D. 44 pounds



Problems 4–5: The mean of Data Set A is 30.1 and the standard deviation is 4.3. The mean of Data Set B is 15.9 and the standard deviation is 2.5.

4. Which data set is more spread out? Explain your thinking.

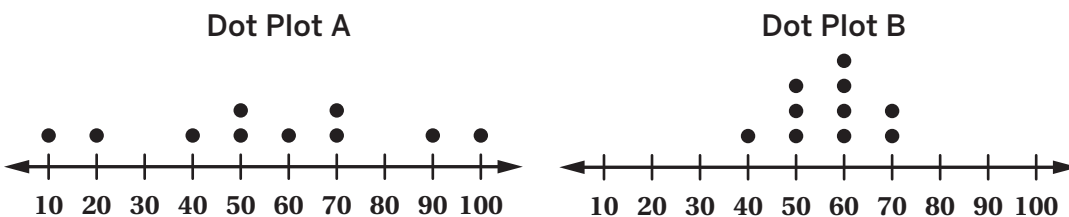
Data set A. Explanations vary. The standard deviation is greater for Data Set A than it is for Data Set B.

5. Complete each statement to describe what differences you would expect to see when comparing dot plots of the two data sets.

Data Set A's dot plot will have most of the data centered around **30.1**. On average, the data will be **4.3** units above or below that value.

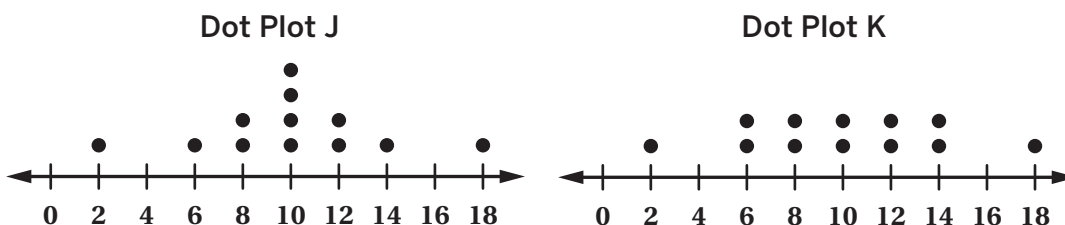
Data Set B's dot plot will have most of the data centered around **15.9**. On average, the data will be **2.5** units above or below that value.

6. Refer to the two dot plots. Which dot plot has a larger standard deviation? Explain your thinking.



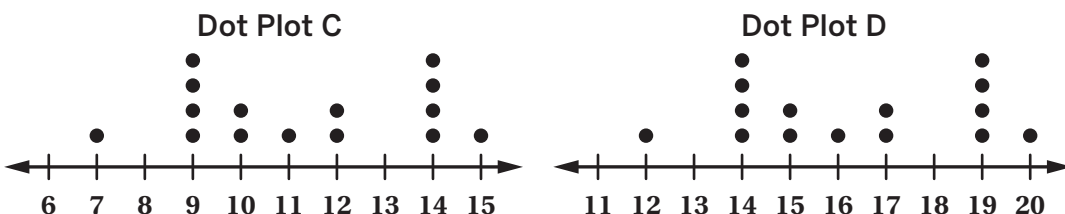
Dot Plot A has a larger standard deviation because the values are not clustered together; they are spread out from 10 to 100. The values of Dot Plot B are all clustered from 40 to 70.

7. Which of the following statements is true about Dot Plots J and K?



- A. The dot plots have the same mean and the same standard deviation.
- B. The dot plots have the same mean, but different standard deviations.**
- C. The dot plots have the same standard deviation, but different means.
- D. The dot plots have different means and different standard deviations.

8. Refer to the two dot plots. Priya claims that the standard deviation of the data in Dot Plot D is greater than the standard deviation of the data in Dot Plot C. Do you agree with Priya? Explain your thinking.



No. Explanations vary. All the data values in Dot Plot D are 5 units greater than the values in Dot Plot C, but the distribution is otherwise the same for the two sets of data.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.08

1. The data set represents the grams of protein in all the breakfast bars in a box.
8, 8, 8, 8, 8, 8, 8, 8

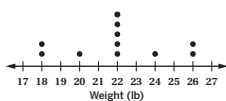
- a. What is the mean?
8
- b. What is the standard deviation?
0

2. Which of the following data sets has the same mean as the data set 3, 3, 3, 3, 3?

- A. -3, -3, -3, -3, -3
B. -2, -2, -2, 4, 4
C. 1, 1, 1, 5, 5
D. 1, 4, 1, 4, 1, 4

3. The dot plot represents the weights of 11 watermelons. Determine which of the following best estimates the standard deviation of the weights of the watermelons.

- A. 2 pounds** C. 22 pounds
B. 10 pounds D. 44 pounds



Problems 4–5: The mean of Data Set A is 30.1 and the standard deviation is 4.3. The mean of Data Set B is 15.9 and the standard deviation is 2.5.

4. Which data set is more spread out? Explain your thinking.

Data set A. Explanations vary. The standard deviation is greater for Data Set A than it is for Data Set B.

5. Complete each statement to describe what differences you would expect to see when comparing dot plots of the two data sets.

Data Set A's dot plot will have most of the data centered around **30.1**. On average, the data will be **4.3** units above or below that value.

Data Set B's dot plot will have most of the data centered around **15.9**. On average, the data will be **2.5** units above or below that value.

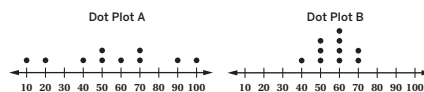
Unit 3 Lesson 8

61

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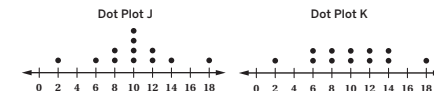
Name: _____ Date: _____ Period: _____

6. Refer to the two dot plots. Which dot plot has a larger standard deviation? Explain your thinking.



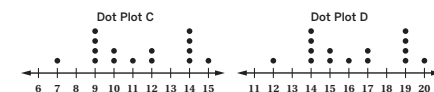
Dot Plot A has a larger standard deviation because the values are not clustered together; they are spread out from 10 to 100. The values of Dot Plot B are all clustered from 40 to 70.

7. Which of the following statements is true about Dot Plots J and K?



- A. The dot plots have the same mean and the same standard deviation.
B. The dot plots have the same mean, but different standard deviations.
C. The dot plots have the same standard deviation, but different means.
D. The dot plots have different means and different standard deviations.

8. Refer to the two dot plots. Priya claims that the standard deviation of the data in Dot Plot D is greater than the standard deviation of the data in Dot Plot C. Do you agree with Priya? Explain your thinking.



No. Explanations vary. All the data values in Dot Plot D are 5 units greater than the values in Dot Plot C, but the distribution is otherwise the same for the two sets of data.

Unit 3 Lesson 8

62

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.A.2
2	1	HSS.ID.A.2
3	2	HSS.ID.A.2
4	2	HSS.ID.A.2
5	2	HSS.ID.A.2
6	2	HSS.ID.A.2
7	3	HSS.ID.A.2
8	3	HSS.ID.A.2

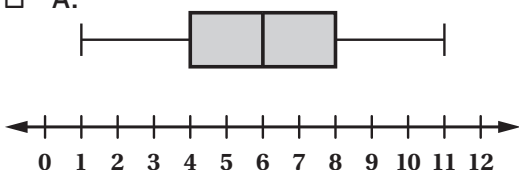
Notes:

Additional Practice

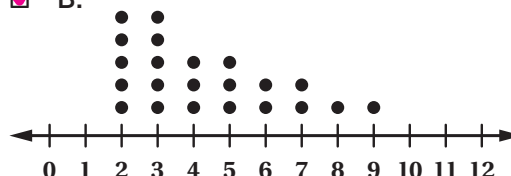
A1.3.10

- A set of data has values that are clustered closely near the center. Which of the following is a *true* statement?
 - The distribution is uniform.
 - The distribution is skewed.
 - The distribution has high variability.
 - ☒ The distribution has low variability.
- For which of the following data sets would you use IQR as a measure of spread over standard deviation? Select *all* that apply.

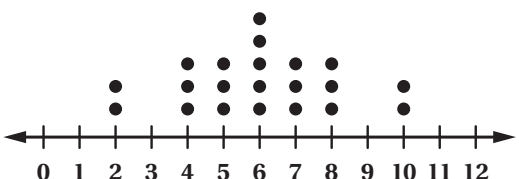
☐ A.



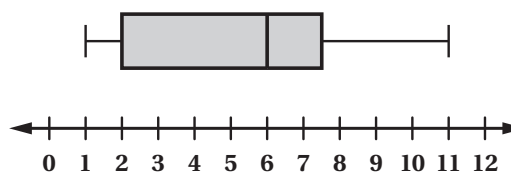
☒ B.



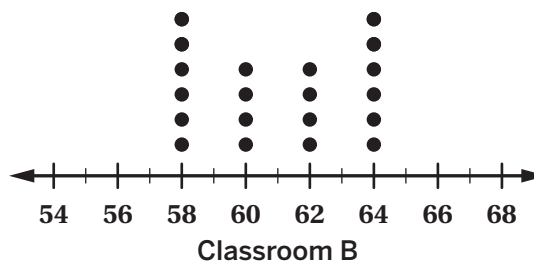
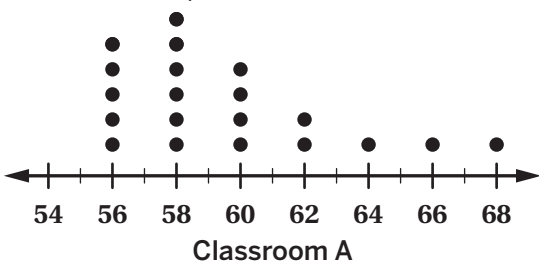
☐ C.



☒ D.



Problems 3–5: The heights, in inches, of 20 students from two different classrooms are shown in the dot plots.

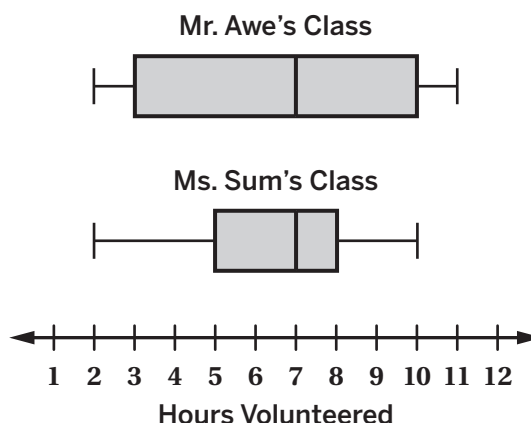


- What are the appropriate measures of center and variability to use with each data set? Explain your thinking.
Classroom A: Median and IQR; Classroom B: Mean and standard deviation; Explanations vary. The median and IQR are most appropriate for the Classroom A data set because the distribution is skewed. The mean and standard deviation are most appropriate for the Classroom B data set because the distribution is symmetric.
- Which classroom shows a greater typical height? Explain your thinking.
Classroom B. Explanations vary. The data for Classroom B has a greater mean.
- Which classroom shows a greater variability? Explain your thinking.
Classroom A. Explanations vary. It has a greater standard deviation.

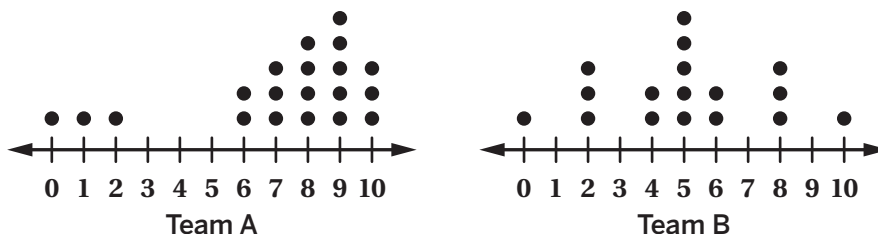
6. Here is data collected from two homeroom classes that shows the amount of hours students volunteered per semester. What does the IQR tell you about each class's volunteer hours?

Responses vary.

- The amount of hours volunteered by Mr. Awe's class has more variability.
- The amount of hours volunteered in Ms. Sum's class is more consistent.



Problems 7–8: Here are two dot plots that show the number of free throw shots made in each game by two basketball teams.



7. Which team generally scored more free throws? Explain your thinking.

Team A. Explanations vary. Team A's points are skewed to the right or to the higher values.

8. For which team would the most appropriate measure of spread be the IQR? Explain your thinking.

Team A. Explanations vary. The IQR is the appropriate measure of spread for Team A because their data is skewed.

Name: _____ Date: _____ Period: _____

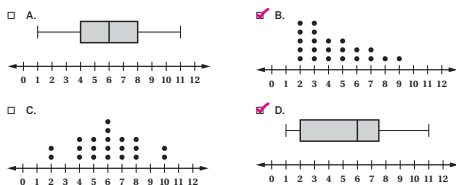
Additional Practice

A1.3.10

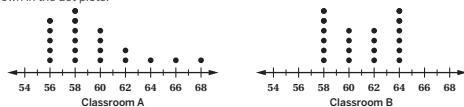
1. A set of data has values that are clustered closely near the center. Which of the following is a true statement?

A. The distribution is uniform. B. The distribution is skewed.
C. The distribution has high variability. D. The distribution has low variability.

2. For which of the following data sets would you use IQR as a measure of spread over standard deviation? Select *all* that apply.



Problems 3–5: The heights, in inches, of 20 students from two different classrooms are shown in the dot plots.



3. What are the appropriate measures of center and variability to use with each data set? Explain your thinking.

Classroom A: Median and IQR; Classroom B: Mean and standard deviation; *Explanations vary.* The median and IQR are most appropriate for the Classroom A data set because the distribution is skewed. The mean and standard deviation are most appropriate for the Classroom B data set because the distribution is symmetric.

4. Which classroom shows a greater typical height? Explain your thinking.

Classroom B. *Explanations vary.* The data for Classroom B has a greater mean.

5. Which classroom shows a greater variability? Explain your thinking.

Classroom A. *Explanations vary.* It has a greater standard deviation.

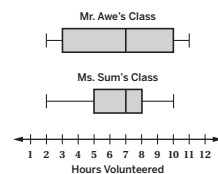
Unit 3 Lesson 10

65

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Name: _____ Date: _____ Period: _____

6. Here is data collected from two homeroom classes that shows the amount of hours students volunteered per semester. What does the IQR tell you about each class's volunteer hours?



Responses vary.

- The amount of hours volunteered by Mr. Awe's class has more variability.
- The amount of hours volunteered in Ms. Sum's class is more consistent.

Problems 7–8: Here are two dot plots that show the number of free throw shots made in each game by two basketball teams.



7. Which team generally scored more free throws? Explain your thinking.

Team A. *Explanations vary.* Team A's points are skewed to the right or to the higher values.

8. For which team would the most appropriate measure of spread be the IQR? Explain your thinking.

Team A. *Explanations vary.* The IQR is the appropriate measure of spread for Team A because their data is skewed.

Unit 3 Lesson 10

66

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.A.2
2	1	HSS.ID.A.3
3	2	HSS.ID.A.2
4	2	HSS.ID.A.2
5	2	HSS.ID.A.2
6	3	HSS.ID.A.3
7	2	HSS.ID.A.3

Notes:

Additional Practice

A1.3.11

- Here is a data set: 0, 0, 1, 1, 2, 4, 4, 5, 6, 20.
Which measure of center is greater: the mean or the median?

Mean

Problems 2–4: Here is a data set: 30, 30, 32, 33, 35, 35, 35, 36, 38, 90.

- Determine the mean and median.

Mean: **39.4**

Median: **35**

- Eliminate the greatest value, 90, from the data set. Determine the mean and median.

Mean: **33.8**

Median: **35**

- Which measure of center was most affected by eliminating the greatest value in the data set? Explain your thinking.

Mean. Explanations vary. Because the greatest value is an outlier it has the greatest affect on the mean or average. The median did not change.

Problems 5–6: Here is a data set: 70, 70, 70, 75, 75, 75, 78, 78, 79, 80.

- Do the mean and standard deviation of the data set increase or decrease if the 80 is changed to an 8?

The mean decreases from 75 to 67.8 and the standard deviation increases from about 3.86 to about 21.29.

- If the 80 is changed to an 8, would the median or the mean be a more appropriate measure of center? Explain your thinking.

Median. Explanations vary. The data set with the 8 is a skewed distribution.

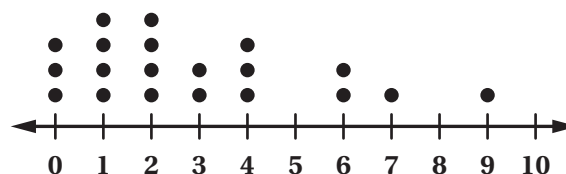
7. Here is a data set: $-8, 5, 5, 6, 8, 8, 9, 10, 10, 10$.

If the least value is eliminated from the data set, which measure is affected more: the mean or the median? Explain your thinking.

Mean. *Explanations vary.* If -8 is eliminated, the mean increases from 6.3 to about 7.9 , but the median does not change.

Problems 8–10: Here is a dot plot.

8. Which measure of center do you think is more appropriate for this data set? Explain your thinking.



Median. *Explanations vary.* The median is a more appropriate measure of center because the distribution appears to be skewed.

9. Which measure of spread do you think is more appropriate for this data set? Explain your thinking.

IQR. *Explanations vary.* The IQR is a more appropriate measure of variability because the distribution appears to be skewed.

10. Does this data set include an outlier? Explain your thinking.

Yes. *Explanations vary.* The data point 9 is considered an outlier in this data set. The IQR for the data is 3 , and the value of $Q3$ is $4.4 + 1.5(3) = 8.5$, and 9 is greater than 8.5 .

Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.11

1. Here is a data set: 0, 0, 1, 1, 2, 4, 4, 5, 6, 20.
Which measure of center is greater: the mean or the median?
Mean

Problems 2–4: Here is a data set: 30, 30, 32, 33, 35, 35, 35, 36, 38, 90.

2. Determine the mean and median.

Mean: 39.4

Median: 35

3. Eliminate the greatest value, 90, from the data set. Determine the mean and median.

Mean: 33.8

Median: 35

4. Which measure of center was most affected by eliminating the greatest value in the data set? Explain your thinking.

Mean. Explanations vary. Because the greatest value is an outlier it has the greatest affect on the mean or average. The median did not change.

Problems 5–6: Here is a data set: 70, 70, 70, 75, 75, 75, 78, 78, 79, 80.

5. Do the mean and standard deviation of the data set increase or decrease if the 80 is changed to an 8?

The mean decreases from 75 to 67.8 and the standard deviation increases from about 3.86 to about 21.29.

6. If the 80 is changed to an 8, would the median or the mean be a more appropriate measure of center? Explain your thinking.

Median. Explanations vary. The data set with the 8 is a skewed distribution.

Unit 3 Lesson 11

67

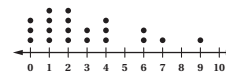
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Name: _____ Date: _____ Period: _____

7. Here is a data set: -8, 5, 5, 6, 8, 8, 9, 10, 10, 10.
If the least value is eliminated from the data set, which measure is affected more: the mean or the median? Explain your thinking.

Mean. Explanations vary. If -8 is eliminated, the mean increases from 6.3 to about 7.9, but the median does not change.

Problems 8–10: Here is a dot plot.



8. Which measure of center do you think is more appropriate for this data set? Explain your thinking.

Median. Explanations vary. The median is a more appropriate measure of center because the distribution appears to be skewed.

9. Which measure of spread do you think is more appropriate for this data set? Explain your thinking.

IQR. Explanations vary. The IQR is a more appropriate measure of variability because the distribution appears to be skewed.

10. Does this data set include an outlier? Explain your thinking.

Yes. Explanations vary. The data point 9 is considered an outlier in this data set. The IQR for the data is 3, and the value of Q3 is $4.4 + 1.5(3) = 8.5$, and 9 is greater than 8.5.

Unit 3 Lesson 11

68

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.A.2
2	1	HSS.ID.A.3
3	1	HSS.ID.A.3
4	1	HSS.ID.A.2
5	3	HSS.ID.A.2, HSS.ID.A.3
6	2	HSS.ID.A.2
7	2	HSS.ID.A.3
8	2	HSS.ID.A.2
9	2	HSS.ID.A.2
10	3	HSS.ID.A.3

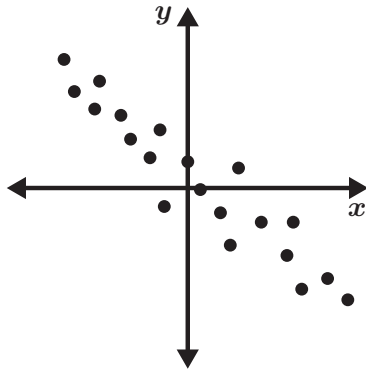
Notes:

Additional Practice

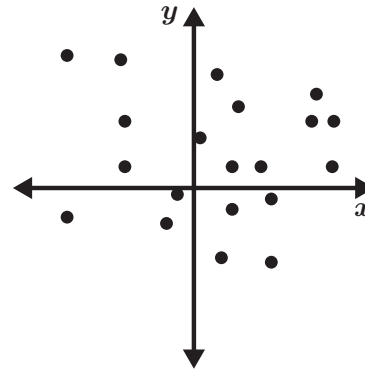
A1.3.13

Problems 1–4: Determine whether each scatterplot has a strong linear relationship, weak linear relationship, or no linear relationship. Circle your choice for each problem.

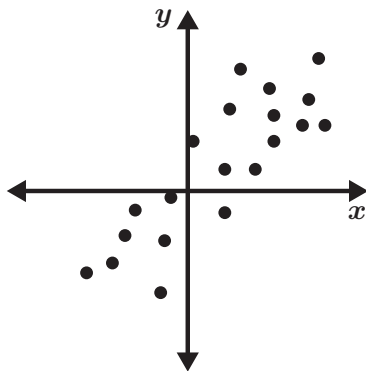
1. **Strong** Weak None



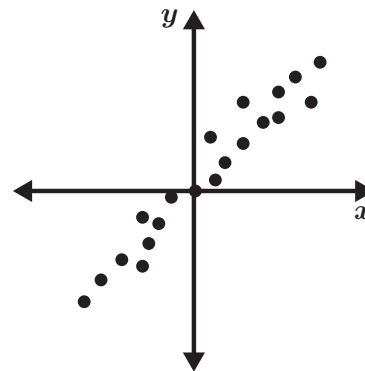
2. Strong Weak **None**



3. Strong **Weak** None



4. **Strong** Weak None



5. A scatter plot is found to have a correlation coefficient of $r = 0.88$. What does this tell you about the data? Select *all* that apply.

- ☐ A. The data has a weak association.
- ☒ B. The data has a strong association.
- ☒ C. The trend of the data has a positive association.
- ☐ D. The trend of the data has a negative association.

6. Which correlation coefficient indicates the *weakest* association?

- A. $r = 0.75$
- B. $r = 0.23$**
- C. $r = -0.32$
- D. $r = -0.94$

7. Marilyn was talking about the strength of linear associations. She said that all data with negative r -values are weak associations and all data with positive r -values are strong associations. What is incorrect about Marilyn's thinking? How would you convince her otherwise?

Responses vary. Marilyn is incorrect because the closer the correlation coefficient, r , is to -1 or 1 , the stronger the association. So, you can have a strong negative association r -value, like $r = -0.99$, and a weak positive association, like $r = 0.50$.

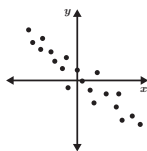
Name: _____ Date: _____ Period: _____

Additional Practice

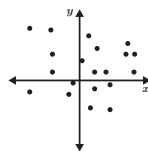
A1.3.13

Problems 1–4: Determine whether each scatterplot has a strong linear relationship, weak linear relationship, or no linear relationship. Circle your choice for each problem.

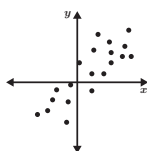
1. **Strong** Weak None



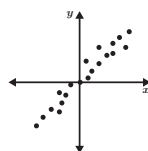
2. Strong Weak **None**



3. Strong **Weak** None



4. **Strong** Weak None



5. A scatter plot is found to have a correlation coefficient of $r = 0.88$. What does this tell you about the data? Select *all* that apply.

- ☐ A. The data has a weak association.
- ☒ B. The data has a strong association.
- ☒ C. The trend of the data has a positive association.
- ☐ D. The trend of the data has a negative association.

Unit 3 Lesson 13

71

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Name: _____ Date: _____ Period: _____

6. Which correlation coefficient indicates the weakest association?

- A. $r = 0.75$
- B. $r = 0.23$**
- C. $r = -0.32$
- D. $r = -0.94$

7. Merilyn was talking about the strength of linear associations. She said that all data with negative r -values are weak associations and all data with positive r -values are strong associations. What is incorrect about Merilyn's thinking? How would you convince her otherwise?

Responses vary. Merilyn is incorrect because the closer the correlation coefficient, r , is to -1 or 1 , the stronger the association. So, you can have a strong negative association r -value, like $r = -0.99$, and a weak positive association, like $r = 0.50$.

Unit 3 Lesson 13

72

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.A.1, HSS.ID.A.3
2	1	HSS.ID.A.1, HSS.ID.A.3
3	1	HSS.ID.A.1, HSS.ID.A.3
4	1	HSS.ID.A.1, HSS.ID.A.3
5	1	HSS.ID.C.8
6	1	HSS.ID.C.8
7	2	HSS.ID.C.8

Notes:

Additional Practice

A1.3.14

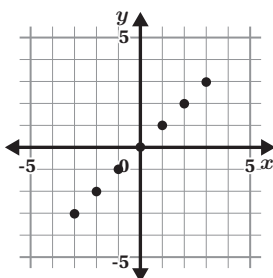
1. Match each scatter plot to its correlation coefficient, r .

a. $r = 0.92$.

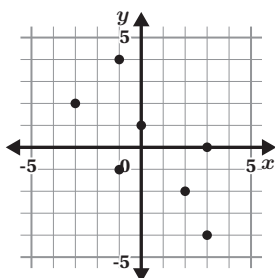
b. $r = 0.16$

c. $r = 1$

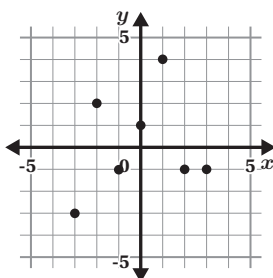
d. $r = -0.68$



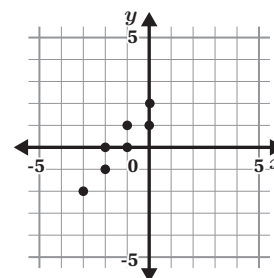
c



d



b



a

2. A scatter plot is found to have a correlation coefficient of $r = -0.49$. Select *all* the conclusions you can make about the data.

- ☒ A. There is a weak association between the two variables.
- ☐ B. There is a strong association between the two variables.
- ☐ C. There is no association between the two variables.
- ☐ D. As one variable increases, the other increases.
- ☒ E. As one variable increases, the other decreases.

3. The data in the table represents the number of cans students collected during a school food drive. Elena graphs the data and claims the correlation coefficient for the data would likely be a number between 0.5 and 1. Lin argues that the correlation coefficient would likely be a number between 0 and 0.5. Who do you agree with? Explain your thinking.

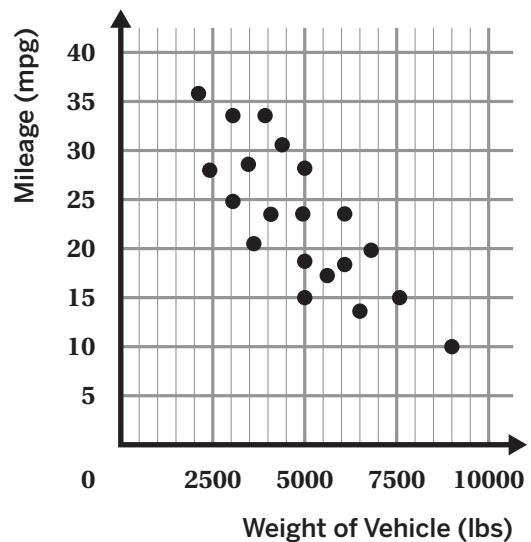
Day	1	2	3	4	5	6	7	8
Number of Cans	20	50	45	53	60	62	71	80

Elena. *Explanations vary.* As the days progress, the number of cans increases. There are some data points, like (2, 50), that fall out of the positive pattern but overall, this pattern would result in a r -value between 0.5 and 1.

Problems 4–6: Marcus is interested in buying a new car that gets good gas mileage. He found data about two variables:

- Weight of vehicle (in pounds)
- Mileage of vehicle (miles per gallon)
- $r = -0.78$

Here is a scatter plot of the data.



4. Based on the r -value of $r = -0.88$, what relationship is there between the variables? Circle one.

Positive

Negative

Neither

5. What is the strength of the relationship? Circle one.

Weak

Strong

6. Which of the following claims can be made about the relationship between the weight of a car and its average mileage?

A. The lighter the vehicle, the better its gas mileage.

B. The heavier the vehicle, the better its gas mileage.

C. There is a negative association between the weight of a vehicle and its average gas mileage.

D. There is a positive association between the weight of a vehicle and its average gas mileage.

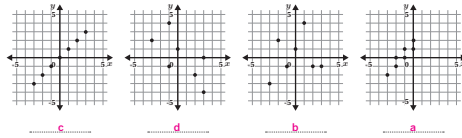
Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.14

1. Match each scatter plot to its correlation coefficient, r .

a. $r = 0.92$ b. $r = 0.16$ c. $r = 1$ d. $r = -0.68$



2. A scatter plot is found to have a correlation coefficient of $r = -0.49$. Select all the conclusions you can make about the data.

- ☒ A. There is a weak association between the two variables.
☐ B. There is a strong association between the two variables.
☐ C. There is no association between the two variables.
☐ D. As one variable increases, the other increases.
☒ E. As one variable increases, the other decreases.

3. The data in the table represents the number of cans students collected during a school food drive. Elena graphs the data and claims the correlation coefficient for the data would likely be a number between 0.5 and 1. Lin argues that the correlation coefficient would likely be a number between 0 and 0.5. Who do you agree with? Explain your thinking.

Day	1	2	3	4	5	6	7	8
Number of Cans	20	50	45	53	60	62	71	80

Elena. Explanations vary. As the days progress, the number of cans increases. There are some data points, like (2, 50), that fall out of the positive pattern but overall, this pattern would result in a r -value between 0.5 and 1.

Unit 3 Lesson 14

73

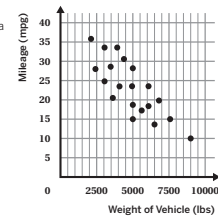
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Name: _____ Date: _____ Period: _____

Problems 4–6: Marcus is interested in buying a new car that gets good gas mileage. He found data about two variables:

- Weight of vehicle (in pounds)
- Mileage of vehicle (miles per gallon)
- $r = -0.78$

Here is a scatter plot of the data.



4. Based on the r -value of $r = -0.88$, what relationship is there between the variables? Circle one.

Positive **Negative** Neither

5. What is the strength of the relationship? Circle one.

Weak **Strong**

6. Which of the following claims can be made about the relationship between the weight of a car and its average mileage?

- A. The lighter the vehicle, the better its gas mileage.
 B. The heavier the vehicle, the better its gas mileage.
☒ C. There is a negative association between the weight of a vehicle and its average gas mileage.
 D. There is a positive association between the weight of a vehicle and its average gas mileage.

Unit 3 Lesson 14

74

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSS.ID.C.8
2	1	HSS.ID.C.8
3	2	HSS.ID.C.8, HSS.ID.A.3
4	1	HSS.ID.C.8
5	1	HSS.ID.C.8
6	2	HSS.ID.C.8, HSS.ID.A.3

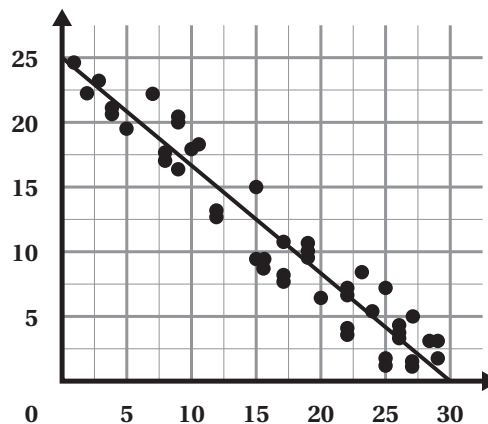
Notes:

Additional Practice

A1.3.15

1. Which is the best estimate for the slope of the line?

- A. $-\frac{5}{6}$
 B. $-\frac{1}{2}$
 C. $\frac{1}{2}$
 D. $\frac{5}{6}$



Problems 2–5: The scatter plot shows the amount of a discount on a product and the number of items that are sold in a store.

2. The scatter plot includes a point at (8, 17). Describe what this point means in context.

Explanations vary. The point (8, 17) means that when the discount is \$8, then 17 items are sold.

3. An item is discounted by \$12. Approximately, how many items will be sold after the discount?

24 items. Responses between 23 and 25 are considered correct.

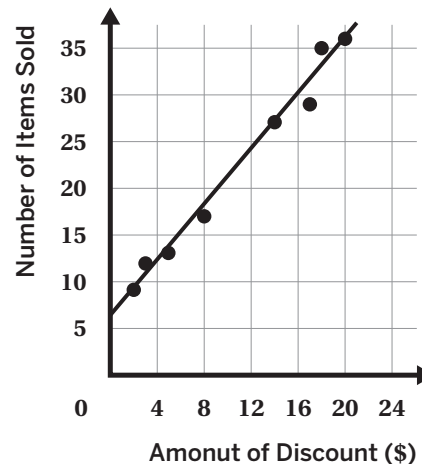
4. The equation for the line that best fits this data is $y = 1.48x + 6.15$. What do the numbers 1.48 and 6.15 mean in this context?

Explanations vary. The 1.48 means that for every \$1 discount, 1.48 additional items are sold. The 6.15 means that about 6 items are sold without any discounts.

5. The company decides to collect the same data at their other two stores. The equation of their lines of best fit are shown in the table.

Which store sells the *fewest* amount of additional products with each \$1 of discount? Explain your thinking.

Store B. Explanations Vary. They sell 0.85 additional products for every \$1 discount. The other stores sell more additional products per \$1 discount.



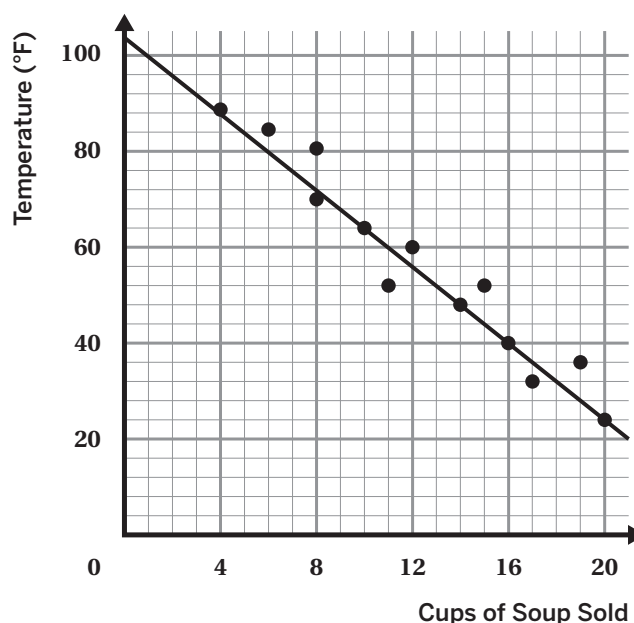
Store	Line of Best Fit
A	$y = 1.4x + 6.15$
B	$y = 0.85x + 5$
C	$y = 1.72x + 5.5$

6. The scatter plot shows the the outside temperature and the number of cups of soup sold at a deli. The equation for the line of fit is given by $y = -4x + 105.6$, where y represents the outside temperature in degrees Fahrenheit and x represents number of cups of soup sold. The slope of the line is -4 and the y -intercept is 105.6 .

Which of the following statements can be made based on the equation of the line of fit?

Select *all* that apply.

- ☐ A. The amount of soup sold increases with the temperature.
- ☒ B. The amount of soup sold increases as the temperature decreases.
- ☐ C. A rise in temperature by 4°F leads to 1 more cup of soup sold.
- ☒ D. No cups of soup are sold when the temperature is 105.6°F .
- ☐ E. Approximately 106 cups of soup are sold when the temperature is 0°F .



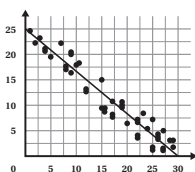
Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.15

1. Which is the best estimate for the slope of the line?

- A. $-\frac{5}{6}$
B. $-\frac{1}{2}$
C. $\frac{1}{2}$
D. $\frac{5}{6}$



Problems 2–5: The scatter plot shows the amount of a discount on a product and the number of items that are sold in a store.

2. The scatter plot includes a point at (8, 17). Describe what this point means in context.

Explanations vary. The point (8, 17) means that when the discount is \$8, then 17 items are sold.

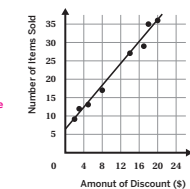
3. An item is discounted by \$12. Approximately, how many items will be sold after the discount?

24 items. Responses between 23 and 25 are considered correct.

4. The equation for the line that best fits this data is $y = 1.48x + 6.15$. What do the numbers 1.48 and 6.15 mean in this context?

Explanations vary. The 1.48 means that for every \$1 discount, 1.48 additional items are sold. The 6.15 means that about 6 items are sold without any discounts.

5. The company decides to collect the same data at their other two stores. The equation of their lines of best fit are shown in the table.



Store	Line of Best Fit
A	$y = 1.4x + 6.15$
B	$y = 0.85x + 5$
C	$y = 1.72x + 5.5$

Which store sells the fewest amount of additional products with each \$1 of discount? Explain your thinking.

Store B. Explanations Vary. They sell 0.85 additional products for every \$1 discount. The other stores sell more additional products per \$1 discount.

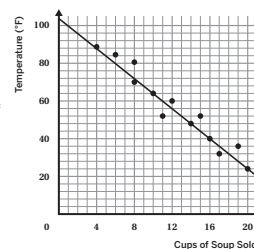
Unit 3 Lesson 15

75

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Name: _____ Date: _____ Period: _____

6. The scatter plot shows the outside temperature and the number of cups of soup sold at a deli. The equation for the line of fit is given by $y = -4x + 105.6$, where y represents the outside temperature in degrees Fahrenheit and x represents number of cups of soup sold. The slope of the line is -4 and the y -intercept is 105.6.



Which of the following statements can be made based on the equation of the line of fit? Select *all* that apply.

- ☐ A. The amount of soup sold increases with the temperature.
☒ B. The amount of soup sold increases as the temperature decreases.
☐ C. A rise in temperature by 4°F leads to 1 more cup of soup sold.
☒ D. No cups of soup are sold when the temperature is 105.6°F.
☐ E. Approximately 106 cups of soup are sold when the temperature is 0°F.

Unit 3 Lesson 15

76

Additional Practice

Practice Problem Analysis

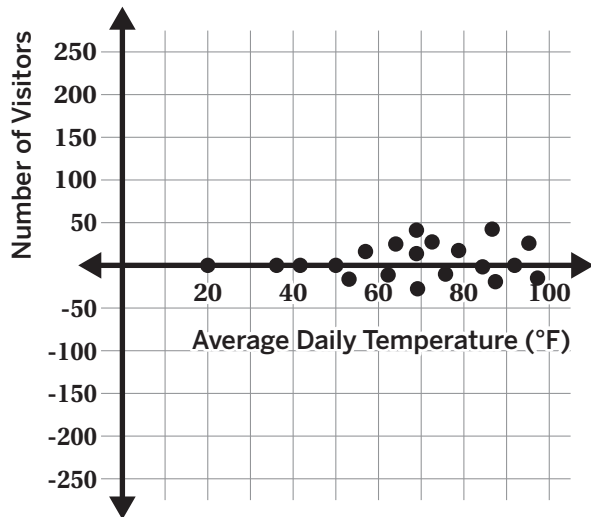
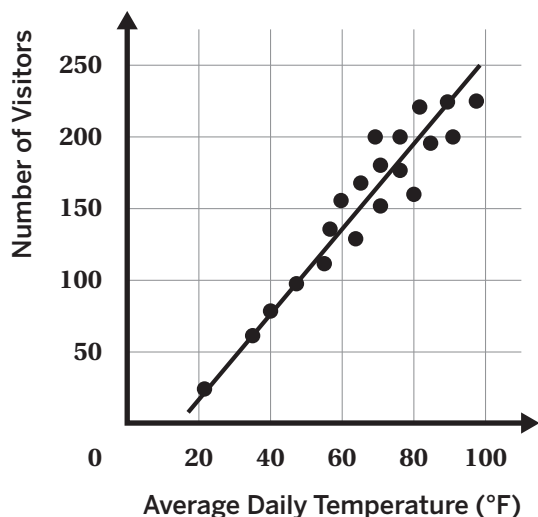
Problem	DOK	Standard(s)
1	1	HSS.ID.B.6
2	1	HSS.ID.C.7
3	1	HSS.ID.C.7
4	1	HSS.ID.C.7
5	2	HSS.ID.C.7
6	2	HSS.ID.C.7

Notes:

Additional Practice

A1.3.16

Problems 1–2: The scatterplot shows the number of visitors to a local beach, the average daily temperature, and a line that best fits the data. The residual plot is also shown.



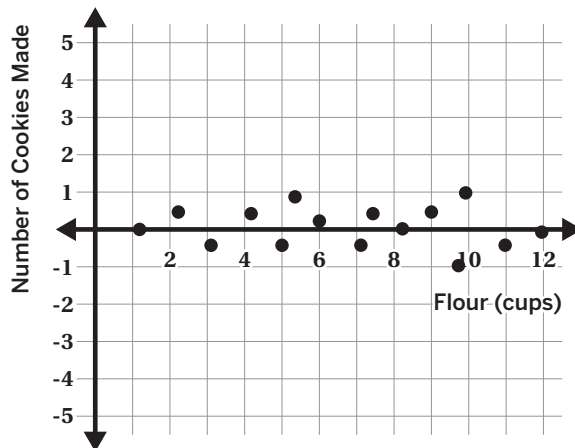
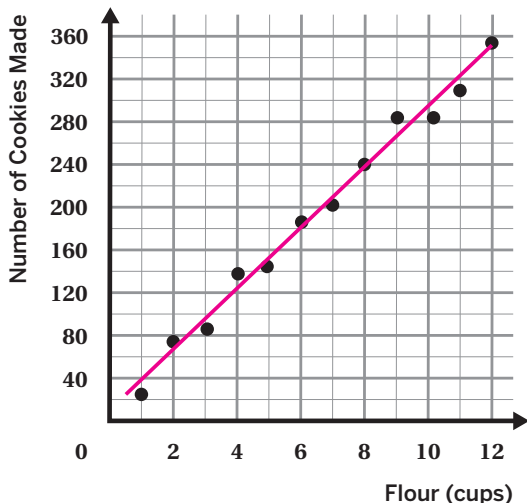
1. Predict the number of visitors at the beach when it is 70°F.

Responses between 160 and 180 are considered correct.

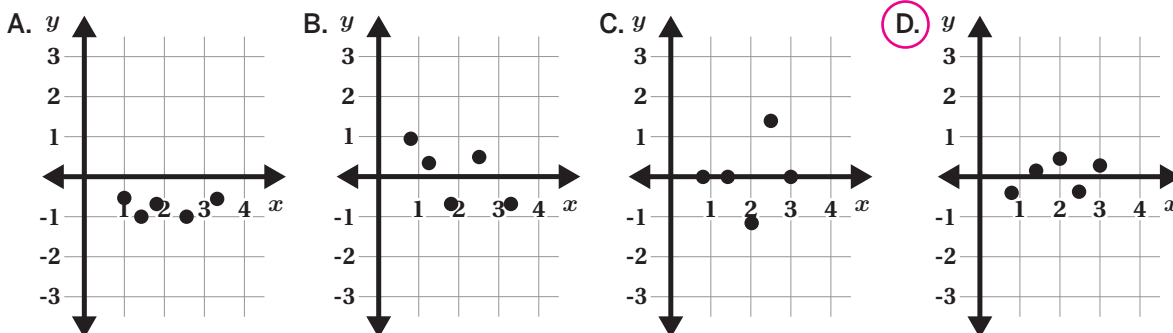
2. How can you tell that the graphed line is a good fit for the data? Use the residual plot if it helps your thinking.

Response vary. The line of fit follows the same direction as the data and the data fit closely around the line, both on the scatterplot and in the residual plot.

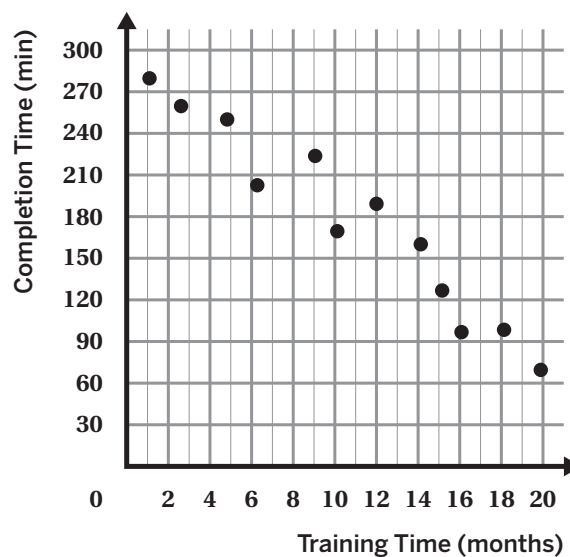
3. Here is a scatter plot and its corresponding residual plot.
Draw a line of fit on the scatter plot that corresponds to the residual plot.



4. These residual plots are from the same set of data, but each one represents a different line of fit. Which residual plot shows the best line of fit?



Problems 5–6: This scatter plot shows the relationship between the number of months training for a marathon and the completion time (in minutes).



5. Which r -value could represent the correlation coefficient for this data?
- A. 0.82 **B. -0.82**
 C. 0.26 D. -0.26
6. Which equation could represent the line of best fit?
- A. $y = 290x + 12$ B. $y = 12x + 290$
 C. $y = -290x + 12$ **D. $y = -12x + 290$**

Explain your thinking.

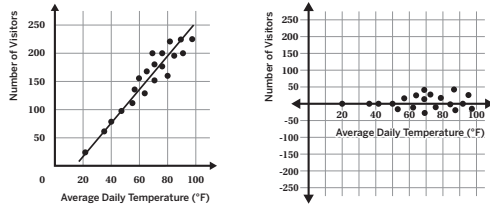
Explanations vary. The y -intercept of the line of best fit is between 270 and 300 and the slope of the line is negative.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.16

Problems 1-2: The scatterplot shows the number of visitors to a local beach, the average daily temperature, and a line that best fits the data. The residual plot is also shown.



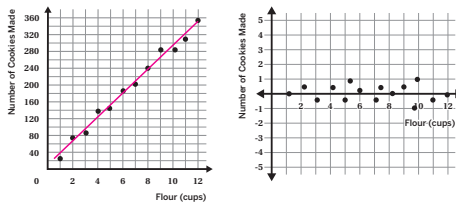
1. Predict the number of visitors at the beach when it is 70°F.

Responses between 160 and 180 are considered correct.

2. How can you tell that the graphed line is a good fit for the data? Use the residual plot if it helps your thinking.

Response vary. The line of fit follows the same direction as the data and the data fit closely around the line, both on the scatterplot and in the residual plot.

3. Here is a scatter plot and its corresponding residual plot. Draw a line of fit on the scatter plot that corresponds to the residual plot.



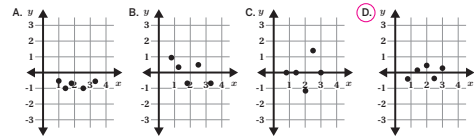
Unit 3 Lesson 16

77

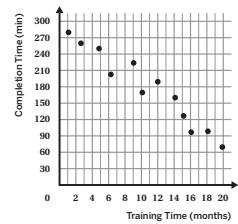
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Name: _____ Date: _____ Period: _____

4. These residual plots are from the same set of data, but each one represents a different line of fit. Which residual plot shows the best line of fit?



Problems 5-6: This scatter plot shows the relationship between the number of months training for a marathon and the completion time (in minutes).



5. Which r -value could represent the correlation coefficient for this data?

A. 0.82
B. -0.82
C. 0.26
D. -0.26

6. Which equation could represent the line of best fit?

A. $y = 290x + 12$
B. $y = 12x + 290$
C. $y = -290x + 12$
D. $y = -12x + 290$

Explain your thinking.

Explanations vary. The y -intercept of the line of best fit is between 270 and 300 and the slope of the line is negative.

Unit 3 Lesson 16

78

Additional Practice

Practice Problem Analysis

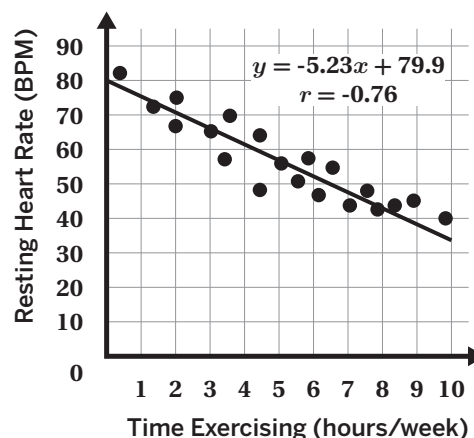
Problem	DOK	Standard(s)
1	2	HSS.ID.B.6.A
2	2	HSS.ID.B.6.B
3	2	HSS.ID.B.6.C
4	1	HSS.ID.B.6.B
5	1	HSS.ID.C.8
6	1	HSS.ID.C.7

Notes:

Additional Practice

A1.3.17

Problems 1–3: This scatter plot shows the relationship between the number of hours a person exercises each week and their resting heart rate, in beats per minute.



1. The equation for the line of best fit is $y = -5.23x + 79.9$.

- a What does the 79.9 mean in this situation?

Responses vary. The resting heart rate of someone who doesn't exercise.

- b What does the -5.23 mean in this situation?

Responses vary. The decrease in resting heart rate expected with every extra hour of exercise a week.

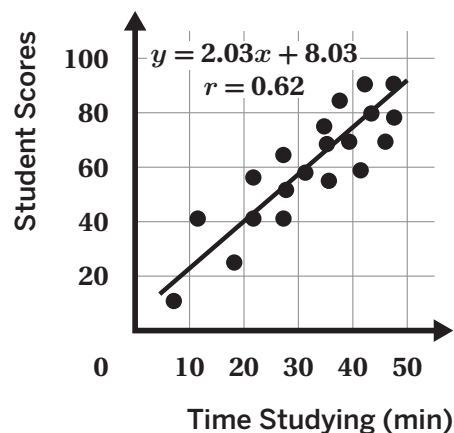
2. Nina exercised for 4 hours this week. Use the equation of line of best fit to predict her resting heart rate.

$$y = -5.23(4) + 79.9 = 58.98 \text{ BPM}$$

3. Do you think the prediction is accurate? Use the r -value to explain your thinking.

Responses vary. Looking at the data for 4 hours of exercise, there is almost 20 BPM difference between those two values. However, I think it is a close estimate to the line of best fit because the r -value is strong.

Problems 4–6: The scatter plot shows the relationship between the amount of time spent studying for a recent math test (in minutes) and the scores the students earned on the test.



4. The equation for the best fit line is $y = 2.03x + 8.03$. What does the 2.03 mean in this situation?

For every one minute spent studying, the test score will go up by 2.03 points.

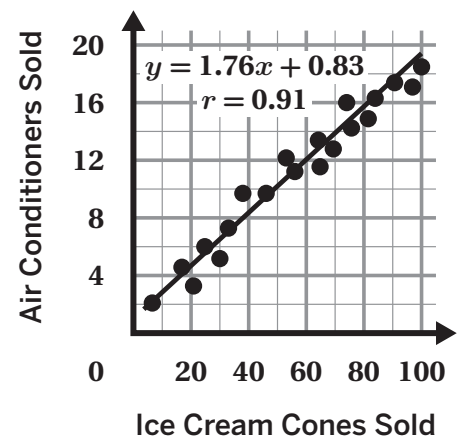
5. Rajah studied for this test for 40 minutes. Use the equation of line of best fit to predict his score on the test.

$$y = 2.03(40) + 8.03 = 89.23 \text{ points}$$

6. Do you think the prediction is accurate? Use the r -value to explain your thinking.

Explanations vary. Looking at the data near 40 minutes of studying, there is almost a 25-point difference between the largest and smallest value. However, I think each of those values is close to the line of best fit because the r -value is strong.

Problems 7–9: The scatter plot shows the relationship between the number of ice cream cones and the number of air conditioners sold in a month.



7. Describe the relationship between the number of ice cream cones sold and the number of air conditioners sold in a month.

Explanations vary. There is a strong positive association between the number of ice cream cones and the number of air conditioners sold in a month.

8. Do you think one variable causes the other? Explain your thinking.

Explanations vary. No. Buying ice cream does not lead to buying air conditioners.

9. What else might affect this relationship? Explain your thinking.

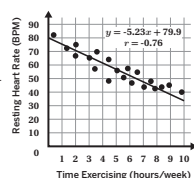
Explanations vary. The temperature. Both of these sales increase when the temperature rises in the summer months.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.3.17

Problems 1–3: This scatter plot shows the relationship between the number of hours a person exercises each week and their resting heart rate, in beats per minute.



1. The equation for the line of best fit is $y = -5.23x + 79.9$.

a. What does the 79.9 mean in this situation?

Responses vary. The resting heart rate of someone who doesn't exercise.

b. What does the -5.23 mean in this situation?

Responses vary. The decrease in resting heart rate expected with every extra hour of exercise a week.

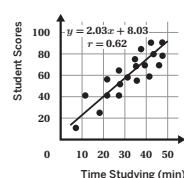
2. Nina exercised for 4 hours this week. Use the equation of line of best fit to predict her resting heart rate.

$$y = -5.23(4) + 79.9 = 58.98 \text{ BPM}$$

3. Do you think the prediction is accurate? Use the r -value to explain your thinking.

Responses vary. Looking at the data for 4 hours of exercise, there is almost 20 BPM difference between those two values. However, I think it is a close estimate to the line of best fit because the r -value is strong.

Problems 4–6: The scatter plot shows the relationship between the amount of time spent studying for a recent math test (in minutes) and the scores the students earned on the test.



4. The equation for the best fit line is $y = 2.03x + 8.03$. What does the 2.03 mean in this situation?

For every one minute spent studying, the test score will go up by 2.03 points.

5. Rajah studied for this test for 40 minutes.

Use the equation of line of best fit to predict his score on the test.

$$y = 2.03(40) + 8.03 = 89.23 \text{ points}$$

6. Do you think the prediction is accurate? Use the r -value to explain your thinking.

Explanations vary. Looking at the data near 40 minutes of studying, there is almost a 25-point difference between the largest and smallest value. However, I think each of those values is close to the line of best fit because the r -value is strong.

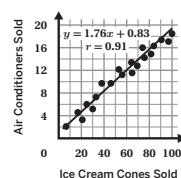
Unit 3 Lesson 17

79

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Name: _____ Date: _____ Period: _____

Problems 7–9: The scatter plot shows the relationship between the number of ice cream cones and the number of air conditioners sold in a month.



7. Describe the relationship between the number of ice cream cones sold and the number of air conditioners sold in a month.

Explanations vary. There is a strong positive association between the number of ice cream cones and the number of air conditioners sold in a month.

8. Do you think one variable causes the other? Explain your thinking.

Explanations vary. No. Buying ice cream does not lead to buying air conditioners.

9. What else might affect this relationship? Explain your thinking.

Explanations vary. The temperature. Both of these sales increase when the temperature rises in the summer months.

Unit 3 Lesson 17

80

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSS.ID.C.7
2	1	HSS.ID.B.6.A
3	2	HSS.ID.C.8
4	2	HSS.ID.C.7
5	1	HSS.ID.B.6.A
6	2	HSS.ID.C.8
7	2	HSS.ID.B.6
8	2	HSS.ID.B.6
9	2	HSS.ID.C.9

Notes:

Additional Practice

A1.4.01

- Here is Rule A.

Rule A

Input	2	3	2	6
Output	A	C	B	F

Is Rule A a function? Explain your thinking.

No. Explanations vary. Rule A is not a function because one of the inputs has multiple outputs. The input 2 has an output of A and B.

- Problems 2–3:** Here is Rule B.

Rule B

Input	3	6	9	12
Output	2	4	6	?

- Is Rule B a function? Explain your thinking.

Yes. Explanations vary. Each input has exactly one output.

- Predict what the output could be when the input is 12.

Responses vary. The output could be 8.

Problems 4–5: Here is Rule C. Rule C adds 3 to the input and then multiplies by 2 to get the output.

Rule C

Input	1	9	11	20
Output	8	?	28	46

4. Is Rule C a function? Explain your thinking.

Yes. Explanations vary. Each input has exactly one output.

5. Predict the output when the input is 9.

24

6. Here is Rule D.

Rule D

Input	2	2	3	4
Output	3	4	4	6

Is Rule D a function? Explain your thinking.

No. Explanations vary. Rule D is not a function because one of the inputs has multiple outputs. The input 2 has an output of 3 and 4.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.O1

1. Here is Rule A.

Rule A				
Input	2	3	2	6
Output	A	C	B	F

Is Rule A a function? Explain your thinking.

No. Explanations vary. Rule A is not a function because one of the inputs has multiple outputs. The input 2 has an output of A and B.

- Problems 2–3: Here is Rule B.

Rule B				
Input	3	6	9	12
Output	2	4	6	?

2. Is Rule B a function? Explain your thinking.

Yes. Explanations vary. Each input has exactly one output.

3. Predict what the output could be when the input is 12.

Responses vary. The output could be 8.

Unit 4 Lesson 1

85

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Name: _____ Date: _____ Period: _____

- Problems 4–5: Here is Rule C. Rule C adds 3 to the input and then multiplies by 2 to get the output.

Rule C				
Input	1	9	11	20
Output	8	?	28	46

4. Is Rule C a function? Explain your thinking.

Yes. Explanations vary. Each input has exactly one output.

5. Predict the output when the input is 9.

24

6. Here is Rule D.

Rule D				
Input	2	2	3	4
Output	3	4	4	6

Is Rule D a function? Explain your thinking.

No. Explanations vary. Rule D is not a function because one of the inputs has multiple outputs. The input 2 has an output of 3 and 4.

Unit 4 Lesson 1

86

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	HSF.IF.A.1
2	1	HSF.IF.A.1
3	2	HSF.IF.A.1
4	2	HSF.IF.A.1
5	1	HSF.IF.A.1
6	2	HSF.IF.A.1

Notes:

Additional Practice

A1.4.02

1. The function $c(t)$ models the cost of attending a carnival, after purchasing t tickets. What does $c(3) = 21$ represent in this model?

The cost of 3 tickets is \$21

2. The function $f(t)$ represents the amount of meters Elena and her dog walk after t minutes. Which equation represents her walking the dog 280 meters in 4 minutes?

A. $f(2) = 140$

C. $f(140) = 2$

B. $f(4) = 280$

D. $f(280) = 4$

3. The function $r(t)$ represents the height of a roller coaster t seconds after its launch. Match each verbal statement with its corresponding function notation. Not all of the function notation equations will be used.

Verbal statement

- a. The roller coaster is n feet above the ground after t seconds.

- b. After 75 seconds, the roller coaster is 130 feet above the ground.

- c. The roller coaster is 75 feet above the ground after 130 seconds.

- d. After 110 seconds, the roller coaster is at ground level.

Function notation

d $r(110) = 0$

b $r(75) = 130$

$r(0) = 110$

c $r(130) = 75$

a $r(t) = n$

$r(n) = t$

Problems 4–5: A dessert shop sells ice cream, milkshakes, and smoothies. They use the functions $i(x)$, $m(x)$, and $s(x)$ to represent the cost of their ice cream, milkshakes, and smoothies in dollars with x additional toppings.

4. What does the statement $s(0) = 13$ mean?

Responses vary. A smoothie with no additional toppings costs \$13.

5. Hikari says $i(3) < m(3)$ means that an ice cream with 3 toppings cost more than a milkshake with 3 toppings. Do you agree?

Yes

No

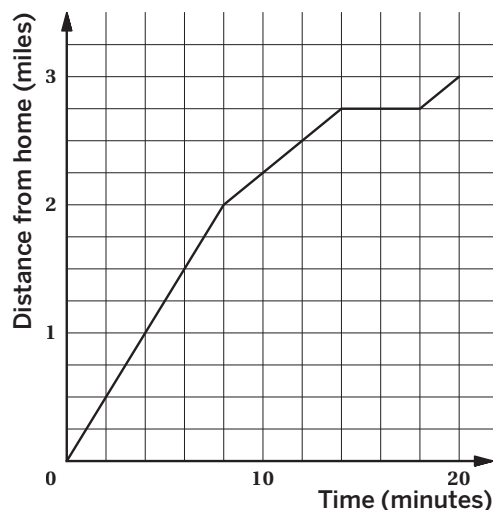
Explain your thinking

The statement $i(3) < m(3)$ means that the cost of a milkshake with 3 toppings is greater than the cost of an ice cream with 3 toppings

6. The function $c(n)$ gives the cost, in dollars, of buying n packs of gum. Which of the following statements are true? Select *all* that apply.

- ☐ A. $c(4) = 6$ means 6 packs of gum cost \$4.
- ☒ B. $c(4) = 6$ means 4 packs of gum cost \$6.
- ☒ C. $c(3)$ represents the cost of 3 packs of gum.
- ☐ D. $c(3)$ represents the packs of gum that cost \$3.
- ☒ E. The equation $c(5) = 7.5$ means that five packs of gum cost \$7.50.

Problems 7–9: Shawn is riding a bike to a friend's house. The graph represents the function $d(t)$, Shawn's distance from home, in miles, after t minutes.



7. How far away from home is Shawn after 10 minutes?

2.25 miles

8. After 20 minutes, Shawn is 3 miles from home. Lin writes this verbal statement in function notation as $d(3) = 20$. Is she correct? Explain your thinking.

No. Explanations vary. Lin switched the input and output values. The correct statement in function notation is $d(20) = 3$.

9. Kiran claims that distance is a function of time, and time is a function of distance. Do you agree? Explain your thinking.

No. Explanations vary. Distance is a function of time, because for each input, there is exactly one output. Time is not a function of distance because there are multiple output values when the input is 2.75 miles.

Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.02

1. The function $c(t)$ models the cost of attending a carnival, after purchasing t tickets. What does $c(3) = 21$ represent in this model?

The cost of 3 tickets is \$21

2. The function $f(t)$ represents the amount of meters Elena and her dog walk after t minutes. Which equation represents her walking the dog 280 meters in 4 minutes?

- A. $f(2) = 140$ C. $f(140) = 2$
B. $f(4) = 280$ D. $f(280) = 4$

3. The function $r(t)$ represents the height of a roller coaster t seconds after its launch. Match each verbal statement with its corresponding function notation. Not all of the function notation equations will be used.

Verbal statement	Function notation
a. The roller coaster is n feet above the ground after t seconds.	d. $r(110) = 0$
b. After 75 seconds, the roller coaster is 130 feet above the ground.	b. $r(75) = 130$
c. The roller coaster is 75 feet above the ground after 130 seconds.	c. $r(130) = 75$
d. After 110 seconds, the roller coaster is at ground level.	a. $r(t) = n$
	$r(n) = t$

Unit 4 Lesson 2

87

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Name: _____ Date: _____ Period: _____

Problems 4–5: A dessert shop sells ice cream, milkshakes, and smoothies. They use the functions $i(x)$, $m(x)$, and $s(x)$ to represent the cost of their ice cream, milkshakes, and smoothies in dollars with x additional toppings.

4. What does the statement $s(0) = 13$ mean?

Responses vary. A smoothie with no additional toppings costs \$13.

5. Hikari says $i(3) < m(3)$ means that an ice cream with 3 toppings cost more than a milkshake with 3 toppings. Do you agree?

Yes

No

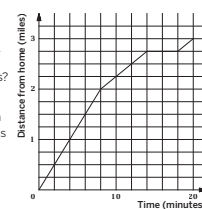
Explain your thinking

The statement $i(3) < m(3)$ means that the cost of a milkshake with 3 toppings is greater than the cost of an ice cream with 3 toppings

6. The function $c(n)$ gives the cost, in dollars, of buying n packs of gum. Which of the following statements are true? Select all that apply.

- ☐ A. $c(4) = 6$ means 6 packs of gum cost \$4.
☒ B. $c(4) = 6$ means 4 packs of gum cost \$6.
☒ C. $c(3)$ represents the cost of 3 packs of gum.
☐ D. $c(3)$ represents the packs of gum that cost \$3.
☒ E. The equation $c(5) = 7.5$ means that five packs of gum cost \$7.50.

Problems 7–9: Shawn is riding a bike to a friend's house. The graph represents the function $d(t)$. Shawn's distance from home, in miles, after t minutes.



7. How far away from home is Shawn after 10 minutes?

2.25 miles

8. After 20 minutes, Shawn is 3 miles from home. Lin writes this verbal statement in function notation as $d(3) = 20$. Is she correct? Explain your thinking.

No. Explanations vary. Lin switched the input and output values. The correct statement in function notation is $d(20) = 3$.

9. Kiran claims that distance is a function of time, and time is a function of distance. Do you agree? Explain your thinking.

No. Explanations vary. Distance is a function of time, because for each input, there is exactly one output. Time is not a function of distance because there are multiple output values when the input is 2.75 miles.

Unit 4 Lesson 2

88

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.A.2
2	1	HSF.IF.A.2
3	2	HSF.IF.A.2
4	2	HSF.IF.A.2
5	2	HSF.IF.A.2
6	3	HSF.IF.A.2
7	3	HSF.IF.A.2
8	3	HSF.IF.A.2
9	3	HSF.IF.A.2

Notes:

Additional Practice

A1.4.03

1. Let $f(x) = 2x - 6$.

Calculate the value of each function notation expression.

Expression	Value
$f(-2)$	-10
$f(4)$	2
$f(1.5)$	-3
$f(6)$	6

2. Match each statement with a description of the function it represents.

Statement	Description
a. $f(x) = 5x - 3$	c To get the output value, subtract 3 from the input value, and then multiply the result by 5.
b. $g(x) = 3(x - 5)$	b To get the output value, subtract 5 from the input value, and then multiply the result by 3.
c. $h(x) = 5(x - 3)$	a To get the output value, multiply the input value by 5, and then subtract 3 from the result.
d. $j(x) = 3x - 5$	d To get the output value, multiply the input value by 3, and then subtract 5 from the result.

3. One tomato plant costs \$6. The function $c(x)$ represents the cost, in dollars, of x tomato plants, where the cost of 1 tomato plant is \$6.

- a Complete the table.

x	0	1	2	3	4	5	6
$c(x)$	0	6	12	18	24	30	36

- b Write a function notation statement to represent the function $c(x)$.
 $c(x) = 6x$

4. Consider the function $b(x) = 7.5x + 30$. What is the value of $b(4)$?

- A. 30
B. 40.5
C. 41.5
D. 60

Problems 5–7: A local swimming pool offers membership Plans Q and R as described in the table. The function representing each plan gives the total cost, in dollars, for m months of membership.

Plan Q	Plan R
\$50 per month, plus a \$25 application fee $q(m) = 50m + 25$	\$400 per year $r(m) = 400$

5. Complete the following to describe the meaning of the statement $q(3) = 175$.

The total cost of Plan Q is \$ 175 after 3 months.

6. Which is greater, $q(6)$ or $r(6)$?

$r(6)$

7. Which is less, $q(10)$ or $r(10)$?

$r(10)$

8. A dolphin is swimming at a constant speed of 3 miles per hour. The total distance the dolphin has traveled in t hours can be represented by the function $d(t) = 3t$.

Determine the value of $d(2.5)$ and explain what it means in this situation.

$d(2.5) = 7.5$. **Explanations vary. The total distance traveled by the dolphin after 2.5 hours is 7.5 miles.**

Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.03

1. Let $f(x) = 2x - 6$.

Calculate the value of each function notation expression.

Expression	Value
$f(-2)$	-10
$f(4)$	2
$f(1.5)$	-3
$f(6)$	6

2. Match each statement with a description of the function it represents.

Statement	Description
a. $f(x) = 5x - 3$	c. To get the output value, subtract 3 from the input value, and then multiply the result by 5.
b. $g(x) = 3(x - 5)$	b. To get the output value, subtract 5 from the input value, and then multiply the result by 3.
c. $h(x) = 5(x - 3)$	a. To get the output value, multiply the input value by 5, and then subtract 3 from the result.
d. $j(x) = 3x - 5$	d. To get the output value, multiply the input value by 3, and then subtract 5 from the result.

3. One tomato plant costs \$6. The function $c(x)$ represents the cost, in dollars, of x tomato plants, where the cost of 1 tomato plant is \$6.

a. Complete the table.

x	0	1	2	3	4	5	6
$c(x)$	0	6	12	18	24	30	36

b. Write a function notation statement to represent the function $c(x)$.
 $c(x) = 6x$

Unit 4 Lesson 3

89

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Name: _____ Date: _____ Period: _____

4. Consider the function $h(x) = 7.5x + 30$. What is the value of $h(4)$?

- A. 30
B. 40.5
C. 41.5
D. 60

Problems 5–7: A local swimming pool offers membership Plans Q and R as described in the table. The function representing each plan gives the total cost, in dollars, for m months of membership.

Plan Q	Plan R
\$50 per month, plus a \$25 application fee $q(m) = 50m + 25$	\$400 per year $r(m) = 400$

5. Complete the following to describe the meaning of the statement $q(3) = 175$.

The total cost of Plan Q is \$175 after 3 months.

6. Which is greater, $q(6)$ or $r(6)$?

$r(6)$

7. Which is less, $q(10)$ or $r(10)$?

$r(10)$

8. A dolphin is swimming at a constant speed of 3 miles per hour. The total distance the dolphin has traveled in t hours can be represented by the function $d(t) = 3t$.

Determine the value of $d(2.5)$ and explain what it means in this situation.

$d(2.5) = 7.5$. Explanations vary. The total distance traveled by the dolphin after 2.5 hours is 7.5 miles.

Unit 4 Lesson 3

90

Additional Practice

Practice Problem Analysis

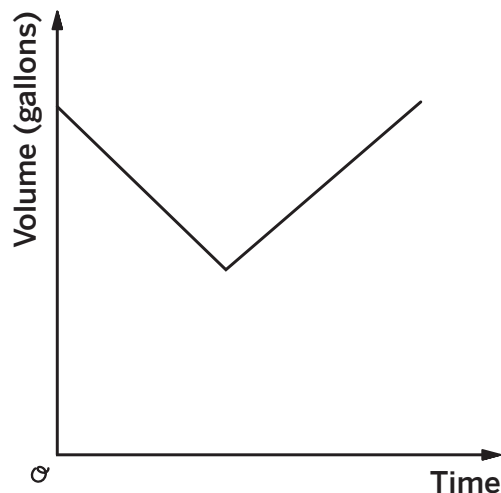
Problem	DOK	Standard(s)
1	1	HSF.IF.A.2
2	2	HSF.BF.A.1.A
3	2	HSF.BF.A.1.A, HSF.IF.A.1
4	1	HSF.IF.A.2
5	2	HSF.IF.A.2
6	3	HSF.IF.A.2
7	2	HSF.IF.A.2
8	3	HSF.IF.A.2

Notes:

Additional Practice

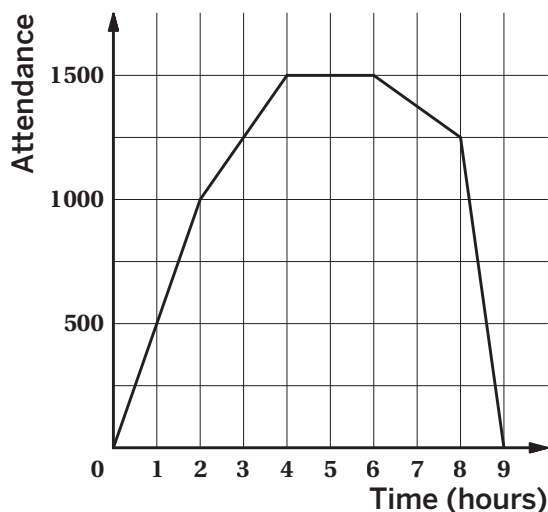
A1.4.04

1. The graph represents the volume of water in a tank as a function of time. Which of the descriptions matches the graph?



- A. A 2,000-gallon water tank starts out empty. It is filled for 5 hours, slowly at first, and faster later.
- ☒ B. A full 10-gallon water tank is drained for 30 seconds until it is half full. Afterwards, it gets refilled.
- C. An empty 20-gallon water tank is filled at a constant rate for 3 minutes until it is half full. Then it is emptied at a constant rate for 3 minutes.
- D. An empty 100-gallon water tank is filled in 50 minutes. Then a dog jumps in and splashes around for 10 minutes, letting 7 gallons of water out. The tank is refilled afterwards.

2. The graph shows the attendance at an arts festival as a function of time in hours. Which of the following statements about the graph are true? Select *all* that apply.



- ☒ A. The number of people increases the first 4 hours, then stays the same for 2 hours, and then decreases the last 3 hours.
- ☐ B. The number of people decreases the first 4 hours, then stays the same for 2 hours, and then increases the last 3 hours.
- ☐ C. There were 1500 people in attendance for 3 hours.
- ☒ D. In the fourth hour there are 1500 people at the arts festival.
- ☒ E. The number of people in attendance dips by 250 people between hour 6 and hour 8.

Problems 3–5: Lin goes for a run every morning. Her fitness tracker made a graph to represent the distance she runs over time.

3. At what time does Lin take a rest?

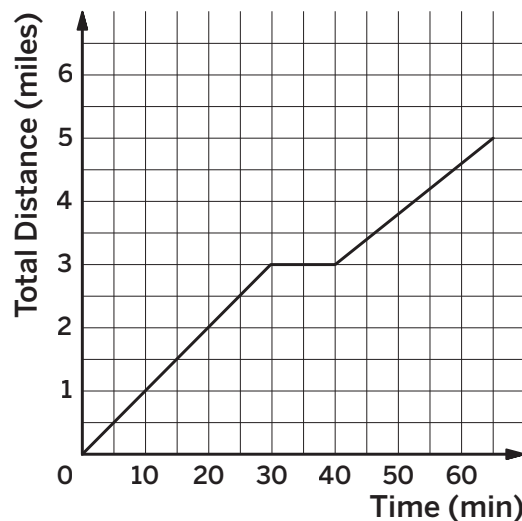
Responses vary. 35 minutes.

4. How long is Lin's running routine?

65 minutes.

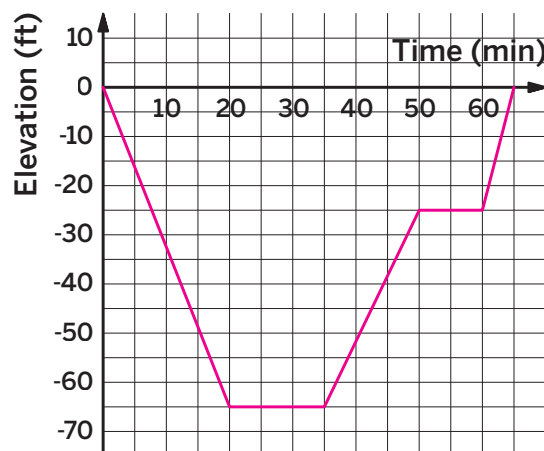
5. How many miles does Lin run?

5 miles.



6. Priya goes on a tour of a cave. The tour starts at ground level and then descends for 20 minutes to a depth of 65 feet below ground level. The tour stays at this level for 15 minutes, and then ascends for 15 minutes to a depth of 25 feet below ground level. The tour stays at this level for 10 minutes, and then spends the last 5 minutes ascending to ground level.

Sketch a possible graph describing Priya's elevation (distance relative to ground level) as a function of time.



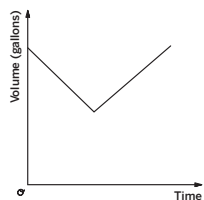
Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.04

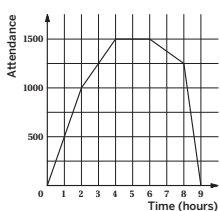
1. The graph represents the volume of water in a tank as a function of time. Which of the descriptions matches the graph?

- A. A 2,000-gallon water tank starts out empty. It is filled for 5 hours, slowly at first, and faster later.
- B. A full 10-gallon water tank is drained for 30 seconds until it is half full. Afterwards, it gets refilled.**
- C. An empty 20-gallon water tank is filled at a constant rate for 3 minutes until it is half full. Then it is emptied at a constant rate for 3 minutes.
- D. An empty 100-gallon water tank is filled in 50 minutes. Then a dog jumps in and splashes around for 10 minutes, letting 7 gallons of water out. The tank is refilled afterwards.



2. The graph shows the attendance at an arts festival as a function of time in hours. Which of the following statements about the graph are true? Select all that apply.

- ☒ A. The number of people increases the first 4 hours, then stays the same for 2 hours, and then decreases the last 3 hours.
- ☐ B. The number of people decreases the first 4 hours, then stays the same for 2 hours, and then increases the last 3 hours.
- ☐ C. There were 1500 people in attendance for 3 hours.
- ☒ D. In the fourth hour there are 1500 people at the arts festival.
- ☒ E. The number of people in attendance dips by 250 people between hour 6 and hour 8.



Unit 4 Lesson 4

91

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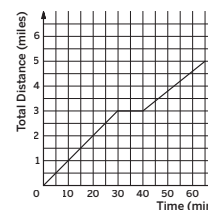
Name: _____ Date: _____ Period: _____

- Problems 3–5: Lin goes for a run every morning. Her fitness tracker made a graph to represent the distance she runs over time.

3. At what time does Lin take a rest?
Responses vary, 35 minutes.

4. How long is Lin's running routine?
65 minutes.

5. How many miles does Lin run?
5 miles.



6. Priya goes on a tour of a cave. The tour starts at ground level and then descends for 20 minutes to a depth of 65 feet below ground level. The tour stays at this level for 15 minutes, and then ascends for 15 minutes to a depth of 25 feet below ground level. The tour stays at this level for 10 minutes, and then spends the last 5 minutes ascending to ground level.

Sketch a possible graph describing Priya's elevation (distance relative to ground level) as a function of time.



Unit 4 Lesson 4

92

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.B.4
2	2	HSF.IF.B.4, HSF.IF.B.6
3	2	HSF.IF.B.4, HSF.IF.B.6
4	3	HSF.IF.B.4, HSF.IF.B.6
5	2	HSF.IF.B.4, HSF.IF.B.6
6	2	HSF.IF.B.4, HSF.IF.B.6

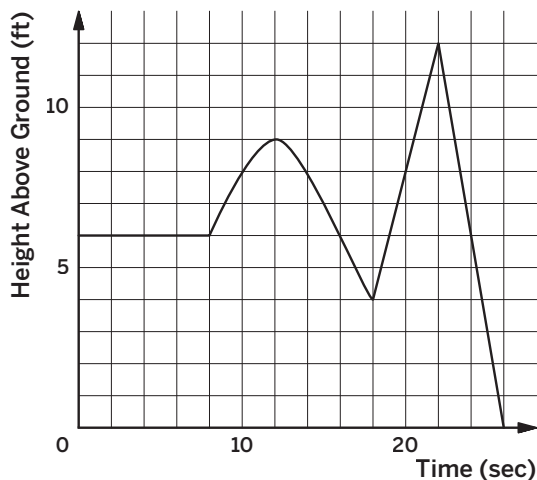
Notes:

Additional Practice

A1.4.05

1. The graph represents Clare's height above ground as she completes an obstacle course. Determine which of the following statements are true. Select *all* that apply.

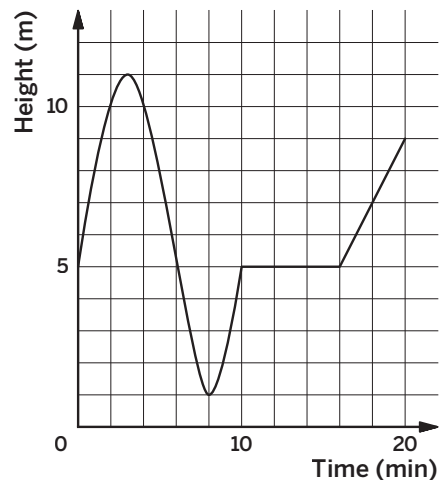
- ☐ A. The minimum of the graph is located at (18, 4).
- ☒ B. The graph is always positive.
- ☒ C. From 12 seconds to 18 seconds, Clare is moving closer the ground.
- ☒ D. The graph has a maximum at (22, 12).
- ☐ E. Clare reached the highest point at about 26 seconds.



2. Nakia was riding her bike around the neighborhood with her friends. The graph represents Nakia's distance from home as a function of time.

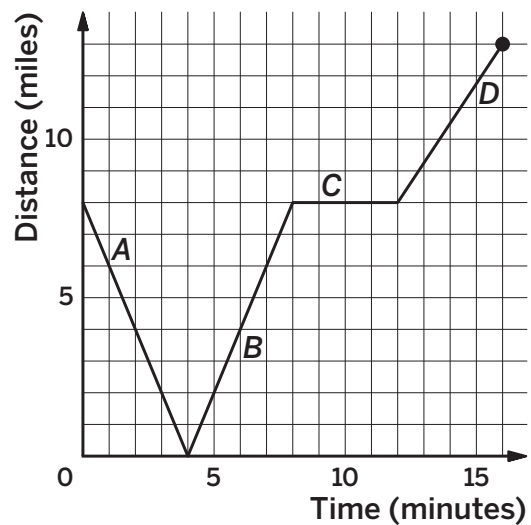
Describe Nakia's distance from home over time. Use terms that you learned in this lesson.

Responses vary. When Nakia is biking with her neighbor her distance from home increases and decreases. After about 3 minutes she is the 11 meters from home. This is her maximum distance from home on this bike ride. After that, her distance from home decreases to 1 meter, her minimum distance from home on this ride. She then increases her distance from home and returns to her neighbors house. She spends 6 minutes there before riding down the street.



3. The graph represents the function $d(t)$, which gives the distance that Kiran is away from home as a function of time t . For which part of the graph does Kiran's distance from home decrease?

- A.** Part A
 B. Part B
 C. Part C
 D. Part D

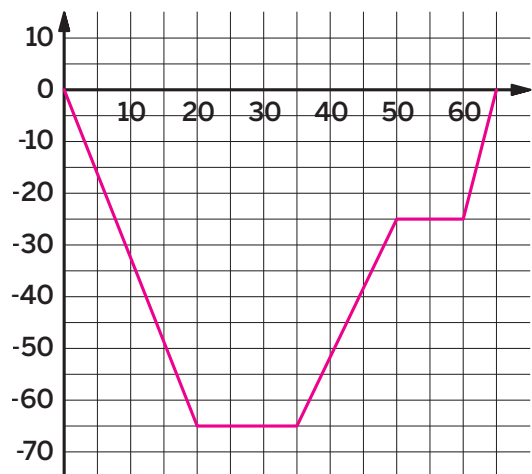


4. Here are three statements:

- The function is always negative.
- The function has a minimum of -65 .
- The function is increasing from 60 to 65.

Sketch a graph that makes all of the statements true.

Graphs vary.



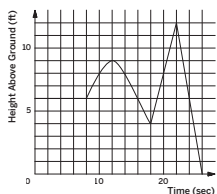
Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.05

1. The graph represents Clare's height above ground as she completes an obstacle course. Determine which of the following statements are true. Select all that apply.

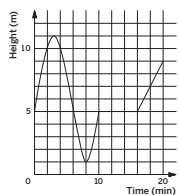
- ☐ A. The minimum of the graph is located at (18, 4).
☒ B. The graph is always positive.
☒ C. From 12 seconds to 18 seconds, Clare is moving closer the ground.
☒ D. The graph has a maximum at (22, 12).
☐ E. Clare reached the highest point at about 26 seconds.



2. Nakia was riding her bike around the neighborhood with her friends. The graph represents Nakia's distance from home as a function of time.

Describe Nakia's distance from home over time. Use terms that you learned in this lesson.

Responses vary. When Nakia is biking with her neighbor her distance from home increases and decreases. After about 3 minutes she is the 11 meters from home. This is her maximum distance from home on this bike ride. After that, her distance from home decreases to 1 meter, her minimum distance from home on this ride. She then increases her distance from home and returns to her neighbors house. She spends 6 minutes there before riding down the street.



Unit 4 Lesson 5

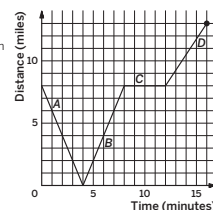
93

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Name: _____ Date: _____ Period: _____

3. The graph represents the function $d(t)$, which gives the distance that Kiran is away from home as a function of time t . For which part of the graph does Kiran's distance from home decrease?

- ☒ A. Part A
☐ B. Part B
☐ C. Part C
☐ D. Part D

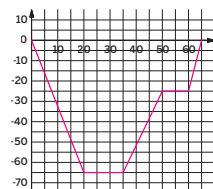


4. Here are three statements:

- The function is always negative.
- The function has a minimum of -65.
- The function is increasing from 60 to 65.

Sketch a graph that makes all of the statements true.

Graphs vary.



Unit 4 Lesson 5

94

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.B.4
2	2	HSF.IF.B.4
3	1	HSF.IF.B.4
4	3	HSF.IF.B.4, HSF.IF.B.6

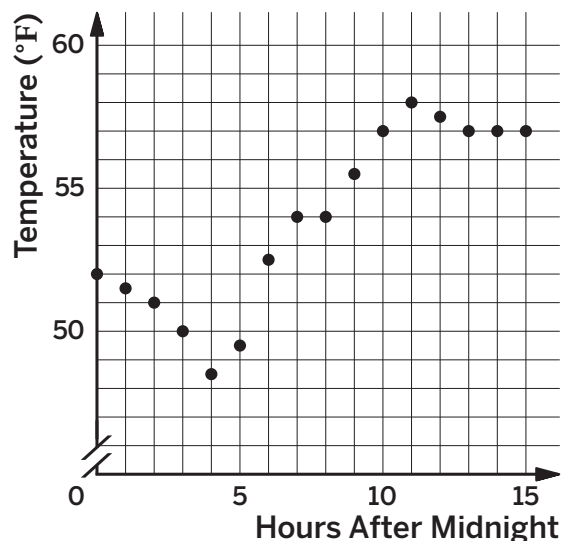
Notes:

Additional Practice

A1.4.06

1. The temperature was recorded at several times during the day. The function $t(n)$ represents the temperature in degrees Fahrenheit given the number of hours since midnight n . Use the graph to determine if the average rate of change for each interval is *positive*, *negative*, or *zero*. Place a check in the appropriate column.

	+	-	0
$n = 1$ to $n = 4$		✓	
$n = 4$ to $n = 7$	✓		
$n = 7$ to $n = 8$			✓
$n = 11$ to $n = 13$		✓	
$n = 13$ to $n = 15$			✓



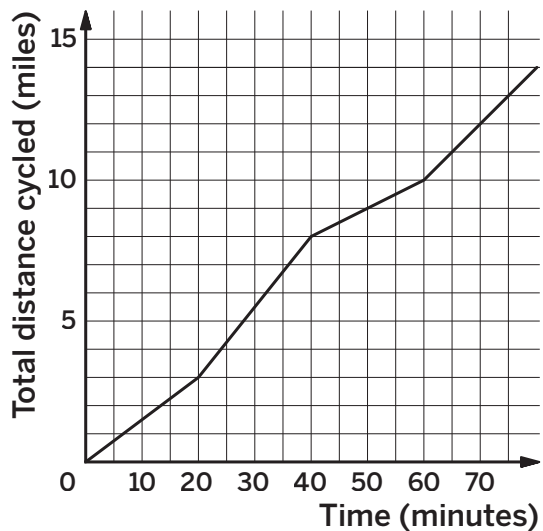
Problems 2–3: The graph shows the total distance, in miles, that Kiran cycled as a function of time, in minutes.

2. Was Kiran cycling faster between 20 and 35 minutes or between 50 and 65 minutes? Explain your thinking.

Between 20 and 35 minutes. Explanations vary. Kiran's average rate of change was greater.

3. Was Kiran cycling faster between 30 and 50 minutes or between 60 and 80 minutes? Explain your thinking.

Between 60 and 80 minutes. Explanations vary. Kiran's average rate of change was greater.



Problems 4–5: This table shows the height of a tree throughout the years. The function $h(t)$ gives the height of the tree in Year t .

Year	0	3	6	9	12	15	18	21
Height (ft)	2	4.5	8.1	11.4	15.6	19.2	23.1	26.4

4. Determine the average rate of change for $h(t)$ between Years 3 and 9.

1.15 ft per year

5. Is the average rate of change for $h(t)$ between Years 12 and 21 *greater than*, *less than*, or *equal to* the rate of change between Years 3 and 9? Explain your thinking.

Greater than. Explanations vary. The average rate of change between Years 12 and 21 is 1.2 ft per year, which is greater than 1.15 ft per year.

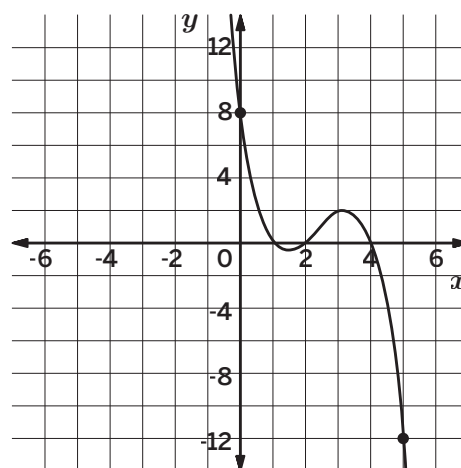
6. Use the graph to determine the average rate of change between $x = 0$ and $x = 5$.

A. -4

B. 4

C. -5

D. 5



7. The number of people who visited an aquarium each day for a week is shown in the table. The function $f(d)$ gives the number of visitors on Day d . Shawn claims that the average rate of change between Days 1 and 3 is greater than the average rate of change between Days 3 and 6. Do you agree? Explain your thinking.

Day	1	2	3	4	5	6	7
Visitors (thousands)	4.5	4.8	5.2	3.9	4.4	5.6	6.1

Yes. Explanations vary. The average rate of change between Days 1 and 3 is an increase of 350 people per day. The average rate of change between Days 3 and 6 is an increase of about 133 people per day.

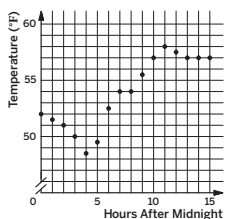
Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.06

1. The temperature was recorded at several times during the day. The function $t(n)$ represents the temperature in degrees Fahrenheit given the number of hours since midnight n . Use the graph to determine if the average rate of change for each interval is *positive*, *negative*, or *zero*. Place a check in the appropriate column.

	+	-	0
$n = 1$ to $n = 4$		✓	
$n = 4$ to $n = 7$	✓		
$n = 7$ to $n = 8$			✓
$n = 11$ to $n = 13$		✓	
$n = 13$ to $n = 15$			✓



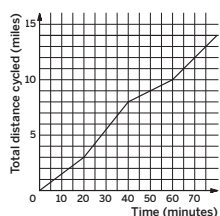
Problems 2–3: The graph shows the total distance, in miles, that Kiran cycled as a function of time, in minutes.

2. Was Kiran cycling faster between 20 and 35 minutes or between 50 and 65 minutes? Explain your thinking.

Between 20 and 35 minutes. Explanations vary. Kiran's average rate of change was greater.

3. Was Kiran cycling faster between 30 and 50 minutes or between 60 and 80 minutes? Explain your thinking.

Between 60 and 80 minutes. Explanations vary. Kiran's average rate of change was greater.



Unit 4 Lesson 6

95

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Name: _____ Date: _____ Period: _____

Problems 4–5: This table shows the height of a tree throughout the years. The function $h(t)$ gives the height of the tree in Year t .

Year	0	3	6	9	12	15	18	21
Height (ft)	2	4.5	8.1	11.4	15.6	19.2	23.1	26.4

4. Determine the average rate of change for $h(t)$ between Years 3 and 9.

1.15 ft per year

5. Is the average rate of change for $h(t)$ between Years 12 and 21 *greater than*, *less than*, or *equal to* the rate of change between Years 3 and 9? Explain your thinking.

Greater than. Explanations vary. The average rate of change between Years 12 and 21 is 1.2 ft per year, which is greater than 1.15 ft per year.

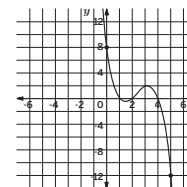
6. Use the graph to determine the average rate of change between $x = 0$ and $x = 5$.

A. -4

B. 4

C. -5

D. 5



7. The number of people who visited an aquarium each day for a week is shown in the table. The function $f(d)$ gives the number of visitors on Day d . Shawn claims that the average rate of change between Days 1 and 3 is greater than the average rate of change between Days 3 and 6. Do you agree? Explain your thinking.

Day	1	2	3	4	5	6	7
Visitors (thousands)	4.5	4.8	5.2	3.9	4.4	5.6	6.1

Yes. Explanations vary. The average rate of change between Days 1 and 3 is an increase of 350 people per day. The average rate of change between Days 3 and 6 is an increase of about 133 people per day.

Unit 4 Lesson 6

96

Additional Practice

Practice Problem Analysis

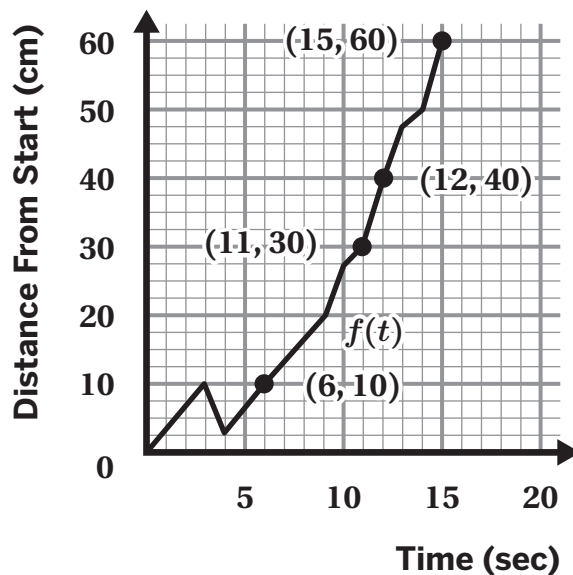
Problem	DOK	Standard(s)
1	1	HSF.IF.B.6
2	2	HSF.IF.B.6
3	2	HSF.IF.B.6
4	2	HSF.IF.B.6
5	2	HSF.IF.B.6
6	2	HSF.IF.B.6
7	3	HSF.IF.B.6

Notes:

Additional Practice

A1.4.07

Problems 1–2: Fatima built a model train for a competition in her technology class. $f(t)$ represents the distance of Fatima's model train, in centimeters, after t seconds.



1. Use the graph to determine the missing value in each function statement.

$$f(11) = \underline{30}$$

$$f(\underline{3 \text{ or } 6}) = 10$$

$$f(12) = \underline{40}$$

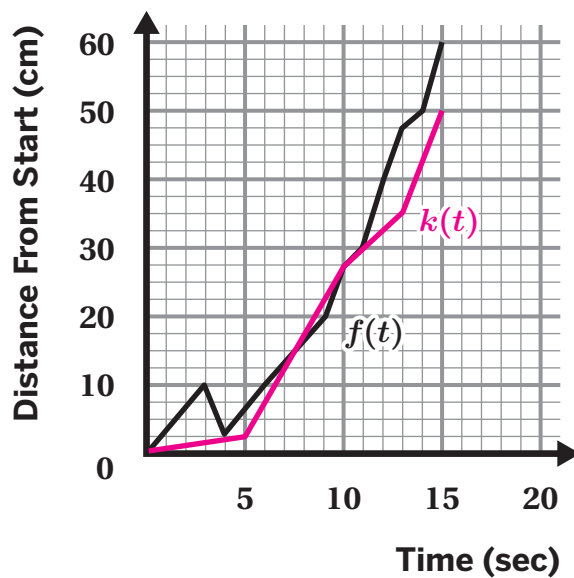
$$f(\underline{15}) = 60$$

2. Over what interval did Fatima's model train travel the slowest?

- ☒ a. 0 to 3 seconds
- ☐ b. 4 to 6 seconds
- ☐ c. 11 to 12 seconds
- ☐ d. 12 to 15 seconds

3. Kyler built a model train to race against Fatima. Use this information to make a graph that could represent the distance of Kyler's model train, $k(t)$, after t seconds:

- $k(5) < f(5)$
- $k(10) = f(10)$
- The average rate of change of $k(t)$ and $f(t)$ is the same from $t = 10$ to $t = 11$.
- $f(t)$ has a greater maximum than $k(t)$.

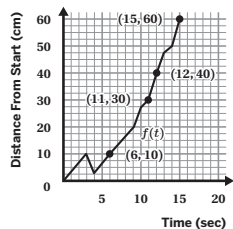


Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.07

Problems 1-2: Fatima built a model train for a competition in her technology class. $f(t)$ represents the distance of Fatima's model train, in centimeters, after t seconds.



1. Use the graph to determine the missing value in each function statement.

$f(11) = \underline{30}$

$f(\underline{3 \text{ or } 6}) = 10$

$f(12) = \underline{40}$

$f(\underline{15}) = 60$

2. Over what interval did Fatima's model train travel the slowest?

- ☒ a. 0 to 3 seconds
- ☐ b. 4 to 6 seconds
- ☐ c. 11 to 12 seconds
- ☐ d. 12 to 15 seconds

Unit 4 Lesson 7

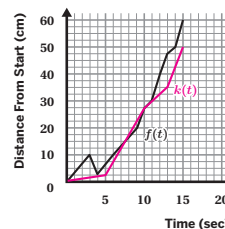
97

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Name: _____ Date: _____ Period: _____

3. Kyler built a model train to race against Fatima. Use this information to make a graph that could represent the distance of Kyler's model train, $k(t)$, after t seconds:

- $k(5) < f(5)$
- $k(10) = f(10)$
- The average rate of change of $k(t)$ and $f(t)$ is the same from $t = 10$ to $t = 11$.
- $f(t)$ has a greater maximum than $k(t)$.



Unit 4 Lesson 7

98

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSA.REI.D.11, HSF.IF.A.2, HSF.IF.B.4, HSF.IF.B.6
2	1	HSA.REI.D.11, HSF.IF.A.2, HSF.IF.B.4, HSF.IF.B.6
3	2	HSA.REI.D.11, HSF.IF.A.2, HSF.IF.B.4, HSF.IF.B.6

Notes:

Additional Practice

A1.4.09

Problems 1–2: Carlos and Madeline disagree about the domain of $f(x)$.

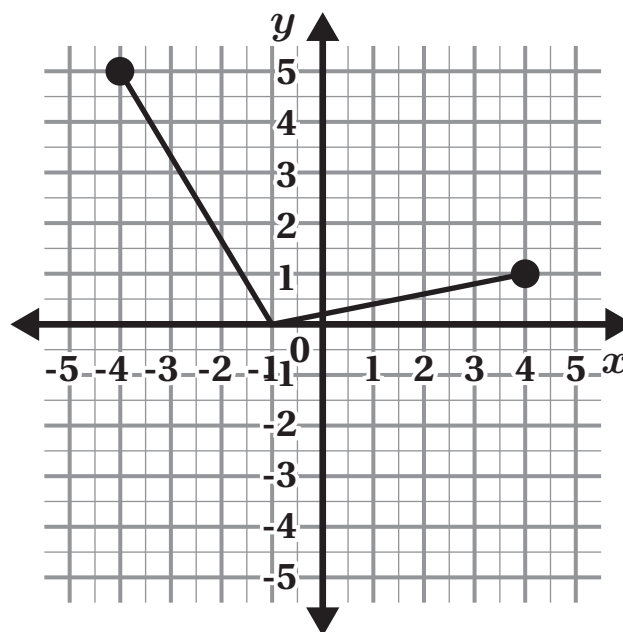
- Carlos says the domain is $0 \leq x \leq 5$
- Madeline says the domain is $-4 \leq f(x) \leq 4$

1. Whose answer is correct? Circle one.

Carlos Madeline Neither

2. Explain your thinking.

Explanations vary. Carlos is incorrect because his inequality includes numbers that would represent the range of $f(x)$. Madeline is incorrect because she used $f(x)$ in her inequality.



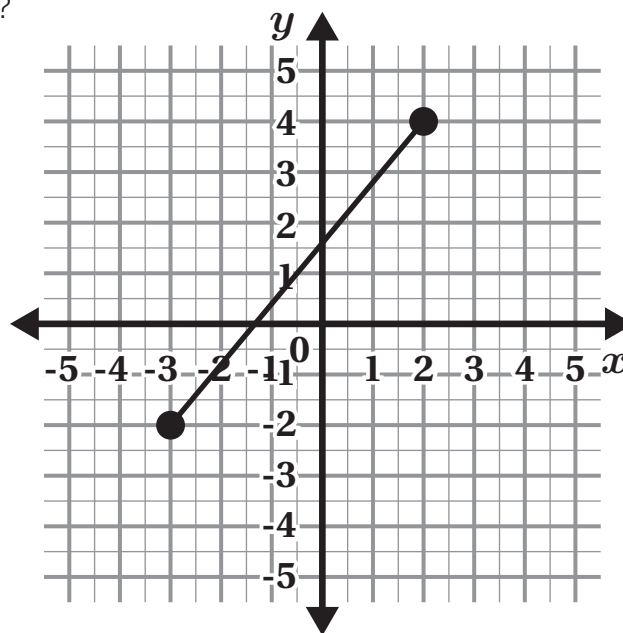
Problems 3–4: You are given the graph of a function.

3. What is the domain of the function shown?

- A. $-3 \leq x \leq 2$
- B. $-3 \leq x < 2$
- C. $-2 \leq x < 4$
- D. $-2 < x \leq 4$

4. What is the range of the function shown?

- A. $-3 < f(x) \leq 2$
- B. $-3 \leq f(x) < 2$
- C. $-2 \leq f(x) \leq 4$
- D. $-2 < f(x) \leq 4$



Problems 5–8: A squirrel is searching for food on the ground and in trees. The graph represents the function, $h(t)$, which represents the height of the squirrel as a function of time t .

5. Write a compound inequality to describe the domain of $h(t)$.

$$0 \leq t \leq 10$$

6. What does the domain represent in this situation?

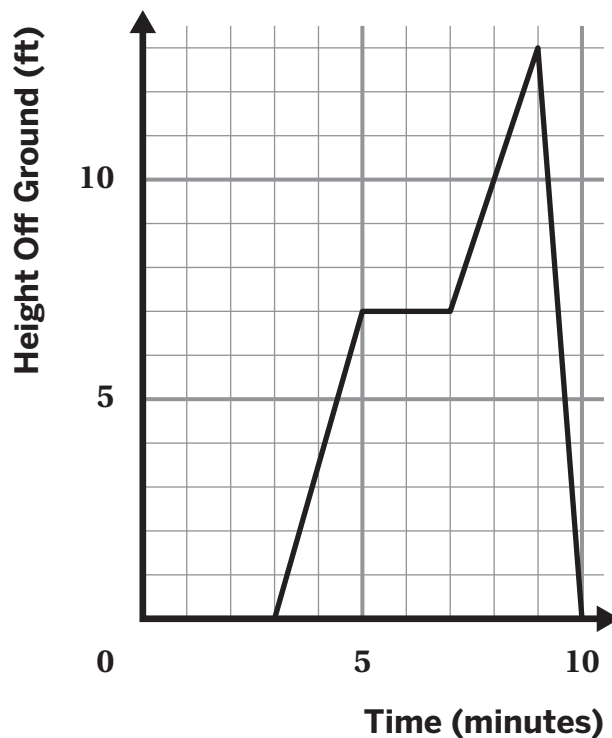
The domain represents the time the squirrel is searching for food.

7. Write a compound inequality to describe the range of $h(t)$.

$$0 \leq h(t) \leq 13$$

8. What does the range represent in this situation?

The range represents the heights, or distances from the ground, at which the squirrel is located.



Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.09

Problems 1–2: Carlos and Madeline disagree about the domain of $f(x)$.

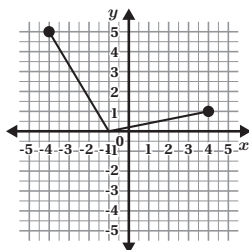
- Carlos says the domain is $0 \leq x \leq 5$
- Madeline says the domain is $-4 \leq f(x) \leq 4$

1. Whose answer is correct? Circle one.

Carlos Madeline **Neither**

2. Explain your thinking.

Explanations vary. Carlos is incorrect because his inequality includes numbers that would represent the range of $f(x)$. Madeline is incorrect because she used $f(x)$ in her inequality.



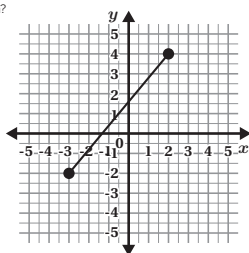
Problems 3–4: You are given the graph of a function.

3. What is the domain of the function shown?

- A.** $-3 \leq x \leq 2$
- B. $-3 \leq x < 2$
- C. $-2 \leq x < 4$
- D. $-2 < x \leq 4$

4. What is the range of the function shown?

- A. $-3 < f(x) \leq 2$
- B. $-3 \leq f(x) < 2$
- C.** $-2 \leq f(x) \leq 4$
- D. $-2 < f(x) \leq 4$



Unit 4 Lesson 9

101

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Name: _____ Date: _____ Period: _____

Problems 5–8: A squirrel is searching for food on the ground and in trees. The graph represents the function, $h(t)$, which represents the height of the squirrel as a function of time t .

5. Write a compound inequality to describe the domain of $h(t)$.

$0 \leq t \leq 10$

6. What does the domain represent in this situation?

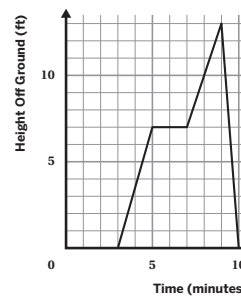
The domain represents the time the squirrel is searching for food.

7. Write a compound inequality to describe the range of $h(t)$.

$0 \leq h(t) \leq 13$

8. What does the range represent in this situation?

The range represents the heights, or distances from the ground, at which the squirrel is located.



Unit 4 Lesson 9

102

Additional Practice

Practice Problem Analysis

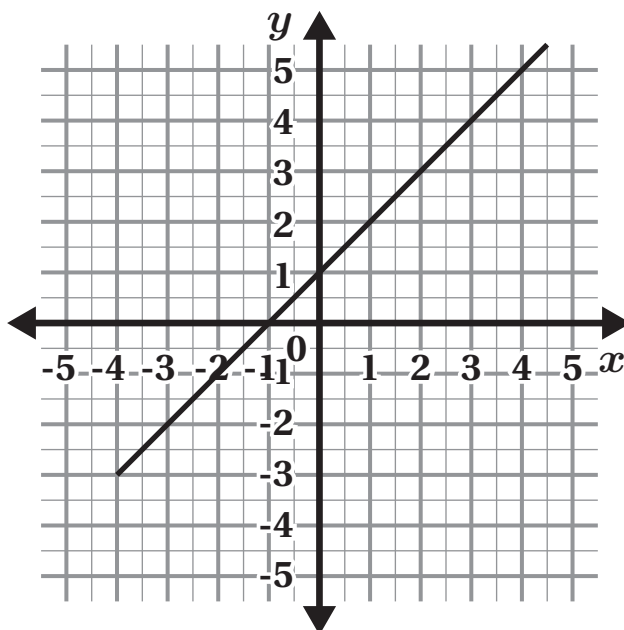
Problem	DOK	Standard(s)
1	2	HSF.IF.B.5
2	2	HSF.IF.B.5
3	1	HSF.IF.B.5
4	1	HSF.IF.B.5
5	1	HSF.IF.B.5
6	2	HSF.IF.B.5
7	1	HSF.IF.B.5
8	2	HSF.IF.B.5

Notes:

Additional Practice

A1.4.10

Problems 1–2: Here is the graph of $f(x)$.



1. What is the domain of $f(x)$?

a. $x \geq 0$

b. $x \geq -4$

c. $x \leq 0$

d. $x \leq -4$

2. What is the range of $f(x)$?

a. $f(x) \geq -3$

b. $f(x) \geq 0$

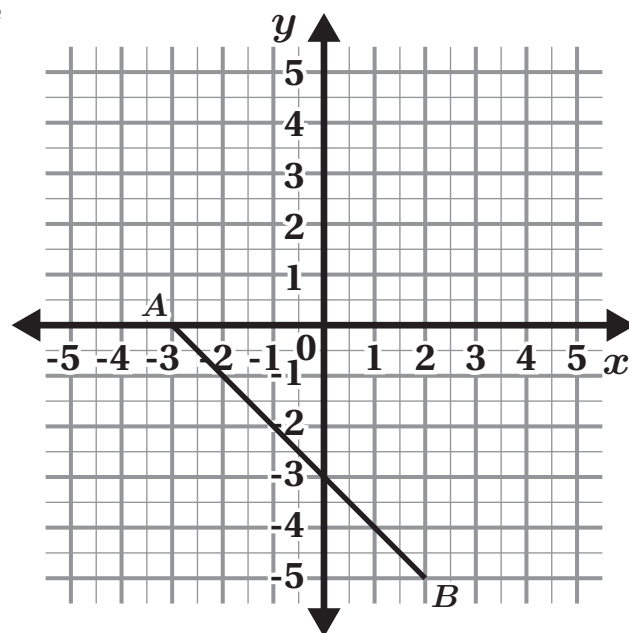
c. $f(x) \leq 3$

d. $f(x) \leq 0$

3. Fill in the blanks for the domain and range of $y = -x - 3$ from point A to point B .

$$\underline{-3} \leq x \leq \underline{2}$$

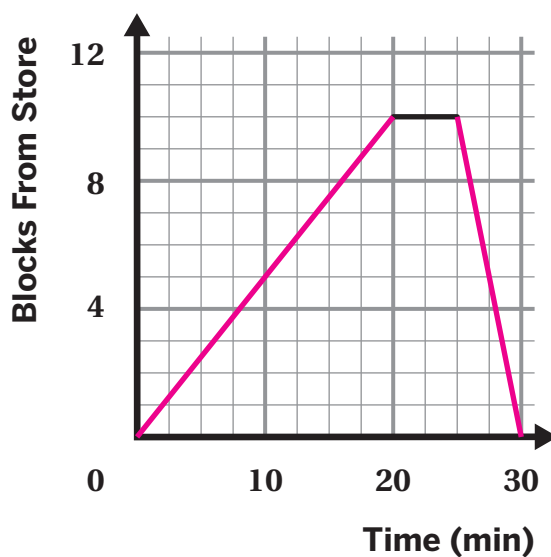
$$\underline{-5} \leq y \leq \underline{0}$$



4. Lia leaves her home to go to the grocery store. This is her path:

- She walks to the store, which is 10 blocks away, at a speed of half a block per minute.
- She is in the store for 5 minutes.
- She runs back home at a speed of 2 blocks per minute.

The graph shows part of her path. Sketch the graph of the missing pieces of Lia's path.

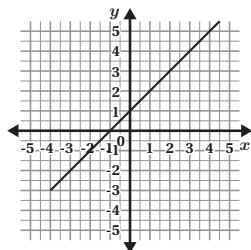


Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.10

Problems 1-2: Here is the graph of $f(x)$.



1. What is the domain of $f(x)$?

- a. $x \geq 0$
- b. $x \geq -4$
- c. $x \leq 0$
- d. $x \leq -4$

2. What is the range of $f(x)$?

- a. $f(x) \geq -3$
- b. $f(x) \geq 0$
- c. $f(x) \leq 3$
- d. $f(x) \leq 0$

Unit 4 Lesson 10

103

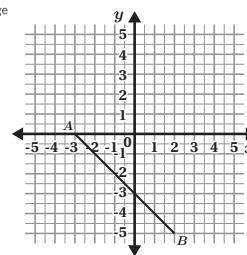
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Name: _____ Date: _____ Period: _____

3. Fill in the blanks for the domain and range of $y = -x - 3$ from point A to point B.

$-3 \leq x \leq 2$

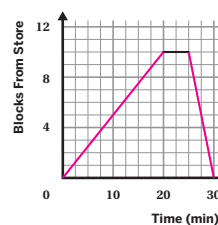
$-5 \leq y \leq 0$



4. Lia leaves her home to go to the grocery store. This is her path:

- She walks to the store, which is 10 blocks away, at a speed of half a block per minute.
- She is in the store for 5 minutes.
- She runs back home at a speed of 2 blocks per minute.

The graph shows part of her path. Sketch the graph of the missing pieces of Lia's path.



Unit 4 Lesson 10

104

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.B.5
2	1	HSF.IF.B.5
3	1	HSF.IF.B.5
4	2	HSF.IF.B.5

Notes:

Additional Practice

A1.4.11

Problems 1–3: Determine the value of each piecewise-defined function, $f(x)$.

1. $f(0) = \underline{\hspace{1cm} 1 \hspace{1cm}}$

2. $f(4) = \underline{\hspace{1cm} 9 \hspace{1cm}}$

3. $f(6) = \underline{\hspace{1cm} 9 \hspace{1cm}}$

$$f(x) = \begin{cases} 2x + 1, & 0 \leq x < 4, \\ 9 & x \geq 4 \end{cases}$$

Problems 4–5: A car rental company charges \$55 to rent a car for the first day, \$110 for up to two days, and \$170 for more than two days but less than a week. Let R represent the dollar price of renting a car for d days. Complete the table.

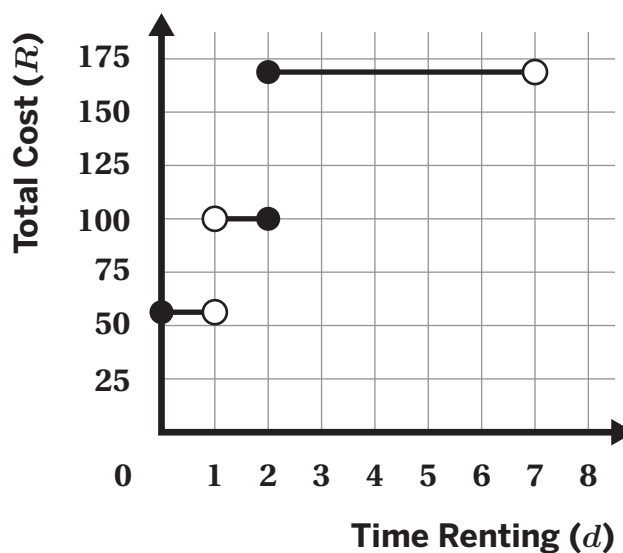
4. Complete the table.

Time renting (d)	0	0.75	1	1.25	2.5
Total cost (R)	0	55	55	110	170

5. The rental company tried to represent their pricing with this graph. What is correct and what should change to make the graph more accurate?

Responses vary.

- The horizontal lines for the time range for renting for 0–1 day and over 2 days are on the correct line. However, the point at (1, 55) should be filled in, and the one at (2, 170) should be open.
- The horizontal line at $y = 100$ is incorrect; it should be at $y = 110$.



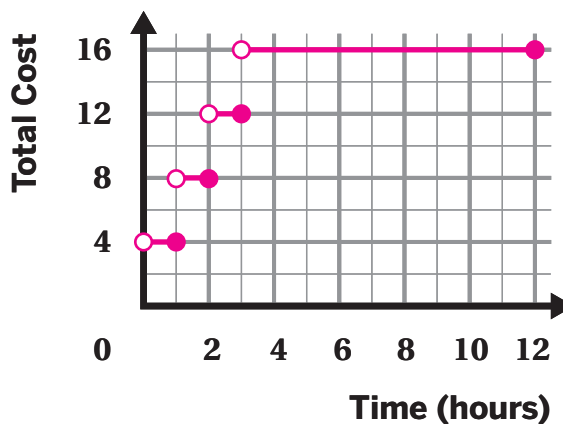
Problems 6–7: The cost of parking at the beach is \$4 for the first hour, \$8 for up to two hours, \$12 for up to three hours, and \$16 for more than three hours.

Let $c(t)$ represent the dollar price of parking for t hours.

6. Complete the table.

Hours parking, t	Total cost, $c(t)$
0	0
0.75	4
1	4
1.25	8
2	8
2.5	12
4	16

7. Graph the function $c(t)$ for $0 < t \leq 12$



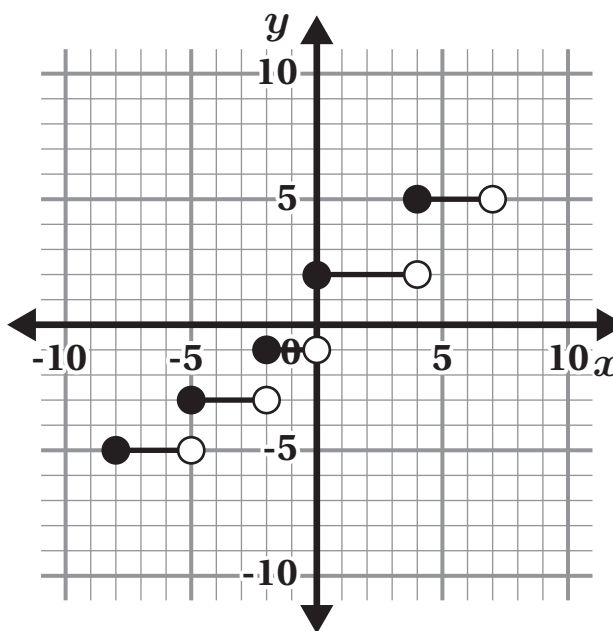
Problems 8–9: Here is the graph of $r(x)$.

8. Determine the following values:

- a $r(-5) = -3$
 b $r(0) = 2$
 c $r(6) = 5$

9. Complete $r(x)$ so that it matches the graph

$$r(x) = \begin{cases} -5, & -8 \leq x < -5 \\ -3, & -5 \leq x < -2 \\ -1, & -2 \leq x < 0 \\ 2, & 0 \leq x < 4 \\ 5, & 4 \leq x < 7 \end{cases}$$



Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.11

Problems 1–3: Determine the value of each piecewise-defined function, $f(x)$.

1. $f(0) =$ 1

2. $f(4) =$ 9

3. $f(6) =$ 9

$$f(x) = \begin{cases} 2x + 1 & 0 \leq x < 4 \\ 9 & x \geq 4 \end{cases}$$

Problems 4–5: A car rental company charges \$55 to rent a car for the first day, \$110 for up to two days, and \$170 for more than two days but less than a week. Let R represent the dollar price of renting a car for d days. Complete the table.

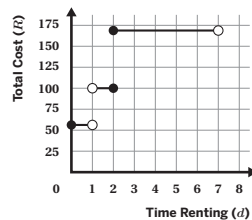
4. Complete the table.

Time renting (d)	0	0.75	1	1.25	2.5
Total cost (R)	0	55	55	110	170

5. The rental company tried to represent their pricing with this graph. What is correct and what should change to make the graph more accurate?

Responses vary.

- The horizontal lines for the time range for renting for 0–1 day and over 2 days are on the correct line. However, the point at (1, 55) should be filled in, and the one at (2, 170) should be open.
- The horizontal line at $y = 100$ is incorrect; it should be at $y = 110$.



Unit 4 Lesson 11

105

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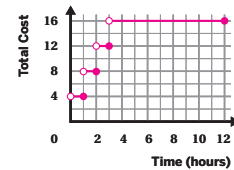
Problems 6–7: The cost of parking at the beach is \$4 for the first hour, \$8 for up to two hours, \$12 for up to three hours, and \$16 for more than three hours.

Let $c(t)$ represent the dollar price of parking for t hours.

6. Complete the table.

Hours parking, t	Total cost, $c(t)$
0	0
0.75	4
1	4
1.25	8
2	8
2.5	12
4	16

7. Graph the function $c(t)$ for $0 < t \leq 12$.



Problems 8–9: Here is the graph of $r(x)$.

8. Determine the following values:

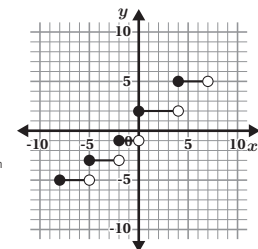
a. $r(-5) =$ -3

b. $r(0) =$ 2

c. $r(6) =$ 5

9. Complete $r(x)$ so that it matches the graph

$$r(x) = \begin{cases} -5, & -8 \leq x < -5 \\ -3, & -5 \leq x < -2 \\ -1, & -2 \leq x < 0 \\ 2, & 0 \leq x < 4 \\ 5, & 4 \leq x < 7 \end{cases}$$



Unit 4 Lesson 11

106

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.A.1
2	1	HSF.IF.A.1
3	1	HSF.IF.A.1
4	1	HSF.IF.C.7.B
5	2	HSF.IF.C.7.B
6	1	HSF.IF.C.7.B
7	1	HSF.IF.C.7.B
8	1	HSF.IF.A.1
9	1	HSF.IF.C.7.B

Notes:

Additional Practice

A1.4.12

Problems 1–2: A shipping company charges shipping fees depending on the weight of the package.

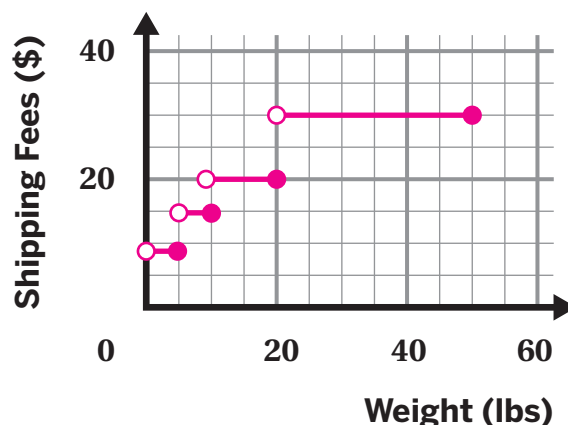
- Packages weighing up to 5 pounds cost \$8.
- Packages weighing from more than 5 pounds and up to 10 pounds cost \$15.
- Packages weighing more than 10 pounds and up to 20 pounds cost \$20.
- Packages weighing more than 20 pounds and up to 50 pounds cost \$30.

Let $c(x)$ represent the shipping fee for an order that weighs x pounds.

1. Complete the table.

Weight (lbs)	Shipping Fee (\$)
3	8
5	8
10	15
19	20
20	20
30	30
40	30
50	30

2. Make a graph that represents the function $c(x)$.



3. Sandra wrote a piecewise-defined function, $f(x)$, that models the price of a photo session lasting x minutes.

- \$50 for 60 minutes or less.
- \$75 for sessions longer than 60 and up to 90 minutes.
- \$1 per minute for session over 90 minutes and lasting up to 3 hours.

$$f(x) = \begin{cases} 50 & 0 < x \leq 60 \\ 75 & 60 < x < 90 \\ x & 90 < x \leq 3 \end{cases}$$

Describe two things that are incorrect in Sandra's function.

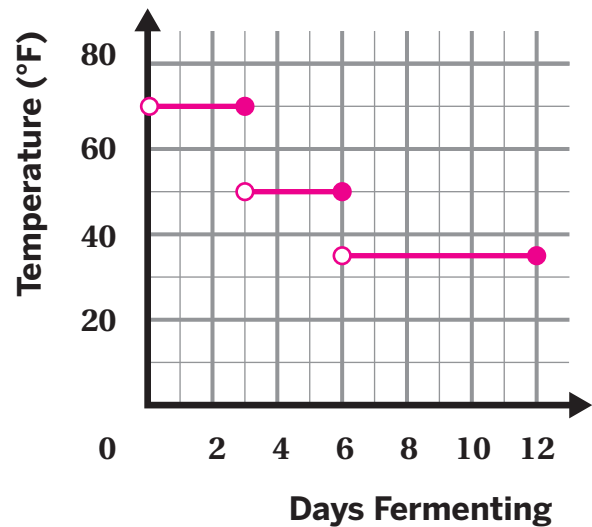
Responses vary. The second inequality should be written $60 < x \leq 90$, because this time frame includes 90-minute sessions. The third inequality should be written $90 < x \leq 180$, because both amounts of time need to be in minutes and that is how many minutes are in 3 hours.

Problems 4–7: Bethany is making kimchi, a tangy cabbage dish, with her grandmother. The directions for making kimchi are:

- Ferment the cabbage for the first 3 days at a temperature of 70°F
- Store the dish at a temperature of 50°F for the next 3 days
- Store the dish in a 35°F refrigerator for the next 6 days

4. Here is part of the piecewise function, $f(t)$, and the graph that models this situation. Fill in the missing numbers and symbols.
5. Complete the graph to represent this situation.

$$f(t) = \begin{cases} \boxed{70}, & 0 < t \leq \boxed{3} \\ 50, & \boxed{3} < t \leq 6 \\ 35, & \boxed{6} < t \leq 12 \end{cases}$$



6. What does $f(9)$ represent in this situation?
 $f(9)$ represent the temperature the kimchi is stored at on the 9th day.
7. What is the value of $f(9)$?
35°F

Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.12

Problems 1–2: A shipping company charges shipping fees depending on the weight of the package.

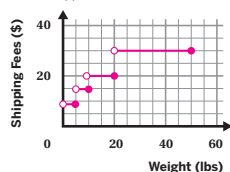
- Packages weighing up to 5 pounds cost \$8.
- Packages weighing from more than 5 pounds and up to 10 pounds cost \$15.
- Packages weighing more than 10 pounds and up to 20 pounds cost \$20.
- Packages weighing more than 20 pounds and up to 50 pounds cost \$30.

Let $c(x)$ represent the shipping fee for an order that weighs x pounds.

1. Complete the table.

Weight (lbs)	Shipping Fee (\$)
3	8
5	8
10	15
19	20
20	20
30	30
40	30
50	30

2. Make a graph that represents the function $c(x)$.



3. Sandra wrote a piecewise-defined function, $f(x)$, that models the price of a photo session lasting x minutes.

- \$50 for 60 minutes or less.
- \$75 for sessions longer than 60 and up to 90 minutes.
- \$1 per minute for session over 90 minutes and lasting up to 3 hours.

$$f(x) = \begin{cases} 50 & 0 < x \leq 60 \\ 75 & 60 < x \leq 90 \\ x & 90 < x \leq 3 \end{cases}$$

Describe two things that are incorrect in Sandra's function.

Responses vary. The second inequality should be written $60 < x \leq 90$, because this time frame includes 90-minute sessions. The third inequality should be written $90 < x \leq 180$, because both amounts of time need to be in minutes and that is how many minutes are in 3 hours.

Unit 4 Lesson 12

107

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Name: _____ Date: _____ Period: _____

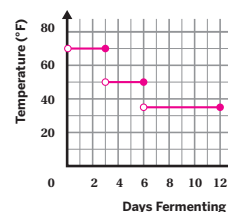
Problems 4–7: Bethany is making kimchi, a tangy cabbage dish, with her grandmother. The directions for making kimchi are:

- Ferment the cabbage for the first 3 days at a temperature of 70°F
- Store the dish at a temperature of 50°F for the next 3 days
- Store the dish in a 35°F refrigerator for the next 6 days

4. Here is part of the piecewise function, $f(t)$, and the graph that models this situation. Fill in the missing numbers and symbols.

$$f(t) = \begin{cases} 70 & 0 < t \leq 3 \\ 50 & 3 < t \leq 6 \\ 35 & 6 < t \leq 12 \end{cases}$$

5. Complete the graph to represent this situation.



6. What does $f(9)$ represent in this situation?

$f(9)$ represent the temperature the kimchi is stored at on the 9th day.

7. What is the value of $f(9)$?

35°F

Unit 4 Lesson 12

108

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.C.7.B
2	1	HSF.IF.C.7.B
3	2	HSF.IF.C.7.B
4	1	HSF.IF.C.7.B
5	1	HSF.IF.C.7.B
6	2	HSF.IF.A.1
7	1	HSF.IF.A.1

Notes:

Additional Practice

A1.4.13

Problems 1–3: Let $p(n)$ represent the value of term n in this sequence: 3, 8, 13, 18, 23, ...
Write the number that makes each equation true.

1. $p(1) = \underline{\hspace{1cm} 3 \hspace{1cm}}$ 2. $p(9 - 1) = \underline{\hspace{1cm} 38 \hspace{1cm}}$ 3. $p(n) = p(n - 1) + \underline{\hspace{1cm} 5 \hspace{1cm}}$

4. Match each sequence with one of the definitions.

Sequence

Definition

a. 2, 6, 18, 54, ...

$\underline{\hspace{1cm} b \hspace{1cm}}$ $f(1) = 2$
 $f(n) = f(n - 1) + 15$

b. 2, 17, 32, 47, ...

$\underline{\hspace{1cm} c \hspace{1cm}}$ $f(1) = 2$
 $f(n) = \frac{3}{2} \cdot f(n - 1)$

c. 2, 3, $\frac{9}{2}$, $\frac{27}{4}$, ...

$\underline{\hspace{1cm} a \hspace{1cm}}$ $f(1) = 2$
 $f(n) = 3 \cdot f(n - 1)$

Problems 5–6: Write the first four terms of each sequence.

5. $a(1) = \frac{1}{4}$
 $a(n) = 2 \cdot a(n - 1)$
 $\underline{\hspace{1cm} \frac{1}{4} \hspace{1cm}} \rightarrow \underline{\hspace{1cm} \frac{1}{2} \hspace{1cm}} \rightarrow \underline{\hspace{1cm} 1 \hspace{1cm}} \rightarrow \underline{\hspace{1cm} 2 \hspace{1cm}}$

6. $c(1) = \frac{1}{4}$
 $c(n) = c(n - 1) + 2$
 $\underline{\hspace{1cm} \frac{1}{4} \hspace{1cm}} \rightarrow \underline{\hspace{1cm} 2\frac{1}{4} \hspace{1cm}} \rightarrow \underline{\hspace{1cm} 4\frac{1}{4} \hspace{1cm}} \rightarrow \underline{\hspace{1cm} 6\frac{1}{4} \hspace{1cm}}$

Problems 7–8: Here are the first five terms of some sequences. Write a recursive definition for each one.

7. 30, 25, 20, 15, 10

$f(1) = \underline{\hspace{1cm} 30 \hspace{1cm}}$
 $f(n) = \underline{\hspace{1cm} f(n - 1) - 5 \hspace{1cm}}$

8. 4, 12, 36, 108, 324

$g(1) = \underline{\hspace{1cm} 4 \hspace{1cm}}$
 $g(n) = \underline{\hspace{1cm} 3 \cdot g(n - 1) \hspace{1cm}}$

9. An arithmetic sequence $a(n)$ and geometric sequence $g(n)$ both have the same first and third term. Determine a recursive definition for each. *Responses vary.*

Arithmetic Sequence

$$a(1) = 2$$

$$a(n) = f(n - 1) + 3$$

Geometric Sequence

$$g(1) = 2$$

$$g(n) = 2 \cdot g(n - 1)$$

10. Write a recursive definition for the sequence in this table.

$$f(1) = \underline{500}$$

$$f(n) = \underline{\frac{1}{2} \cdot f(n - 1)}$$

Term	Value
1	500
2	250
3	125
4	62.5
5	31.25

Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.13

Problems 1–3: Let $p(n)$ represent the value of term n in this sequence: 3, 8, 13, 18, 23, ...
Write the number that makes each equation true.

1. $p(1) = \underline{3}$ 2. $p(9-1) = \underline{38}$ 3. $p(n) = p(n-1) + \underline{5}$

4. Match each sequence with one of the definitions.

Sequence	Definition
a. 2, 6, 18, 54, ...	$f(1) = 2$ $f(n) = f(n-1) + 15$ b
b. 2, 17, 32, 47, ...	$f(1) = 2$ $f(n) = \frac{3}{2} \cdot f(n-1)$ c
c. $2, 3, \frac{9}{2}, \frac{27}{4}, \dots$	$f(1) = 2$ $f(n) = 3 \cdot f(n-1)$ a

Problems 5–6: Write the first four terms of each sequence.

5. $a(1) = \frac{1}{4}$
 $a(n) = 2 \cdot a(n-1)$
 $\underline{\frac{1}{4}} \quad \underline{\frac{1}{2}} \quad \underline{1} \quad \underline{2}$
6. $c(1) = \frac{1}{4}$
 $c(n) = c(n-1) + 2$
 $\underline{\frac{1}{4}} \quad \underline{2\frac{1}{4}} \quad \underline{4\frac{1}{4}} \quad \underline{6\frac{1}{4}}$

Problems 7–8: Here are the first five terms of some sequences. Write a recursive definition for each one.

7. 30, 25, 20, 15, 10
 $f(1) = \underline{30}$
 $f(n) = \underline{f(n-1) - 5}$
8. 4, 12, 36, 108, 324
 $g(1) = \underline{4}$
 $g(n) = \underline{3 \cdot g(n-1)}$

Unit 4 Lesson 13

109

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Name: _____ Date: _____ Period: _____

9. An arithmetic sequence $a(n)$ and geometric sequence $g(n)$ both have the same first and third term. Determine a recursive definition for each. **Responses vary.**

Arithmetic Sequence

$$a(1) = 2$$

$$a(n) = f(n-1) + 3$$

Geometric Sequence

$$g(1) = 2$$

$$g(n) = 2 \cdot g(n-1)$$

10. Write a recursive definition for the sequence in this table.

$$f(1) = \underline{500}$$

$$f(n) = \underline{\frac{1}{2} \cdot f(n-1)}$$

Term	Value
1	500
2	250
3	125
4	62.5
5	31.25

Unit 4 Lesson 13

110

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.A.3
2	1	HSF.IF.A.3
3	1	HSF.BF.A.2
4	2	HSF.BF.A.2
5	1	HSF.BF.A.2
6	1	HSF.BF.A.2
7	2	HSF.BF.A.2
8	2	HSF.BF.A.2
9	2	HSF.BF.A.2
10	2	HSF.BF.A.2

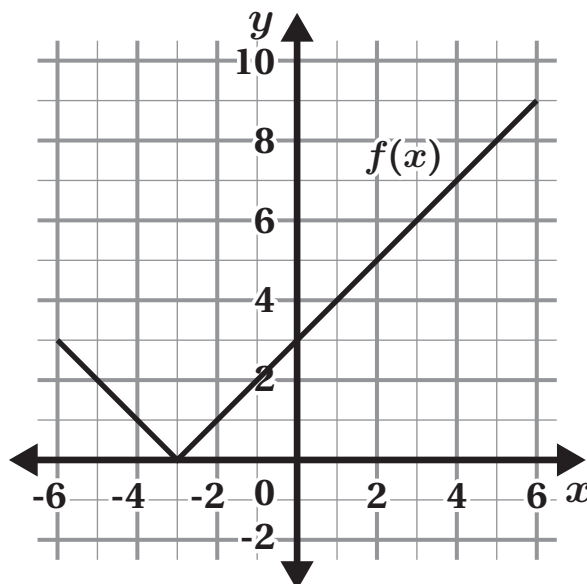
Notes:

Additional Practice

A1.4.15

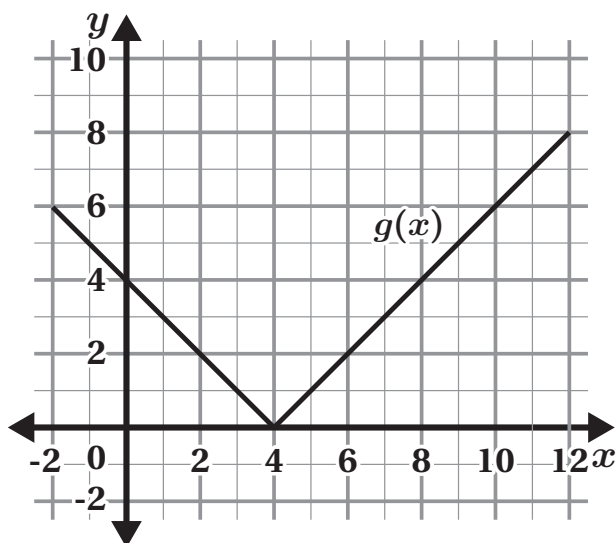
Problems 1–3: Use the graph of $f(x)$ to determine each value.

1. $f(0) = \underline{\hspace{1cm} 3 \hspace{1cm}}$
2. $f(2) = \underline{\hspace{1cm} 5 \hspace{1cm}}$
3. $f(-3) = \underline{\hspace{1cm} 0 \hspace{1cm}}$

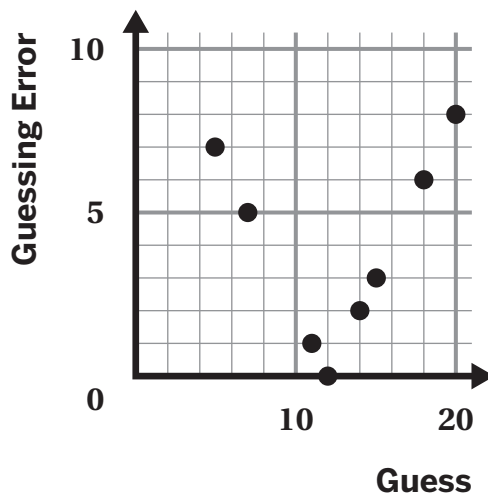


4. Which equation represents the graph of $g(x)$? Circle your choice.

- ☒ A. $g(x) = |x - 4|$
- B. $g(x) = |x + 4|$
- C. $g(x) = |x| - 4$
- D. $g(x) = |x| + 4$



5. A group of 8 friends played a number guessing game. They were asked to select a number between 1 and 20. The graph shows the guesses made by each of the 8 friends. The actual number was 12.



What is the greatest guessing error?

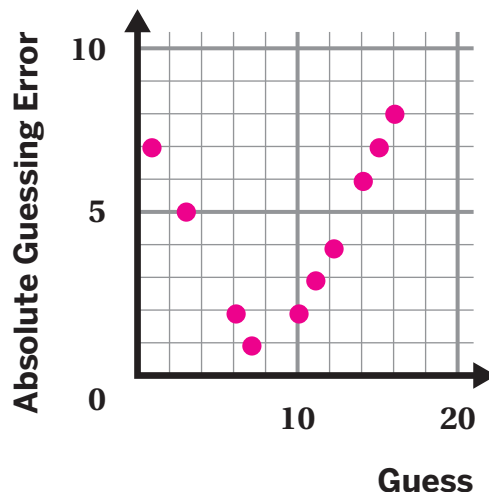
8

Problems 6–8: A group of 10 friends played a number guessing game. They were asked to select a number between 1 and 20. The person closest to the target number wins. The table below shows the guesses made by each of the 10 friends.

6. The actual number was 8. Complete the table with the absolute guessing errors.

Guess	15	3	10	14	16	12	1	7	6	11
Guessing Error	7	5	2	6	8	4	7	1	2	3

7. Graph each guess, x , and its corresponding absolute guessing error, $g(x)$ on the coordinate plane.



8. Nadine writes $g(20) = 12$.

What does her equation mean?

If someone guessed the number 20, they were 12 away from the actual number.

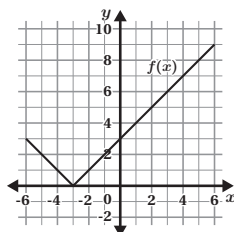
Name: _____ Date: _____ Period: _____

Additional Practice

A1.4.15

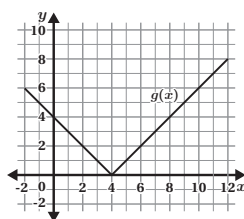
Problems 1–3: Use the graph of $f(x)$ to determine each value.

1. $f(0) =$ 3
2. $f(2) =$ 5
3. $f(-3) =$ 0



4. Which equation represents the graph of $g(x)$? Circle your choice.

- ☒ A. $g(x) = |x - 4|$
☐ B. $g(x) = |x + 4|$
☐ C. $g(x) = |x| - 4$
☐ D. $g(x) = |x| + 4$



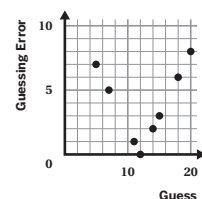
Unit 4 Lesson 15

113

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Name: _____ Date: _____ Period: _____

5. A group of 8 friends played a number guessing game. They were asked to select a number between 1 and 20. The graph shows the guesses made by each of the 8 friends. The actual number was 12.



What is the greatest guessing error?

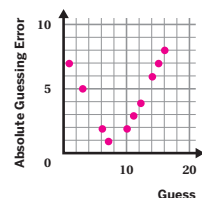
8

Problems 6–8: A group of 10 friends played a number guessing game. They were asked to select a number between 1 and 20. The person closest to the target number wins. The table below shows the guesses made by each of the 10 friends.

6. The actual number was 8. Complete the table with the absolute guessing errors.

Guess	15	3	10	14	16	12	1	7	6	11
Guessing Error	7	5	2	6	8	4	7	1	2	3

7. Graph each guess, x , and its corresponding absolute guessing error, $g(x)$, on the coordinate plane.



8. Nadine writes $g(20) = 12$.

What does her equation mean?

If someone guessed the number 20, they were 12 away from the actual number.

Unit 4 Lesson 15

114

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.IF.A.2
2	1	HSF.IF.A.2
3	1	HSF.IF.A.2
4	1	HSF.IF.C.7.B
5	1	HSF.IF.C.7.B
6	1	HSF.IF.C.7.B
7	1	HSF.IF.C.7.B
8	2	HSF.IF.A.2

Notes:

Additional Practice

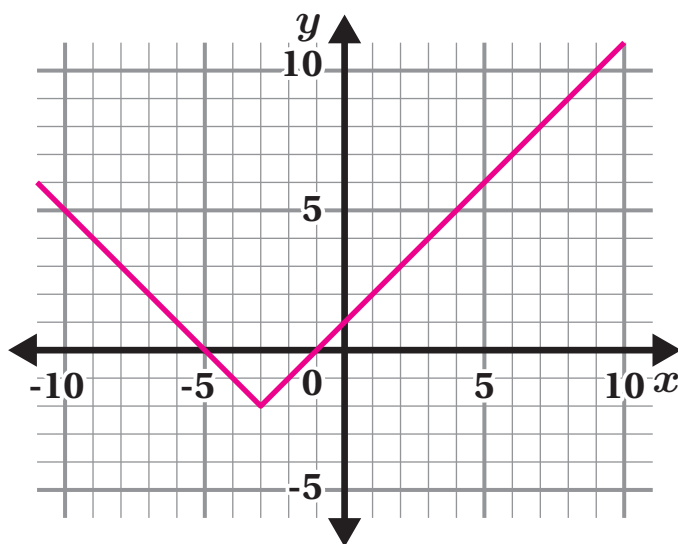
A1.4.16

Problems 1–3: Write each expression as a single integer.

1. $|-5|$ **5** 2. $|12| - 4$ **8** 3. $|-6| + 3$ **9**

4. Graph $f(x) = |x + 3| - 2$. Use the table if it helps your thinking.

x	$f(x)$
-5	0
-3	-2
0	1
1	2
2	3

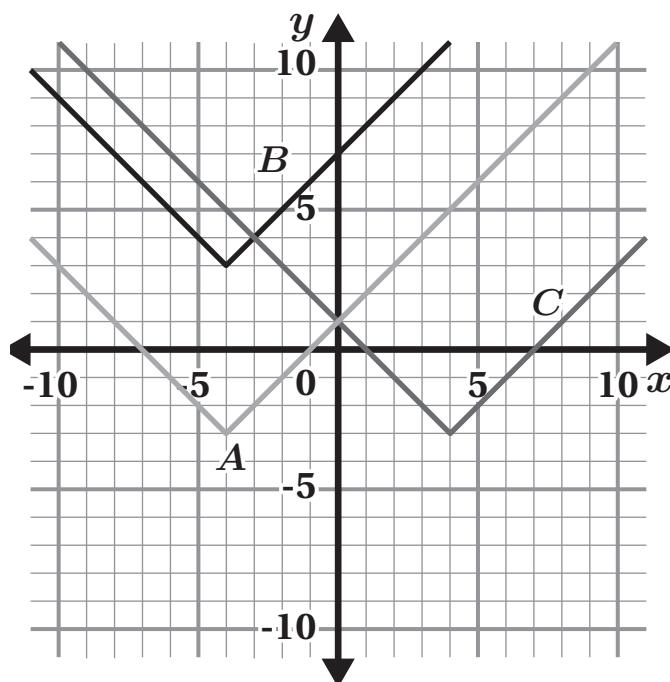


5. Match each function with its graph.

$f(x) = |x - 4| - 3$ **C**

$f(x) = |x + 4| + 3$ **B**

$f(x) = |x + 4| - 3$ **A**

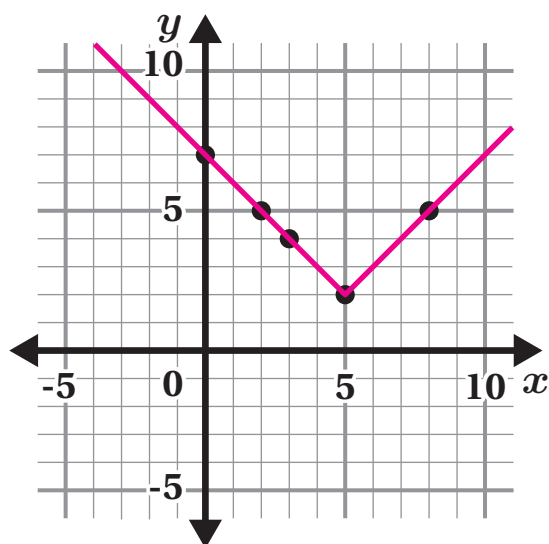


6. Here are some points on the graph of $g(x) = |x - 5| + 2$.

- Sketch a graph of $g(x)$.
- Describe the graph using some of these terms:

positive	maximum	increasing	domain
negative	minimum	decreasing	range
symmetry			
piecewise-defined function			

Responses vary. This graph has a minimum of 2 when $x = 5$. The domain is all numbers and the range is all numbers greater than or equal to 2. The graph is increasing when $x > 5$ and decreasing when $x < 5$.

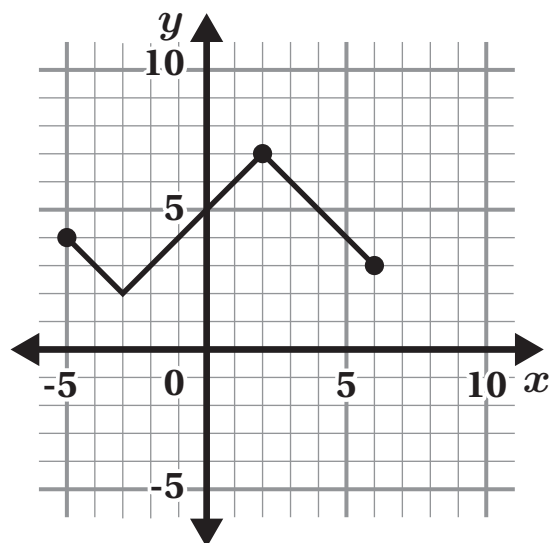


7. Determine two different piecewise-defined functions that could represent this graph.

Responses vary.

$$f(x) = \begin{cases} |x + 3| + 2 & -5 \leq x < 2 \\ -|x - 2| + 7 & 2 \leq x \leq 6 \end{cases}$$

$$f(x) = \begin{cases} |x + 3| + 2 & -5 \leq x \leq 2 \\ -x + 9 & 2 \leq x \leq 6 \end{cases}$$



Name: _____ Date: _____ Period: _____

Additional Practice

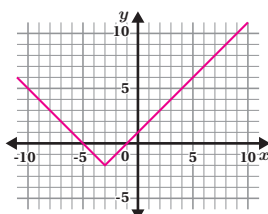
A1.4.16

Problems 1–3: Write each expression as a single integer.

1. $|-5|$ **5** 2. $|12| - 4$ **8** 3. $|-6| + 3$ **9**

4. Graph $f(x) = |x + 3| - 2$. Use the table if it helps your thinking.

x	$f(x)$
-5	0
-3	-2
0	1
1	2
2	3

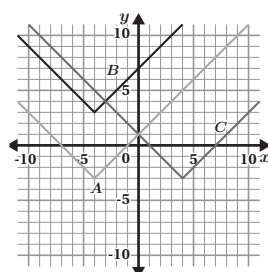


5. Match each function with its graph.

$f(x) = |x - 4| - 3$ **C**

$f(x) = |x + 4| + 3$ **B**

$f(x) = |x + 4| - 3$ **A**



Unit 4 Lesson 16

115

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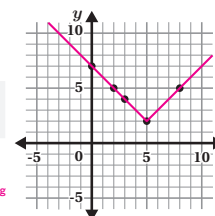
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6. Here are some points on the graph of

$g(x) = |x - 5| + 2$.

- Sketch a graph of $g(x)$.
- Describe the graph using some of these terms:

positive maximum increasing domain
negative minimum decreasing range
symmetry piecewise-defined function



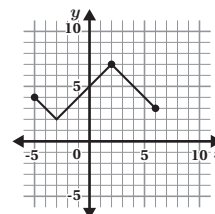
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7. Determine two different piecewise-defined functions that could represent this graph.

Responses vary.

$f(x) = \begin{cases} |x + 3| + 2 & -5 \leq x < 2 \\ -|x - 2| + 7 & 2 \leq x \leq 6 \end{cases}$

$f(x) = \begin{cases} |x + 3| + 2 & -5 \leq x \leq 2 \\ -x + 9 & 2 \leq x \leq 6 \end{cases}$



Unit 4 Lesson 16

116

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.NS.C.7
2	1	6.NS.C.7
3	1	6.NS.C.7
4	1	HSF.IF.C.7.B
5	1	HSF.IF.C.7.B
6	2	HSF.IF.C.7.B, HSF.IF.B.4
7	2	HSF.IF.C.7.B

Notes:

Additional Practice

A1.4.18

1. The graph of function $k(x)$ passes through the point at $(-9, 4)$. The graph of the inverse of $k(x)$ must pass through which of the following points?

A. $(-9, 4)$

B. $(9, -4)$

☒ C. $(4, -9)$

D. $(-4, 9)$

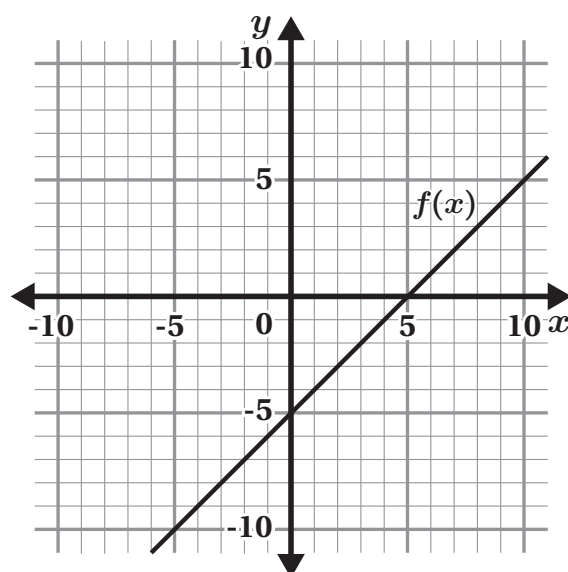
2. Use the graph of $f(x) = x - 5$ to determine the equation for the inverse function.

A. $a(x) = -5x$

☒ B. $b(x) = x + 5$

C. $c(x) = -x + 5$

D. $d(x) = 5 - x$



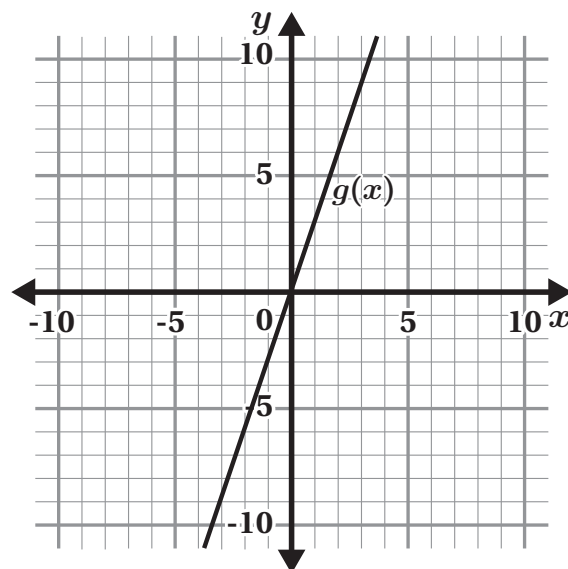
3. Use the graph of $g(x) = 3x$ to determine the equation for the inverse function.

A. $a(x) = -3x$

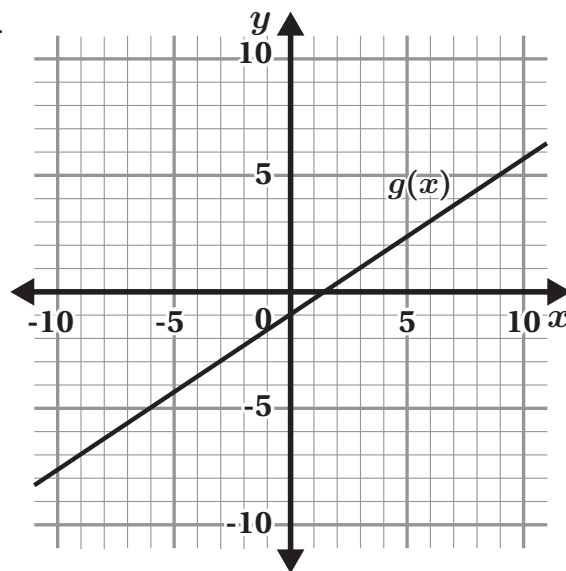
B. $b(x) = x - 3$

C. $c(x) = -x + 3$

☒ D. $d(x) = \frac{1}{3}x$



Problems 4–6: Here is the graph of $g(x) = \frac{2}{3}x - 1$.



4. List two points that the inverse function $n(x)$ will pass through.

Responses vary. $(-1, 0)$ and $(1, 3)$

5. Write an equation for the inverse function $n(x)$.

$y = \frac{3}{2}(x + 1)$ or equivalent.

6. Which of the functions will pass through point $(-3, -3)$? Circle your choice.

$g(x)$

$n(x)$

Both

Neither

Explain your reasoning.

Explanations vary. I substituted $(-3, -3)$ into the equation for $g(x)$ and $n(x)$ to see if they were true statements. Since both $g(-3) = -3$ and $n(-3) = -3$, they both pass through the point $(-3, -3)$.

7. Kendra attempts to determine the inverse of the function $h(x) = -4x + 12$. She graphs the inverse of $h(x)$ and determines the graph of the inverse is a line with a slope of 4 and a vertical intercept of -12 , because the inverse is found by reversing the operations of the original function. Do you agree? Explain your thinking.

No. Explanations vary. Two functions are inverses of each other if their input-output pairs are reversed, not if the operations are reversed.

Name: _____ Date: _____ Period: _____

Additional Practice

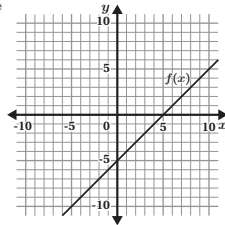
A1.4.18

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 C. $(4, -9)$
 D. $(-4, 9)$

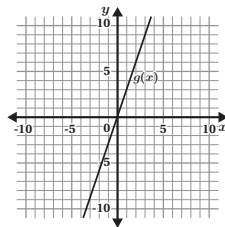
2. Use the graph of $f(x) = x - 5$ to determine the equation for the inverse function.

A. $a(x) = -5x$
 B. $b(x) = x + 5$
 C. $c(x) = -x + 5$
 D. $d(x) = 5 - x$



3. Use the graph of $g(x) = 3x$ to determine the equation for the inverse function.

A. $a(x) = -3x$
 B. $b(x) = x - 3$
 C. $c(x) = -x + 3$
 D. $d(x) = \frac{1}{3}x$



Unit 4 Lesson 18

119

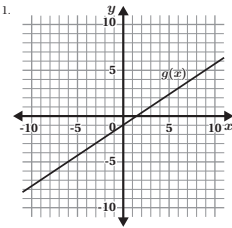
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Name: _____ Date: _____ Period: _____

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$g(x)$ $n(x)$ **Both** Neither

Explain your reasoning.

Explanations vary. I substituted $(-3, -3)$ into the equation for $g(x)$ and $n(x)$ to see if they were true statements. Since both $g(-3) = -3$ and $n(-3) = -3$, they both pass through the point $(-3, -3)$.

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No. Explanations vary. Two functions are inverses of each other if their input-output pairs are reversed, not if the operations are reversed.

Unit 4 Lesson 18

120

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	HSF.BF.B.4
2	1	HSF.BF.B.4
3	1	HSF.BF.B.4
4	1	HSF.BF.B.4
5	2	HSF.BF.B.4
6	2	HSF.BF.B.4
7	2	HSF.BF.B.4

Notes: